Spring 2026: Automorphic Representations

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*based on lectures by Zhiwei Yun

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Administrative Notes

The topic will be introduction to automorphic forms. There will be 2 quizzes. Textbook: "Automorphic Forms and Representations" by Bump.

1 Overview and Tate's Thesis

Let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ acting on the upper half plane $\mathbb{H}$. Modular forms are certain holomorphic functions transformed in certain ways under the action of Γ. Today we study Tate's thesis, for reproving analytic properties of $\zeta(s), L(s,\chi)$ using Fourier analysis. The theory of modular forms are about GL_2, Tate's thesis is about GL_1.

Let F be a global field. Recall the places of F consists of finite places and infinite places for a number field; for a function field, places correspond to closed points on the corresponding curve.

Given v a place of F, let's normalize the norm $|-|_v$ as:

1. If $F = \mathbb{Q}$, then $|p^n a|_p = p^{-n}$, $|x|_\infty = |x|_{\mathbb{R}}$

2. If $F/\mathbb{Q}$ is number field. For any place v, it's either over $\mathbb{Q}_p$ or $\mathbb{R}$. Then we define:

$$|x|_v = |N_{F_v/\mathbb{Q}_p}(x)|_p$$

   or similarly for a place over $\mathbb{R}$.

   Example. *If $F_v = \mathbb{C}$, I have:*

$$|x|_v = |x \cdot \overline{x}|_{\mathbb{R}} = \|x\|^2$$

3. For function field $F = \mathbb{F}_q(X)$, there is no canonical copy of $\mathbb{F}_q(t)$ inside. So alternatively, for a closed point v on X, we have:

$$F_v \simeq k_v((t)), \quad k_v = \mathbb{F}_{q_v}$$

   then we define $|t|_v = q_v^{-1}$

Proposition 1.1 (Product Formula)**.** *For $x \in F^\times$, we have:*

$$\prod_v |x|_v = 1.$$

For number fields, the proof reduces to $\mathbb{Q}$. For a function field $F = \mathbb{F}_q(X)$ and $f \in F$ vanishes to order n at v, then:

$$|f|_v = (q^{\deg(v)})^{-n} = q^{-\deg(v)\cdot \mathrm{ord}_v(f)}$$

The product formula essentially says:

$$\sum_{v \in |X|} \deg(v) \mathrm{ord}_v(f) = 0$$

Let's denote S_∞ be the set of infinite places for a number field.

1.1 Structures of Adèle and Idèle

Definition 1.2 (Adèle, Idèle)**.** The **ring of Adèles** is:

$$\mathbb{A}_F \subset \prod_v F_v, \quad (a_v) \in \mathbb{A}_F \iff \text{for almost all } v, a_v \in \mathcal{O}_v$$

In other words:

$$\mathbb{A}_F = \bigcup_{S \text{ finite set of places } S \supset S_\infty} (\prod_{v \notin S} \mathcal{O}_v) \times (\prod_{v \in S} F_v)$$

Let's put product topology on each piece, and put direct limit topology on $\mathbb{A}_F$. This general procedure is called "restricted product".

The **group of idéles** is defined to be $\mathbb{A}_F^\times$. Concretely, it consists of $(a_v) \in \prod F_v^\times$ such that almost all v, $a_v \in \mathcal{O}_v^\times$.

For v non-archimedean, we have a short exact sequence:

$$1 \to \mathcal{O}_v^\times \to F_v^\times \to \mathbb{Z} \to 0$$

we put profinite topology on the first entry and discrete topology on the third entry. This gives a topology on $F_v^\times$. As a topological space, we have:

$$F_v^\times = \coprod_{n \in \mathbb{Z}} t^n \mathcal{O}_v^\times$$

this is the subspace topology inherited from F_v. We can thus as before give idéle a topology making it a topological group.

Remark. *A neighborhood basis of* 0 *in* $\mathbb{A}_F$ *will be: for S a finite set of places $v \in S$, $U_v \subset F_v$ neighborhood of* 0*, we include*

$$(\prod_{v \notin S} \mathcal{O}_v) \times (\prod_{v \in S} U_v)$$

Modular forms relate to representation theory of $\mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}})$.

Consider the map:

$$\det : \mathrm{GL}_2(\mathbb{A}_F) \to \mathbb{A}_F^\times$$

we must endow $\mathrm{GL}_2(\mathbb{A}_F)$ with a topology making det continuous. Let's embed $\mathrm{GL}_2(\mathbb{A}_F)$ into a 5-dimensional affine space. Thus $\mathrm{GL}_2(\mathbb{A}_F)$ gets subspace topology of $(\mathbb{A}_F)^5$.

We can express $\mathrm{GL}_2(\mathbb{A}_F)$ as a restricted product of $\mathrm{GL}_2(F_v)$ with respect to $\mathrm{GL}_2(\mathcal{O}_v)$ as a topological group. One can show that a representation of $\mathrm{GL}_2(\mathbb{A}_F)$ is equivalent to a collection of representations π_v of $\mathrm{GL}_2(F_v)$, Roughly,

$$\pi \simeq \otimes_v' \pi_v$$

Therefore, we will study (mostly infinite dimensional) representations of

$$\mathrm{GL}_2(\mathbb{Q}_p), \mathrm{GL}_2(\mathbb{R}), \mathrm{GL}_2(\mathbb{C})$$

Tate's thesis is about representations of $\mathrm{GL}_1(\mathbb{A}_F) = \mathbb{A}_F^\times$.

We have diagonal copy $F \subset \mathbb{A}_F$, this is a discrete subgroup, because for every pair $a, b \in F$,

$$\prod |a-b|_v = 1$$

this will imply a, b are far away at some place. Similarly $F^\times \subset \mathbb{A}_F^\times$ is also discrete subgroup.

We're interested in continuous characters

$$\chi : \mathbb{A}_F^\times \to \mathbb{C}^\times$$

We have a canonical copy $F_v^\times \to \mathbb{A}_F^\times$, so that χ gives a character:

$$\chi_v : F_v^\times \to \mathbb{C}^\times$$

for each v. The continuity of χ forces $\chi_v|_{\mathcal{O}_v^\times} = 1$ (i.e. χ_v is unramified) for almost all finite v. In fact this is an iff: if I have a collection of χ_v with almost all of them unramified, I can define:

$$\chi((a_v)) = \prod_v \chi_v(a_v)$$

Remark. *Any unramified character has form*

$$\chi_v(a) = |a|_v^s$$

for some $s \in \mathbb{C}$.

We're particularly interested in the so called **idèle class characters**:

$$F^\times \backslash \mathbb{A}_F^\times \to \mathbb{C}^\times$$

namely continuous characters constant on the diagonal copy. Let's assess the group $F^\times \backslash \mathbb{A}_F^\times$. But before that, let's assess $F \backslash \mathbb{A}_F$

Proposition 1.3 (Approximation theorem)**.** *For a number field $F/\mathbb{Q}$, the groups:*

$$F + \prod_{v \text{ finite}} \mathcal{O}_v + \prod_{v|\infty} F_v = \mathbb{A}_F$$

Remark. *We have $F \otimes_{\mathbb{Q}} \mathbb{R} \simeq \prod_{v|\infty} F_v$.*

For function fields $F = \mathbb{F}_q(X)$, in general:

$$F + \prod_{v} \mathcal{O}_v \subsetneq \mathbb{A}_F$$

except for the case when $F \simeq \mathbb{F}_q(t)$. The failure accounts for the genus of the curve. A fact is that we have a canonical isomorphism:

$$F\backslash\mathbb{A}_F/\prod_v \mathcal{O}_v \simeq H^1(X, \mathcal{O}_X) \simeq \mathbb{F}_q^g$$

In fact, one can see this using the exact sequence:

$$0 \to \mathcal{O}_X \to \mathcal{K} \to \bigoplus_v F_v/\mathcal{O}_v \to 0$$

But if we replace one of $\mathcal{O}_v$ by F_v, we get a trivial quotient. Therefore the two-step complex

$$F \to \mathbb{A}_F/\prod \mathcal{O}_v$$

computing the cohomology of X.

Now return to the number field case. We have a surjection:

$$F \otimes_{\mathbb{Q}} \mathbb{R} \to F\backslash\mathbb{A}_F/\prod_{v \text{ finite}} \mathcal{O}_v$$

The kernel is simply $\mathcal{O}_F$, we have isomorphism as topological groups:

$$(S^1)^d = \mathbb{R}^d/\mathbb{Z}^d \simeq \mathcal{O}_F\backslash(F\otimes_{\mathbb{Q}})\mathbb{R} \simeq F\backslash\mathbb{A}_F/\prod_{v \text{ finite}} \mathcal{O}_v$$

Let's now assess idèle class groups. For $F = \mathbb{F}_q(X)$, we have:

$$F^\times\backslash\mathbb{A}_F^\times/\prod_{v\in|X|} \mathcal{O}_v^\times = F^\times\backslash(\bigoplus_{v\in|X|} F_v^\times/\mathcal{O}_v^\times) = F^\times\backslash\mathrm{Div}(X) = \mathrm{Pic}(X)$$

Recall we have a variety Pic_X a $\mathbb{F}_q$-scheme parametrizing line bundles on X. We can write:

$$\coprod_{n\in\mathbb{Z}} \mathrm{Pic}_X^n, \quad J_X = \mathrm{Pic}_X^0 \text{ is called Jacobian, an abelian variety of } \dim g.$$

We have

$$\mathrm{Pic}(X) = \mathrm{Pic}_X(\mathbb{F}_q) \supset \mathrm{Pic}_X^0(\mathbb{F}_q)$$

It's also useful to write the short exact sequence (track stablizer of the action):

$$1 \to \mathbb{F}_q^\times \to F^\times \to \mathrm{Pic}(X) \to 0$$

For $F/\mathbb{Q}$ a number field, we have

$$\prod_{v|\infty} F_v^\times \to F^\times \backslash \mathbb{A}_F^\times / \prod_{v \text{ finite}} \mathcal{O}_v^\times \to F^\times \backslash \mathbb{A}_{F,f}^\times / \prod \mathcal{O}_v^\times \to 0$$

where $\mathbb{A}_{F,f}^\times$ is idèle without archimedean coordinates. But as before:

$$F^\times \backslash \mathbb{A}_{F,f}^\times / \prod \mathcal{O}_v^\times = F^\times \backslash (\bigoplus_{v \text{ finite}} F_v^\times / \mathcal{O}_v^\times) = F^\times \backslash \mathrm{Div}(\mathcal{O}_F) = \mathrm{Cl}(\mathcal{O}_F)$$

One can also see the kernel of the first map is the unit group $\mathcal{O}_F^\times$. So we have short exact sequence:

$$1 \to \mathcal{O}_F^\times \to \prod_{v|\infty} F_v^\times \to F^\times \backslash \mathbb{A}_F^\times / \prod_{v \text{ finite}} \mathcal{O}_v^\times \to \mathrm{Cl}(\mathcal{O}_F) \to 0$$

Let

$$|-| = \prod_v |-|_v \colon F^\times \backslash \mathbb{A}_F^\times \to \mathbb{R}^{>0}$$

be the norm on $\mathbb{A}_F^\times$. Let's consider the above exact sequence restricting to norm one elements:

$$1 \to \mathcal{O}_F^\times \to (\prod_{v|\infty} F_v^\times)_1 \to F^\times \backslash \mathbb{A}_{F,1}^\times / \prod_{v \text{ finite}} \mathcal{O}_v^\times \xrightarrow{\gamma} \mathrm{Cl}(\mathcal{O}_F) \to 0$$

This extends to a diagram:

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & & & \\
 & & \downarrow & & \downarrow & & & & \\
 & & \mu_\infty(\mathcal{O}_F) & \hookrightarrow & (S^1)^{r_2} & & & & \\
 & & \downarrow & & \downarrow & & & & \\
1 & \longrightarrow & \mathcal{O}_F^\times & \longrightarrow & (\prod_{v|\infty} F_v^\times)_1 & \longrightarrow & F^\times \backslash \mathbb{A}_{F,1}^\times / \prod_{v \text{ finite}} \mathcal{O}_v^\times & \longrightarrow & \mathrm{Cl}(\mathcal{O}_F) \longrightarrow 0 \\
 & & \downarrow & & \downarrow{\scriptstyle (\log|-|_v)_v} & & & & \\
 & & \mathbb{Z}^{r_1+r_2-1} & \hookrightarrow & \mathbb{R}^{r_1+r_2-1} & & & & \\
 & & \downarrow & & \downarrow & & & & \\
 & & 0 & & 0 & & & &
\end{array}$$

with each row and column exact. We conclude that $\ker(\gamma)$ is a compact torus of $\dim = d-1$, so

$$F^\times \backslash \mathbb{A}_{F,1}^\times / \prod_{v \text{ finite}} \mathcal{O}_v^\times$$

is finite direct sum of copies of $(S^1)^{d-1}$.

1.2 Pontryagin Dual

Let X be an abelian locally compact group, the **Pontryagin dual** of X is

$$X^* \coloneqq \mathrm{Hom}_{\mathrm{cts}}(X, S^1)$$

This construction is functorial. Given:

$$1 \to \ker(\phi) \to X \xrightarrow{\phi} Y$$

we have

$$Y^* \xrightarrow{\phi^*} X \to \ker(\phi)^* = \mathrm{coker}\ (\phi^*) \to 1$$

Example. *1.* $\mathbb{Z}^* = S^1$

2. $\mathbb{R}^* = \mathbb{R}$, *by sending* $\xi \mapsto (x \mapsto e^{ix\xi})$

3. More generally, for a real vector space V*, its Pontryagin dual is canonically identified with its linear dual.*

4. $(\mathbb{Z}/n)^* = \mu_n$

5. $(\mathbb{Z}_p)^* = (\varprojlim \mathbb{Z}/p^n)^* = \varinjlim \mu_{p^n} = \mu_{p^\infty} \subset S^1$

6. $\mathbb{Q}_p^* = (\varinjlim \mathbb{Z}_p)^* = \varprojlim \mu_{p^\infty} = \{(x_n) \mid x_n \in \mu_{p^\infty}, x_{n+1}^p = x_n\}$.

7. Recall the definition of Tate twist:

$$\mathbb{Z}_p(1) = \varprojlim \mu_{p^n}$$

So that:

$$\mathbb{Z}_p(1)^* = (\varprojlim \mu_{p^n})^* = \varinjlim \mathbb{Z}/p^n = \mathbb{Q}_p/\mathbb{Z}_p$$

8. We have a canonical diagram:

$$\begin{array}{ccccccccc}
1 & \longrightarrow & (\mathbb{Q}_p/\mathbb{Z}_p)^* = \mathbb{Z}_p(1) \simeq \varprojlim \mu_{p^n} & \longrightarrow & \mathbb{Q}_p^* & \xrightarrow{\pi} & \mathbb{Z}_p^* = \mu_{p^\infty} & \longrightarrow & 1 \\
 & & \uparrow & & \uparrow{\scriptstyle \lambda} & & \uparrow{\scriptstyle \bar{\lambda}} & & \\
1 & \longrightarrow & \mathbb{Z}_p & \longrightarrow & \mathbb{Q}_p & \longrightarrow & \mathbb{Q}_p/\mathbb{Z}_p & \longrightarrow & 1
\end{array}$$

Let's describe each arrow in the diagram. The map π *is simply:*

$$\pi((x_n)) = x_1$$

The map $\overline{\lambda}$ *is given by:*

$$\overline{\lambda}(\sum_{k=-n}^{1} a_k p^k) = \exp(2\pi i \sum_{k=-n}^{1} a_k p^k)$$

The map λ is slightly tricky, we need to get a map:

$$\lambda : \mathbb{Q}_p \to \{(y_n) \mid y_n \in \mu_{p^\infty}, y_{n+1}^p = y_n\}$$

$$\lambda(\sum_{k=-n}^{\infty} a_k p^k) = (\exp(2\pi i \sum_{k=-n}^{1} a_k p^k), \exp(\frac{2\pi i}{p} \sum_{k=-n}^{0} a_k p^k), \exp(\frac{2\pi i}{p^2} \sum_{k=-n}^{1} a_k p^k), \cdots)$$

Essentially, what we construct is the image of 1 *in $\mathbb{Q}_p^*$, call it $\psi_p : \mathbb{Q}_p \to S^1$. Explicitly:*

$$\psi_p = (1, \exp \frac{2\pi i}{p}, \exp \frac{2\pi i}{p^2}, \cdots)$$

Translating back:

$$\psi_p(1/p^n) = \exp \frac{2\pi i}{p^n}, (\psi_p)|_{\mathbb{Z}_p} = 1$$

Using ψ_p, we can recover λ:

$$\lambda : \mathbb{Q}_p \to (\mathbb{Q}_p)^*, \quad x \mapsto (y \mapsto \psi_p(xy))$$

We can do similar procedure to $\mathbb{R}$:

$$\mathbb{R} \to \mathbb{R}^*, x \mapsto (y \mapsto e^{-2\pi i x y} = \psi_\infty(xy))$$

So we've constructed $\psi_p \in \mathbb{Q}_p^*$ for each p, and $\psi_\infty \in \mathbb{R}^*$, this induces a well-defined map:

$$\psi : \mathbb{A}_\mathbb{Q} \to S^1, \quad (x_v) \mapsto \prod_v \psi_v(x_v)$$

since the character ψ_p is trivial on $\mathbb{Z}_p$. Our choice also makes sure $\psi(\mathbb{Q}) = 1$.

1.3 Fourier Transform on Adèle and Idèle

Let's now establish Fourier transform. We wish to have:

$$C(X) \to C(X^*)$$

$$f \mapsto \hat{f}(\xi) = (\xi \mapsto \int_X f(x)\langle x, \xi \rangle dx)$$

To proceed, we need to choose a Haar measure dx on X, unique up to $\mathbb{R}^{>0}$.

By choosing another measure $d\xi$ in X^*, we can form $\hat{\hat{f}}$ is a function on X. We know in the real case, we have:

$$\hat{\hat{f}}(x) = f(-x)$$

if we choose the right Haar measure. In general, we will have:

$$\hat{\hat{f}} = c(dx, d\xi) \cdot f(-x)$$

with c depending continuously on $dx, d\xi$.

We're interested in the case when $X \simeq X^*$, then it suffices to choose dx. Note that

$$c(\lambda dx, \lambda d\xi) = \lambda^2 c(dx, d\xi)$$

so that we have a canonical choice of λ by insiting $c(\lambda dx, \lambda d\xi) = 1$. Hence there is a unique "Fourier self-dual" Haar measure. Hence there will be canonical Haar measure on $\mathbb{Q}_p$ and $\mathbb{R}$:

$$\psi_p : \mathbb{Q}_p \to S^1$$

with the property that $\psi_p|_{\mathbb{Z}_p} = 1, \psi_p|_{p^{-1}\mathbb{Z}_p} \neq 1$, and:

$$\psi_\infty : \mathbb{R} \to S^1, x \mapsto e^{-2\pi i x}$$

We normlize so that for any $a \in \mathbb{Q}$, $\prod_v \psi_v(a) = 1$.

Now let $F/\mathbb{Q}$ be a number field. We can upgrade this character to each F_v. For $v \mid p$, we put:

$$\psi_v : F_v \xrightarrow{\mathrm{Tr}} \mathbb{Q}_p \xrightarrow{\psi_p} S^1$$

For $\mathbb{C}$ we put:

$$\mathbb{C} \xrightarrow{\mathrm{Tr}} \mathbb{R} \xrightarrow{\psi_\infty} S^1$$

Note that for each non-archimedean v, we have $\psi_v(\mathcal{O}_v) = 1$. Therefore for each $(a_v) = a \in \mathbb{A}_F$, the infinite product:

$$\psi = \prod \psi_v(a_v) : \mathbb{A}_F \to S^1$$

is defined. Moreover $\psi(F) = 1$.

Now let's start to put measure on these objects. Note that F_v is Pontryagin self dual, so that we have a canonical isomorphism:

$$F_v \xrightarrow{\sim} (F_v)^*$$

$$x \mapsto (y \mapsto \psi_v(xy))$$

So that there exists a unique Haar measure $d_v x$ on F_v such that Fourier inversion holds.

One can show that for the self dual measure dx on $\mathbb{Z}_p$, we have:

$$\int_{\mathbb{Z}_p} dx = 1$$

One can compute: if $f = \mathbf{1}_{\mathbb{Z}_p}$, then

$$\hat{f}(y) = \int_{\mathbb{Q}_p} \psi_p(xy)\mathbf{1}_{\mathbb{Z}_p}(x)dx = \int_{\mathbb{Z}_p} \psi_p(xy)dx = \mathbf{1}_{\mathbb{Z}_p}$$

In other words, $\mathbf{1}_{\mathbb{Z}_p}$ is of self dual.

In general, for a general local field F_v, we have if $f = \mathbf{1}_{\mathcal{O}_v}$, then:

$$\hat{f} = \mathbf{1}_{\delta_v^{-1}} \int_{\mathcal{O}_v} dx.$$

where δ_v^{-1} is the inverse different ideal.

Note that if $F_v/\mathbb{Q}_p$ is unramified (most of time), we have $\delta_v^{-1} = \mathcal{O}_v$, and for the self-dual measure we still have $\int_{\mathcal{O}_v} dx = 1$. This enables us to equip $\mathbb{A}$ with a measure dx.

The measurable subsets of $\mathbb{A}$ are of form:

for S finite set of places containing archimedean places and ramified places,

$$(\prod_{v \notin S} \mathcal{O}_v) \times X_s$$

where X_s is arbitrary measurable subsets in $\prod_{v \in S} F_v$. We then just insist the left bits have measure 1, and using the local measures on the right bits.

The functions that we intend to integrate has form:

$$f = \otimes_v f_v, \quad f_v : F_v \to \mathbb{C}$$

and for almost all v, $f_v = \mathbf{1}_{\mathcal{O}_v}$.

Now using the canonical additive character on $\mathbb{A}$, we have a canonical identification:

$$\mathbb{A} \simeq \mathbb{A}^*$$

$$x \mapsto (y \mapsto \psi(xy))$$

One can check that the previously constructed dx is of Fourier self-dual.

The function spaces for Fourier transforms to preserve are Schwarz spaces:

$$\mathcal{S}(F_v) = C_c^\infty(F_v, \mathbb{C})$$

$$\mathcal{S}(\mathbb{R}) = \{f \in C^\infty(\mathbb{R}) \mid |x^n (d/dx)^m f(x)| \to \infty, \forall m, n\}$$

and similarly for $\mathcal{S}(\mathbb{C})$. This helps us to define the global Schwarz space as restricted product:

$$\mathcal{S}(\mathbb{A}) = \otimes_v' \mathcal{S}(F_v)$$

This is again preserved under Fourier transform.

Theorem 1.4 (Poisson Summation Formula)**.** *Let $f \in \mathcal{S}(\mathbb{A})$ (with extra conditions). Then:*

$$\sum_{x \in F} f(x) = \sum_{y \in F} \hat{f}(y)$$

under the self dual measure.

In fact Poisson Summation Formula holds in a more general setting. Let (X, dx) be a topological abelian group, with dual X^*. Assume $\Gamma \subset X$ be a discrete cocompact subgroup (think of $F \subset \mathbb{A}_F$). Then we have a subgroup:

$$\Gamma^\perp = \{y \in X^* \mid y|_\Gamma = 1\} \subset X^*$$

But the perp operation exchanges discrete / cocompactness conditions, so that $\Gamma^\perp$ is again discrete and cocompact.

Example. *Consider $\mathbb{Z} \subset \mathbb{R} = X$, with $\mathbb{R}^* \simeq \mathbb{R}$ using $\psi_\infty(x) = e^{-2\pi i x}$. Then $\mathbb{Z}^\perp$ is identified with $\mathbb{Z}$.*

In this setting, we have the general:

Theorem 1.5 (Poisson Summation Formula)**.** *For $f : X \to \mathbb{C}$ and $\hat{f} : X^* \to \mathbb{C}$ be its Fourier transform. Then we have:*

$$\mathrm{vol}(\Gamma\backslash X, dx) \sum_{x \in \Gamma} f(x) = \sum_{y \in \Gamma^\perp} \hat{f}(y)$$

Therefore to deduce our previous version of Poisson summation formula, we must have:

$$\mathrm{vol}(F\backslash \mathbb{A}_F, dx) = 1$$

In fact, more generally if $X \simeq X^*$ and identifies Γ with $\Gamma^\perp$, and dx is self-dual measure on X. Then:

$$\mathrm{vol}(\Gamma\backslash X, dx) = 1$$

One can prove this by plugging in f being an indication function on some fundamental domain of $\Gamma\backslash X$ in the Poisson summation formula.

Lemma 1.6. *Under $\mathbb{A} \simeq \mathbb{A}^*$, we have:*

$$F^\perp = F$$

Proof. Note $F \subset F^\perp$ follows from $\psi(a) = 1$ for all $a \in F$. From homework we know F is a maximal discrete subgroup. Moreover $F^\perp$ is discrete since $F\backslash \mathbb{A}$ is compact. □

Proof. (of Poisson Summation Formula). Given a function f on X, we can produce a Γ-periodic function:

$$\phi(x) = \sum_{\gamma \in \Gamma} f(x + \gamma) : \Gamma\backslash X \to \mathbb{C}$$

We thus have:

$$\hat{\phi} : (\Gamma\backslash X)^* = \Gamma^\perp \to \mathbb{C}$$

Applying Fourier inversion formula to $\hat{\hat{\phi}}$ and evaluate at 0 will give me:

$$\mathrm{vol}(\Gamma\backslash X, dx) \cdot \phi(0) = \sum_{y \in \Gamma^\perp} \hat{\phi}(y)$$

The left hand side is literally what we want, for the right hand side we just check $\hat{\phi}(y) = \hat{f}(y)$. In fact, we compute:

$$\hat{f}(y) = \int_X f(x)\langle x, y\rangle dx = \int_{\Gamma\backslash X} (\sum_{\gamma \in \Gamma} f(x + \gamma))\langle \overline{x}, y\rangle d\overline{x} = \hat{\phi}(y)$$

□

Now let's deal with function fields $F = \mathbb{F}_q(X)$. It's natural to view $\mathbb{A}^*$ as a restricted product:

$$\mathbb{A}_{\omega_F} = \prod{}'(\omega_{F_v}, \omega_{\mathcal{O}_v})$$

Here ω_F is the sheaf of 1-forms on X. The space ω_v is the stalk at place v, explicitly $\omega_v = \mathcal{O}_{X,v} \cdot dt$ for t local coordinate. We put $\omega_{\mathcal{O}_v} = \widehat{\omega_v}$ the completion, this is a free rank 1 module over $\mathcal{O}_v$ with basis dt. We then set $\omega_{F_v} = \omega_{\mathcal{O}_v} \otimes_{\mathcal{O}_v} F_v$ be one-dimensional over F_v.

Let's choose a nonzero character $\psi : \mathbb{F}_p \to S^1$. This induces a character $\mathbb{F}_q \to S^1$ via trace for each q. We thus have a pairing:

$$\mathrm{res}_v : F_v \times \omega_{F_v} \to \mathbb{F}_q(v)$$

$$(f, \theta) \mapsto \mathrm{res}_v(f\theta)$$

where the residue is computed via:

$$\mathrm{res}_v(\sum_{n\in\mathbb{Z}} a_n t^n dt) = a_{-1} \in \mathbb{F}_q(v)$$

Compose with the previous choice of character $\mathbb{F}_q \to S^1$, we get a perfect pairing:

$$\langle -, - \rangle_v : F_v \times \omega_{F_v} \xrightarrow{\mathrm{res}_v} \mathbb{F}_q(v) \xrightarrow{\mathrm{Tr}_{\mathbb{F}_q(v)/\mathbb{F}_p}} \mathbb{F}_p \xrightarrow{\psi} S^1$$

This realizes ω_{F_v} as Pontryagin dual because $\mathcal{O}_v \subset F_v$ are sent to $\omega_{\mathcal{O}_v} = \mathcal{O}_v^\perp \subset \omega_{F_v}$.

Now we can have obtain the global duality:

$$\mathbb{A} \times \mathbb{A}_\omega \xrightarrow{\prod_v \langle \cdot, \cdot \rangle_v} S^1$$

$$((a_v), (\theta_v)) \mapsto \prod_v \langle a_v, \theta_v \rangle_v$$

Therefore $\mathbb{A}_\omega$ is canonically identified with $\mathbb{A}^*$. If we really want to identify $\mathbb{A}$ with $\mathbb{A}_\omega$, we need to chose a $\theta \in \omega_F$ and:

$$\mathbb{A} \simeq \mathbb{A}_\omega$$

by multiplying θ.

Theorem 1.7 (Residue Theorem)**.** *Let $\theta \in \omega_F$, we have:*

$$\sum_v \mathrm{Tr}_{\mathbb{F}_q(v)/\mathbb{F}_q} \mathrm{res}_v(\theta) = 0$$

This enables us to prove:

Lemma 1.8. *Under $\mathbb{A}_\omega \simeq \mathbb{A}^*$, we identify $F^\perp$ with ω_F.*

Since it doesn't make sense to talk about self dual measure, the natural measure on F_v is the one such that $\mathrm{vol}(\mathcal{O}_v, d_v x) = 1$. Now we put:

$$dx = \prod d_v x \text{ on } \mathbb{A}$$

It follows that:

$$\mathrm{vol}(F\backslash\mathbb{A}, dx) = q^{g-1}$$

This comes from the identification:

$$F\backslash\mathbb{A}/\prod \mathcal{O}_v = H^1(X, \mathcal{O}_X) \simeq \mathbb{F}_q^g$$

and there will be one cancellation since this is not a free quotient. By similar reasoning, we see:

$$\mathrm{vol}(\omega_F\backslash\mathbb{A}_\omega, dx) = q^{1-g}$$

with the choice $\mathrm{vol}(\omega_{\mathcal{O}_v}, d_v x) = 1$.

Now we wish to put a measure on $\mathbb{A}^\times$: $(\mathbb{A}^\times, d^\times x)$. We attempt to use the the following measure:

$$(F_v^\times, \frac{d_v x}{|x|_v})$$

this choice makes $d_v x$ a Haar measure on $F_v^\times$. We compute:

$$\int_{\mathbb{A}^\times} \mathbf{1}_{\prod \mathcal{O}_v^\times} f_\infty d^\times x$$

Each local contribution for unramified place is:

$$\int_{\mathcal{O}_v^\times} d_v x = \int_{\mathcal{O}_v} d_v x - \int_{\varpi\mathcal{O}_v} d_v x = 1 - \frac{1}{q_v}$$

Therefore, we the better choice is:

$$(F_v^\times, d_v^\times x = \frac{1}{1 - q_v^{-1}} \frac{d_v x}{|x|_v})$$

So that $\int_{\mathcal{O}_v^\times} d_v^\times x = 1$ at all unramified place. And so that $d^\times x = \prod d_v^\times x$.

Now let's try to compute: $\mathrm{vol}(F^\times\backslash\mathbb{A}^{\times,1}, d^\times x)$. Recall that:

$$F^\times\backslash\mathbb{A}^{\times,1}/\prod_{v\nmid\infty} \mathcal{O}_v^\times (\prod_{v|\infty} F_v^\times)^1 = \mathrm{Cl}(F)$$

Another piece of contribution is the torus:

$$\mathcal{O}_F^\times\backslash(\prod_{v|\infty} F_v^\times)^1$$

It turns out the result is:

$$\frac{2^{r_1}(2\pi)^{r_2} hR}{\sqrt{|d|}w}$$

where h is the class number, R is the regulator, w is the order of the roots of unity $(\mathcal{O}_F^\times)_{\text{torsion}}$. It turns out we can also compute this by the residue of **complete Dedekind Zeta Function**:

$$\text{Res}_{s=1}\Lambda_F(s)$$

It will have functional equation:

$$\Lambda_F(s) = \Lambda_F(1-s)$$

1.4 Tate's Thesis

Definition 1.9. Let F be a number field. The **Dedekind Zeta function** is defined by:

$$\zeta_F(s) = \sum_{I \subset \mathcal{O}_F} \frac{1}{(NI)^s} = \prod_{v \nmid \infty} \frac{1}{1-q_v^{-s}}$$

If we naively plug in $s=1$, it seems we're getting $\prod \frac{1}{1-q_v^{-1}}$ which coincides with our previous normalization:

$$\text{“}\zeta_F(1)\text{”}\frac{dx}{|x|} = d^\times x$$

Definition 1.10. The **L-series** for character $\chi : (\mathbb{Z}/N)^\times \to \mathbb{C}^\times$ is defined to be:

$$L(s,\chi) = \sum_{n \geq 1, (n,N)=1} \frac{\chi(n)}{n^s}$$

If $\chi = 1$, then $L(s,1) = \zeta(s)$ has a pole at $s=1$; if $\chi \neq 1$, we know $L(s,\chi)$ is holomorphic. We wish to generalize this construction. Note that we have

$$\mathbb{A}_\mathbb{Q}^\times \simeq \mathbb{Q}^\times \times \mathbb{R}^{>0} \times \widehat{\mathbb{Z}}^\times \to \widehat{\mathbb{Z}}^\times \to (\mathbb{Z}/N)^\times \xrightarrow{\chi} \mathbb{C}^\times$$

upgrades χ to a character of $\mathbb{A}_\mathbb{Q}^\times$. In general, suppose I have a (continuous) character:

$$\chi : F^\times \backslash \mathbb{A}_F^\times \to \mathbb{C}^\times$$

We call this character a unitary character if the image lies in S^1, in other words this is just an element in the locally compact group $(F^\times \backslash \mathbb{A}_F^\times)^*$. On the other opposite, we have:

$$|-|\colon F^\times \backslash \mathbb{A}_F^\times \to \mathbb{R}^{>0} \subset \mathbb{C}^\times$$

Then any character χ can be uniquely written as:

$$\chi = \chi_0 \cdot |-|^s$$

for χ_0 unitary and $s \in \mathbb{R}$. Therefore we can identify the space of idèle class characters with

$$(F^\times \backslash \mathbb{A}_F^\times)^* \times \mathbb{R}$$

To understand the topology of the first factor, we start with:

$$1 \to F^\times \backslash \mathbb{A}_F^{\times,1} \to F^\times \backslash \mathbb{A}_F^\times \xrightarrow{|-|} \mathbb{R}^{>0} \simeq (\mathbb{R}, +) \to 1$$

The first factor is compact. We take Pontryagin dual:

$$0 \to i\mathbb{R} \to (F^\times \backslash \mathbb{A}_F^\times)^* \to \text{something discrete} \to 0$$

So we can give the idèle class group a manifold structure:

$$(\coprod i\mathbb{R}) \times \mathbb{R} \simeq \coprod \mathbb{C}$$

For function field $F = \mathbb{F}_q(X)$, the similar short exact sequence looks like:

$$1 \to \{\text{compact stuff}\} \to F^\times \backslash \mathbb{A}_F^\times \xrightarrow{(a_v) \mapsto \sum \deg(v) \mathrm{ord}_v(a_v)} \mathbb{Z} \to 1$$

Takes Pontryagin dual:

$$1 \to \mathbb{C}^\times \to \{\text{idèle class characters}\} \to \{\text{discrete}\} \to 1$$

Definition 1.11 (L-series for number fields)**.** Let F be a number field. Fix a idèle class character $\chi : F^\times \backslash \mathbb{A}_F^\times \to \mathbb{C}^\times$. We know $\chi|_{\mathcal{O}_v^\times} = 1$ for almost all v, by the continuity of χ. In this case, we define:

$$L_v(s, \chi) \coloneqq \frac{1}{1 - \chi_v(\varpi) q_v^{-s}}$$

when $\chi|_{\mathcal{O}_v^\times} \neq 1$, we define:

$$L_v(s, \chi_v) = 1$$

Combining these we get the L-series:

$$L(s, \chi) = \prod_{v \nmid \infty} L_v(s, \chi_v) = \sum_{I \subset \mathcal{O}_F, I \text{ prime to the ramified places of } \chi} \frac{\chi(I)}{(NI)^s}$$

where $\chi(\prod \mathfrak{p}_v^{e_v}) = \prod \chi_v(\varpi_v)^{e_v}$.

If χ is unitary and $\Re(s) > 1$, $L(s, \chi)$ is absolute convergent, and it defines an analytic function for $\Re(s) > 1$. Note that we can write:

$$\chi = \chi_0 |-|^t \implies L(s, \chi) = L(s + t, \chi_0)$$

since $\chi_v(\varpi) = \chi_{0,v}(\varpi) q_v^{-t}$. So we lose no information when we restrict to unitary character when talking about L-series.

Theorem 1.12 (Tate's Theorem)**.** *Let $F/\mathbb{Q}$ be a number field. Given a character of idéle class group*

$$\chi : F^\times \backslash \mathbb{A}_F^\times \to \mathbb{C}$$

and a Schwarz funciton $f \in \mathcal{S}(\mathbb{A}_F)$, we define:

$$Z(\chi, f) = \int_{\mathbb{A}_F^\times} \chi(a) f(a) d^\times a$$

Then:

1. *$Z(\chi, f)$ as a function of χ, has a meromorphic continuation.*
2. *$Z(\chi, f) = Z(\chi^{-1} \cdot |-|, \hat{f})$*

Fix χ, f, we define:

$$Z(s, \chi, f) = Z(\chi \cdot |-|^s, f)$$

An equivalence formulation of Tate's theorem is that: for every fixed (f, χ), $Z(s, \chi, f)$ as a funciton $s \in \mathbb{C}$ has a meromorphic continuation. The functional equation reads:

$$Z(s, \chi, f) = Z(1 - s, \chi^{-1}, \hat{f})$$

Now if we take:

$$f = (\mathbf{1}_{\prod_{v \nmid \infty} \mathcal{O}_v}) \cdot (f_\infty)$$

Recall that if v is unramified, then: $\widehat{\mathbf{1}_{\mathcal{O}_v}} = \mathbf{1}_{\mathcal{O}_v}$. At infinity places we have large supply of self-dual functions:

1. If $F_v = \mathbb{R}$, $\chi_v = 1$, we choose $f_v = e^{-\pi x^2}$.
2. If $F_v = \mathbb{C}$, $\chi_v = 1$, we choose $f_v = e^{-2\pi |z|^2}$
3. We generate general case from the above two cases.

We can thus compute:

$$Z(s, 1, f) = \prod_v \int_{F_v^\times} |a_v|^s f_v(a_v) d_v^\times a$$

At non-archimedean unramified places, the contribution is:

$$\int_{\mathcal{O}_v \setminus \{0\}} |a|^s d_v^\times a = \sum_{n \geq 0} (q_v^{-n})^s = \frac{1}{1 - q_v^{-s}} = \zeta_v(s)$$

since $\mathrm{vol}(\mathcal{O}_v^\times) = 1$ and $d_v^\times$ is multiplicative invariant. Therefore we get:

$$Z(s, 1, f) = c \cdot \zeta_F(s) \cdot \prod_{\text{real}} \pi^{-\frac{s}{2}} \Gamma(\frac{s}{2}) \cdot \prod_{v \text{ complex}} 2(2\pi)^{1-s} \Gamma(s)$$

for some constant c coming from the contributions at ramified places.

Now let's deal with function fields.

Definition 1.13 (Zeta function for function fields)**.** Let $F = \mathbb{F}_q(X)$ be a function field. We define the **zeta function** to be:

$$\zeta_X(s) = \prod_v \frac{1}{1 - q_v^{-s}} = \sum_{D \text{ effective divisor}} \frac{1}{q^{\deg D \cdot s}} = \sum_{d \geq 0} q^{-ds} \cdot \#\{\deg d \text{ effective divisors on } X\}$$

Remark. $\#\{\deg d \text{ effective divisors on } X\} = X^{(d)}(\mathbb{F}_q)$ *where* $X^{(d)} = X^d // \Sigma_d$ *is the d-fold symmetric power of* X*.*

Now if

$$\chi : F^\times \backslash \mathbb{A}_F^\times \to \mathbb{C}^\times$$

Suppose χ is unramified, namely it factors through:

$$\chi : \mathrm{Pic}(X) = F^\times \backslash \mathbb{A}_F^\times / \prod \mathcal{O}_v^\times \to \mathbb{C}^\times$$

We can define the L-**series** as:

$$L(s, \chi) = \sum_{D \subset X} q^{-\deg D \cdot s} \chi(\mathcal{O}(D))$$

If $\chi|_{\mathrm{Pic}^0(X)} \neq 1$, then: $L(s, \chi)$ is a polynomial in q^{-s}:

$$L(s, \chi) = \sum_{d \geq 0} q^{-ds} (\sum_{\deg D = d} \chi(\mathcal{O}(D)))$$

We claim that as a polynomial in q^{-s}, it has degree $\leq 2g - 2$. Indeed, for $d > 2g - 2$ we have:

$$X^{(d)} \to \mathrm{Pic}_X^d$$

is a $\mathbb{P}^{d-g}$-bundle, by an argument using complete linear system. Therefore, we have:

$$\sum_{D \in X^{(d)}} \chi(\mathcal{O}(D)) = \sum_{\mathcal{L} \in \mathrm{Pic}^d(X)} \sum_{D : \mathcal{O}_X(D) \simeq \mathcal{L}} \chi(\mathcal{L}) = |\mathbb{P}^{d-g}(\mathbb{F}_q)| \cdot \sum_{\mathcal{L} \in \mathrm{Pic}^d(X)} \chi(\mathcal{L})$$

Choose $\mathcal{L}_0 \in \mathrm{Pic}^d(X)$, we have:

$$\sum_{\mathcal{L} \in \mathrm{Pic}^d(X)} \chi(\mathcal{L}) = \chi(\mathcal{L}_0) \sum_{\mathcal{L}' \in \mathrm{Pic}_0(X)} \chi(\mathcal{L}') = 0$$

This proves our claim. It turns out that the L-series in this case will have degree precisley $2g - 2$.

1.5 Proof of Tate's Theorem

Let $F/\mathbb{Q}$. Given $f \in \mathcal{S}(\mathbb{A})$, we can write:

$$f = \otimes_v f_v, \quad f_v \in \mathcal{S}(F_v)$$

and for almost all $f_v = \mathbf{1}_{\mathcal{O}_v}$. Now given an idèle class character:

$$\chi : F^\times \backslash \mathbb{A}^\times \to \mathbb{C}^\times$$

We defined:

$$Z(s, \chi, f) = \int_{\mathbb{A}^\times} f(a) \chi(a) |a|^s d^\times a$$

The local contribution is

$$\int_{F_v^\times} \mathbf{1}_{\mathcal{O}_v} \chi_v(a) |a|_v^s d_v^\times a = \frac{1}{1 - \chi_v(\varpi_v) q_v^{-s}} = \sum_{n \geq 0} \chi_v(\varpi_v^n) |\varpi_v^n|^s$$

if χ_v is unramified and $\int_{\mathcal{O}_v^\times} d^\times a = 1$.

We defined:

$$L_v(s,\chi_v) \coloneqq \begin{cases} \dfrac{1}{1-\chi_v(\varpi_v)q_v^{-s}}, & \text{if } \chi_v \text{ is unramified;} \\ 1, & \text{otherwise.} \end{cases}$$

$$L^S(s,\chi) = \prod_{v\notin S} L_v(s,\chi_v)$$

Now take $S = S_\infty \cup \{F_v/\mathbb{Q}_p \text{ ramifies}\} \cup \{\chi_v \text{ ramifies}\} \cup \{f_v \neq \mathbf{1}_{\mathcal{O}_v}\}$ be the finite set of bad places. We write:

$$Z(s,\chi,f) = L^S(s,\chi)(\prod_{v\in S} Z_v(s,\chi_v,f_v))$$

The second part has meromorphic continuations. We know $L^S(s,\chi)$ is analytic when $\Re(s) > 1$. The goal is to prove:

1. $Z(s,\chi,f)$ has analytic continuation.
2. $Z(s,\chi,f) = Z(1-s,\chi^{-1},\hat{f})$
3. If χ is trivial, then $Z(s,\chi,f)$ has simple poles at $s = 0,1$ with residues

   $$-f(0)\mathrm{vol}(F^\times\backslash\mathbb{A}^{\times,1}, d^\times a)$$

   at $s = 0$,

   $$\hat{f}(0)\mathrm{vol}(F^\times\backslash\mathbb{A}^{\times,1}, d^\times a)$$

   at $s = 1$.
4. We have obvious relation: $Z(s,\chi\cdot|\cdot|^t,f) = Z(s+t,\chi,f)$. Modulo this relation, the above example in the only instance for $Z(s,\chi,f)$ to have a pole. If $\chi \neq |\cdot|^t$, then $Z(s,\chi,f)$ is analytic for $s\in\mathbb{C}$.

We can write

$$Z(s,\chi,f) = Z_+(s,\chi,f) + Z_-(s,\chi,f)$$

for $Z_+(s,\chi,f) = \int_{|a|>1} f(a)\chi(a)|a|^s d^\times a$, $Z_-(s,\chi,f) = \int_{|a|<1} f(a)\chi(a)|a|^s d^\times a$. The first part is convergent for all $s\in\mathbb{C}$.

To understand the second part, we need to exhibit a symmetry between $Z_-(s,\chi,f)$ and $Z_+(1-s,\chi^{-1},\hat{f})$ via the Poisson summation formula.

We compute:

$$\begin{aligned} Z_-(s,\chi,f) &= \int_{a\in\mathbb{A}^\times,|a|<1} f(a)\chi(a)|a|^s d^\times a \\ &= \int_{F^\times\backslash\mathbb{A}^\times,|a|<1} (\sum_{\gamma\in F^\times} f(\gamma a))\chi(a)|a|^s d^\times a \end{aligned}$$

Now define $f_a(x) = f(xa) \in \mathcal{S}(\mathbb{A})$, for $a \in \mathbb{A}^\times$. Now the above integral becomes:

$$\int_{F^\times\backslash\mathbb{A}^\times,|a|<1} (\sum_{\gamma\in F^\times} f_a(\gamma) - f_a(0))\chi(a)|a|^s d^\times a$$
$$= \int_{F^\times\backslash\mathbb{A}^\times,|a|<1} (\sum_{\gamma\in F} \widehat{f_a}(\gamma))\chi(a)|a|^s d^\times a - f(0)\int_{F^\times\backslash\mathbb{A}^\times,|a|<1} \chi(a)|a|^s d^\times a$$

Compuation shows:

$$\widehat{f_a}(x) = |a|^{-1}\widehat{f}(\frac{x}{a})$$

And choose a splitting $A^\times = \mathbb{A}^{\times,1} \times \mathbb{R}^{>0}$

$$\int_{F^\times\backslash\mathbb{A}^\times,|a|<1} \chi(a)|a|^s d^\times a = \int_0^1 t^s (\int_{F^\times\backslash\mathbb{A}^{\times,1}} \chi(a_1 t^\infty) d^\times a_1) d^\times t = \begin{cases} 0, & \text{if } \chi|_{\mathbb{A}^{\times,1}} \neq 1; \\ \dfrac{\mathrm{vol}(F^\times\backslash\mathbb{A}^{\times,1})}{s}\cdot, & \text{if } \chi = 1 \end{cases}$$

Then the above integral becomes:

$$\int_{F^\times\backslash\mathbb{A}^\times,|a|<1} (\sum_{\gamma\in F} \widehat{f}(\frac{\gamma}{a}))\chi(a)|a|^{s-1} d^\times a - f(0)\cdot \begin{cases} 0, & \text{if } \chi|_{\mathbb{A}^{\times,1}} \neq 1; \\ \dfrac{1}{s}\cdot \mathrm{vol}(F^\times\backslash\mathbb{A}^{\times,1}), & \text{if } \chi = 1 \end{cases}$$

Exchanging $a \to a^{-1}$, we get:

$$\int_{F^\times\backslash\mathbb{A}^\times,|a|>1} \sum_{\gamma\in F} \widehat{f}(\gamma a)\chi(a^{-1})|a|^{1-s} d^\times a - f(0)\cdot c_0(s)$$

$$= Z_+(1-s, \chi^{-1}, \widehat{f}) + \widehat{f}(0)\cdot c_1(s) - f(0)\cdot c_0(s)$$

for $c_1(s) = \begin{cases} 0, & \text{if } \chi|_{\mathbb{A}^{\times,1}} \neq 1; \\ -\mathrm{vol}\cdot\dfrac{1}{1-s}, & \text{if } \chi = 1. \end{cases}$ .We can therefore deduce functional equation from the above calculation, by splitting the cases into $\chi|_{\mathbb{A}^{\times,1}\neq 1}$ or $\chi = 1$.

Finally let's comment on the case for function fields $F = \mathbb{F}_q(X)$. Recall to introduce self duality we need to choose a rational differential form $\theta : \mathbb{A} \to \mathbb{A}_\omega$. However, the choice of self-dual Haar measure doesn't depend on the choice of θ. Then the same proof will work, and we can deduce the functional equation for $\zeta_X(s)$ from the functional equation for $Z(s, \chi, f)$.

In fact we can write:

$$\zeta_X(s) = \frac{P(q^{-s})}{(1-q^{-s})(1-q^{1-s})}$$

for $P(t) = 1 + \cdots + q^g t^{2g} \in \mathbb{Z}[t]$. The functional equation reads:

$$\zeta_X(1-s) = q^{(g-1)(2s-1)}\zeta_X(s)$$

corresponding to $\chi = 1$. The poles are at $q^s = 1$ or $q^{1-s} = 1$.

If χ nontrivial but unramified, then we're getting a character:

$$\chi : \mathrm{Pic}(X) \to \mathbb{C}^\times$$

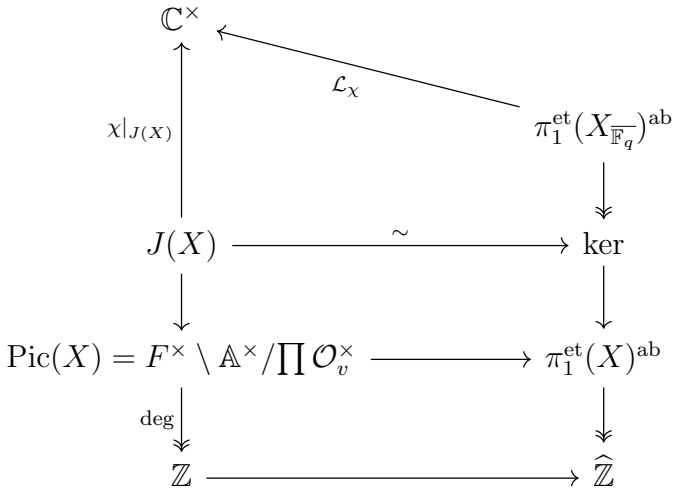
Recall the statement of global class field theory:

$$\begin{array}{ccc} F^\times\backslash\mathbb{A}^\times & \longrightarrow & \mathrm{Gal}(F^{\mathrm{ab}}/F) \\ \downarrow{\scriptstyle \deg} & \lrcorner & \downarrow \\ \mathbb{Z} & \longrightarrow & \mathrm{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q) = \widehat{\mathbb{Z}} \end{array}$$

is Cartesian. It has a unramified version:

$$F^\times\backslash\mathbb{A}^\times/\prod \mathcal{O}_v^\times = \mathrm{Pic}(X) \to \pi_1^{\text{ét}}(X)^{\mathrm{ab}}$$

Now we restrict to the norm one part:



This defines a rank one local system $\mathcal{L}_\chi$ on $X_{\overline{\mathbb{F}_q}}$, extendable to $\pi_1^{\text{ét}}(X)^{\mathrm{ab}}$. Instead, we can view this as a Weil-local system $\mathcal{L}_\chi$ on X of rank 1.

Theorem 1.14.

$$L(s,\chi) = \frac{\det(1 - \sigma \cdot q^{-s} | H^1(X_{\overline{\mathbb{F}_q}}, \mathcal{L}_\chi))}{\det(1 - \sigma q^{-s}|_{H^0}) \det(1 - \sigma q^{-s}|_{H^2})}$$

Now if $\chi|_{J(X)} = 1$, namely χ factors through the degree map, then $\mathcal{L}_\chi$ is a trivial local system on $X_{\overline{\mathbb{F}_q}}$ Then the L function simplifies:

$$L(s,\chi) = \frac{P(\chi(a)q^{-s})}{(1 - \chi(a)q^{-s})(1 - \chi(a)q^{1-s})}$$

for P has degree $2g$.

If $\chi|_{J(X)} \neq 1$, the local system will not have any global section so that the denominator trivializes. And we get:

$$L(s,\chi) = \det(1 - \sigma q^{-s} \mid) = P(q^{-s})$$

for some polynomial P of degree $2g-2$. The functional equation for L-function will become:

$$L(1-s,\chi^{-1}) = q^{(2g-2)} \det(\sigma|_{H^*}) L(s,\chi)$$

In general, for each global field F, we always have functional equation:

$$L(s,\chi) = \varepsilon(s,\chi) L(1-s,\chi^{-1})$$

We computed the epsilon factor for function fields. For number field, it will become:

$$\prod_v \varepsilon_v(s,\chi_v,\psi_v)$$

if we insist $\psi|_F = 1$.

Each local epsilon factor comes from local functional equation:

$$Z_v(1-s,\chi_v^{-1},\hat{f}) = \gamma(s,\chi_v,\psi_v) Z_v(s,\chi_v,f)$$

In homework we saw we can write:

$$Z_v(s,\chi_v,f) = Z_v'(s,\chi_v,f) \cdot L_v(s,\chi_v)$$

where Z_v' is a polynomial in q^{-s}. Therefore we have:

$$Z_v'(1-s,\chi_v^{-1},\hat{f}) L_v(1-s,\chi_v^{-1}) = \gamma(s,\chi_v,\psi_v) Z_v'(s,\chi_v,f) L_v(s,\chi_v)$$

We then define the epsilon factor to be:

$$\varepsilon_v(s,\chi_v,\psi_v) = \frac{Z_v'(1-s,\chi_v^{-1},\hat{f})}{Z_v'(s,\chi_v,f)}$$

2 Representations of p-adic and Real Groups

We've discussed automorphic representation of $\mathrm{GL}_1(\mathbb{A})$:

$$\chi : F^\times \backslash \mathbb{A}^\times \to \mathbb{C}^\times$$

We will be interested in automorphic representations for general G: these are representations of $G(\mathbb{A})$ that appears in:

$$\mathrm{Fun}(G(F)\backslash G(\mathbb{A}))$$

We have:

$$G(\mathbb{A}) = \prod_v{}' G(F_v)$$

It's natural to start with $G(F_v)$-representations. We split into cases for either non-archimedean and archimedean cases. We will study represention of p-adic groups by introducing Hecke algebras; we will study representations of real groups by introducing $(\mathfrak{g}, K)$-modules. Then we will combine them to get:

$$\otimes(\text{local Hecke algebra})$$

and study their representations.

2.1 Basics for p-adic Group Representations

Let F be a local non-archimedean field. Given any affine algebraic group, we have:

$$G \hookrightarrow \mathrm{GL}_N$$

realized as a closed subgroup. Now we have:

$$G(F) \subset \mathrm{GL}_N(F) \subset F^{N^2+1}$$

We endow these with subspace topology. We define:

$$K(\varpi^m) = \{A \in \mathrm{GL}_N(\mathcal{O}_F) \mid A \equiv I_N \bmod \varpi^m\}$$

For $m = 0$, this is just $\mathrm{GL}_N(\mathcal{O}_F)$: all matrix A with coefficients in $\mathcal{O}_F$ with $\det(A) \in \mathcal{O}_F^\times$. It can be shown that $\{K(\varpi^m)\}_{m\geq 0}$ is a neighborhood basis of 1, and each of them is compact and open in $\mathrm{GL}_N(F)$. Compactness is easy to see, to see it's open, we have a short exact sequence:

$$1 \to K(\varpi^m) \to \mathrm{GL}_N(\mathcal{O}_F) \to \mathrm{GL}_N(\mathcal{O}_F/\varpi^m) \to 1$$

for the $\mathrm{GL}_N(\mathcal{O}_F/\varpi^m)$ finite, so that $K(\varpi^m)$ is open in $\mathrm{GL}_N(\mathcal{O}_F)$. But then $\mathrm{GL}_N(\mathcal{O}_F)$ is open in $\mathrm{GL}_N(F)$, since $\mathcal{O}_F^\times \subset F^\times$ is open.

Remark. $\mathrm{GL}_N(F)$ *is an example of a toally disconnected groups. Namely they have a neighborhood basis at* 1 *that are open, compact subgroups.*

Example. *Consider a quadratic space* $(V, q)/F$*. We can talk about:*

$$G = O(V, q) \subset \mathrm{GL}(V)$$

This is a p-adic group.

Given a p-adic group $G(F)$, we study their $\mathbb{C}$-representations $V/\mathbb{C}$:

$$a : G(F) \times V \to V$$

We equip V the discrete topology (and ϖ-adic topology on $G(F)$). We call a is **continuous** if: for every $v \in V$, $\mathrm{stab}_{G(F)}(v)$ is open in $G(F)$. Such V is called a **smooth representation** of $G(F)$.

Example. *Let* $G = \mathrm{GL}_1$*, and a charcter:*

$$\chi : F^\times \to C^\times$$

is smooth iff there exists m *such that* $\chi|_{1+\varpi^n\mathcal{O}} = 1$*.*

Example (Regular representation)**.** *We have an action of* $G(F)$ *on*

$$\mathrm{Fun}(G(F)) = \{f : G(F) \to \mathbb{C}\}$$

by right translation. We need $\mathrm{stab}_{G(F)}(f)$ *to be open. This can be done by the procedure "extracting smooth parts". Indeed, given any representation of* $G(F)$*, we can extract:*

$$V^{\mathrm{sm}} = \{v \in V \mid \mathrm{stab}_{G(F)}(v) \text{ is open}\}$$

a smooth subrepresentation. Recall we have a decreasing sequence of open compact subgroups $\{K_m\}$ *of* $G(F)$ *satisfying* $\cap K_m = \{1\}$*. Note that then:*

$$\bigcup_m \mathrm{Fun}(G(F)/K_m) = \mathrm{Fun}(G(F))^{\mathrm{sm}}$$

This is because:

$$\mathrm{Fun}(G(F)/K_m) \xrightarrow{R_g} \mathrm{Fun}(G(F)/gK_mg^{-1}) \subset \mathrm{Fun}(G(F)/K_{m'})$$

This is the **right regular representation** *for* $G(F)$*.*

There's a smaller space

$$C_c^\infty(G(F)) = \{f : G(F) \to \mathbb{C} \mid \text{compactly supported locally constant}\}$$

called the **Schwarz space**. Note that $f \in C_c^\infty(G(F))$, there exists open subgroup $K_m \subset G(F)$ such that $f(k_1gk_2) = f(g)$, for $k_1, k_2 \in K_m$. It has a smooth action by $G(F) \times G(F)$.

Example. *Any time I have* $J \subset G(F)$ *subgroup, I have:*

$$\mathrm{Fun}(J\backslash G(F))^{\mathrm{sm}}$$

a smooth $G(F)$*-representation.*

Lemma 2.1. *Let* $G(F)$ *be a p-adic group.*

1. *Arbitrary direct sum of smooth representations is smooth.*
2. *For a smooth representation* V*, we have:*

$$V^\vee = (V^*)^{\mathrm{sm}} = \{v^* : V \to \mathbb{C} \mid \exists K \subset G(F) \text{ open}, \langle v^*, k \cdot v\rangle = \langle v^*, v\rangle\}$$

Definition 2.2 (Admissibility)**.** Let V be a smooth representation of $G(F)$. We call V admissible if for all $K \subset G(F)$ open, we have:

$$\dim_{\mathbb{C}} V^K < \infty$$

It is enought to check this for principle compact subgroups.

By definition:

$$(\mathrm{Fun}(G(F))^{\mathrm{sm}})^{K_m} = \mathrm{Fun}(G(F)/K_m)$$

this will not be admissible unless $G(F)$ is compact. To get a admissibile representation for $G(F)$, we seek for J such that:

$$J\backslash G(F)/K_m$$

is finite for all m, then:

$$\bigcup_m \mathrm{Fun}(J\backslash G(F)/K_m)$$

is admissible.

It is hard to check directly that $J\backslash G(F)/K_m$ is finite, but this is equivalent for $J\backslash G(F)$ being compact. The most important instance is:

$$J = P(F)$$

for $P \subset G$ parabolic subgroup.

Lemma 2.3. *If V is admissible, then:*

$$V \to (V^\vee)^\vee$$

is an isomorphism.

Lemma 2.4. *A representation V is smooth iff:*

$$V = \bigcup_{K_m} V^{K_m}$$

Let G be any group, we define:

$$\mathbb{C}[G] = \{f : G \to \mathbb{C} \text{ with finite support}\}$$

this is a unital ring under convolution, with unit δ_1. This makes

$$\{\mathbb{C}\text{-reps of } G\} \simeq \mathbb{C}[G]\text{-mod}$$

We wish to identify smooth representations of $G(F)$ via modules over certain algebra.

Definition 2.5 (Hecke Algebra)**.** Choose a Haar measure dg on $G(F)$, our Hecke algebra is defined to be:

$$\mathcal{H} = C_c^\infty(G(F))$$

given convolution product:

$$(f_1 * f_2)(x) = \int_{G(F)} f_1(g) f_2(g^{-1}x) dg$$

Remark. *We don't need to worry about left / right issue, since most groups we will consider will be unimodular.*

With this definition, $\mathcal{H}$ has no unit unless $G(F)$ is discrete. But it has a lot of idempotents. For each $K \subset G(F)$, then $\mathbf{1}_K \in \mathcal{H}$ has property:

$$\mathbf{1}_K * \mathbf{1}_K = \mathrm{vol}(K, dg)\mathbf{1}_K$$

So that:

$$e_K = \frac{1}{\mathrm{vol}(K, dg)} \mathbf{1}_K$$

is an idempotent. More canonically,

$$\mathcal{H} = \mathrm{Meas}_c^\infty(G(F)) = \{f(g)dg \mid f \in C_c^\infty(G(F))\}$$

It has a multiplication via pushforward:

$$m_* : \mathcal{H} \times \mathcal{H} \to \mathcal{H}$$

$$\mu_1 \boxtimes \mu_2 \mapsto m_*(\mu_1 \boxtimes \mu_2)$$

Now our previous idempotents e_K is the Haar measure on K with volume 1, so that $e_K * e_K = e_K$. In fact:

$$e_K = \frac{\mathbf{1}_K \cdot dg}{\mathrm{vol}(K, dg)}$$

is independent of the choice of the measure dg. Note that e_{K_m} form an "approximate identity" in $\mathcal{H}$.

Now given a smooth $G(F)$-representation V, we can make this a $\mathcal{H}$-module by:

$$\mathcal{H} \otimes V \to V$$

If $\mu = f \cdot dg$

$$\mu \cdot v = \int_{G(F)} \mu(g) g \cdot v = \int_{G(F)} f(g)(g \cdot v) dg$$

Concretely to compute this, we find K such that $v \in V^K$ and $f(gk) = f(g)$ for all $k \in K, g \in G$, then:

$$\int_{G(F)} f(g)(g \cdot v) dg = \sum_{\overline{g} \in G(F)/K} f(g)(g \cdot v) \mathrm{vol}(K, dg)$$

This is a finite summation since f has compact support.

Proposition 2.6. *The functor:*

$$\mathrm{Rep}^{\mathrm{sm}}(G(F)) \to \mathcal{H}\text{-mod}$$

is a fully faithful embedding.

This is essentially because $\mathcal{H}$ has enough functions approximating any element in G.

But this functor is not essentially surjective. Suppose V as a $\mathcal{H}$-module coming from a smooth representation, then:

$$V^K = e_K V$$

in fact

$$e_K : V \to V^K$$

is the averaging map under the K-action. Therefore:

$$V = \bigcup_K V^K = \bigcup_K e_K V$$

Definition 2.7. An $\mathcal{H}$-module M is smooth if $M = \bigcup_K e_K M$. We thus have an equivalence:

$$\mathrm{Rep}_{\mathrm{sm}}(G) \simeq \mathcal{H}\text{-mod}_{\mathrm{sm}}$$

Now if F is a function field, then $G(\mathbb{A})$ is again a totally disconnected group. For $G = \mathrm{GL}_n$, we have:

$$\mathrm{GL}_n(\mathbb{A}) = \prod{}'(\mathrm{GL}_n(F_v), \mathrm{GL}_n(\mathcal{O}_v))$$

Now if D is an effective divisor on X, we have:

$$K_D \subset \prod_v \mathrm{GL}_n(\mathcal{O}_v) \subset \mathrm{GL}_n(\mathbb{A})$$

where the first inclusion is of finite index. Where:

$$K_D = \ker(\prod_v \mathrm{GL}_n(\mathcal{O}_v) \to \mathrm{GL}_n(\mathcal{O}_D))$$

where $\mathrm{GL}_n(\mathcal{O}_D) = \prod_{v, n_v \neq 0} \mathrm{GL}_n(\mathcal{O}_v/\varpi_v^{n_v})$ for $D = \sum n_v v$.

Remark. *Similar procedure can be done for defining $G(\mathbb{A})$ for general G. But we need to make sense of $G(\mathcal{O}_v)$. This requires choosing a model $\mathcal{G} \to \mathrm{Spec}\ R$ for $\mathrm{Spec}\ R \subset \mathrm{Spec}\ \mathcal{O}_F$ of X open . But the good news is for two models $\mathcal{G}, \mathcal{G}'$, we have:*

$$\mathcal{G}(\mathcal{O}_v) = \mathcal{G}'(\mathcal{O}_v) \subset G(F_v)$$

This makes the restricted product uniquely defined.

We can thus define global Hecke algebra for $G(\mathbb{A})$, denoted as $\mathcal{H}_{\mathbb{A}}$. Then:

$$\mathcal{H}_{\mathbb{A}} = \bigotimes{}'(\mathcal{H}_v, e_v), \quad e_v = e_{G(\mathcal{O}_v)}$$

If F is number field, we will define analogue of Hecke algebra for $G(\mathbb{R})$, and we will have some similar result as before. The automorphic representations of $G(\mathbb{A})$ are certain smooth $\mathcal{H}_{\mathbb{A}}$-modules.

From Homework we see for an admisisble representation V and a fixed compact open K, we have:

$$V = \bigoplus_{\sigma \in \mathrm{Irr}(K)} V(\sigma) = \bigoplus_{\sigma \in \mathrm{Irr}(K)} \sigma \otimes \mathrm{Hom}_K(\sigma, V)$$

with each $\mathrm{Hom}_K(\sigma, V)$ finite-dimensional.

Proposition 2.8 (Schur's lemma)**.** *Let V_1, V_2 be irreducible admissible G-reps, then:*

$$\mathrm{Hom}_G(V_1, V_2) = \begin{cases} 0, & \text{if } V_1 \not\simeq V_2; \\ \mathbb{C}, & \text{if } V_1 \simeq V_2. \end{cases}$$

Proof. For any nonzero G-equivariant map $T : V \to V$, it must be isomorphism. So $D = \mathrm{End}_G(V) \supset \mathbb{C}$ is a skew-field. Moreover, $\mathbb{C} \subset D$ is central. Now For any $x \in D \backslash \mathbb{C}$, it must be transcendental. So that $D \supset \mathbb{C}(x)$ - this is uncountable dimensional. However $\mathrm{End}_G(V)$ is countable dimensional over $\mathbb{C}$ if V is admissible. Therefore we conclude $D = \mathbb{C}$. □

Fact: If $G = \underline{G}(F)$ for $\underline{G}$ a reductive group, F non-archimedean local field, then all irreducible smooth representations of G are admissible.

2.2 Parabolic Induction

Let's now set up parabolic induction. Fix $P \subset G$ parabolic, with Levi quotient $L = P/P^u$. Given any smooth representation τ of L with representation space V, then we have naive induction:

$$\mathrm{Ind}_P^G(\mathrm{Inf}_L^P(\tau)) = \{f : G \to V \mid f(pg) = \tau(\overline{p})f(g)\}$$

The group G acts on this space by right translation. We now restrict to only smooth vectors:

$$\{f : G \to V \mid f(pg) = \tau(\overline{p})f(g); \exists k, f(xk) = f(x)\}$$

Moreover let's do another normalization:

$$\mathrm{Ind}_P^G(\tau) = \{f : G \to V \text{ smooth} \mid f(pg) = \delta(p)^{\frac{1}{2}}\tau(\overline{p})f(g)\}$$

where $\delta : P \to \mathbb{R}_{>0}$. The character δ arises in the following way. The right objects to integrate on X are top forms. So I need to look at the line bundle ω_X of top forms, so that I can integrate the sections of ω_X. Given θ a smooth section of ω_X, we wish to define:

$$\int_X |\theta|$$

Now I have partition of unity $\theta = \sum \theta_i$, for θ_i supported on one chart $U_i \subset F^n$. Then I can write $\theta_i = f(x_1, \cdots, x_n)dx_1 \wedge \cdots \wedge dx_n$. Then I define:

$$\int_X |\theta_i| = \int_{F^n} |f(x_1, \cdots, x_n)|_F (dx)^n$$

for a fixed Haar measure dx on F (say of Fourier self-dual). If we change a coordinate to $(y_1, \cdots, y_n)$ with:

$$f(x_1, \cdots, x_n)dx_1 \wedge \cdots \wedge dx_n = g(y_1, \cdots, y_n)dy_1 \wedge \cdots \wedge dy_n$$

we need to check:

$$\int_{F^n} |f(x_1, \cdots, x_n)|_F (dx)^n = \int_{F^n} |f(y_1, \cdots, y_n)|_F (dy)^n$$

Now $TX = \mathfrak{p}\backslash\mathfrak{g} \times^P G$, so that:

$$\omega_X = (\wedge^{\mathrm{top}}((\mathfrak{p}\backslash\mathfrak{g})^*)) \times^P G$$

Then P acts on $\wedge^{\mathrm{top}}((\mathfrak{p}\backslash\mathfrak{g})^*)$ by p acting by $\det(p \mid (\mathfrak{p}\backslash\mathfrak{g})^*) = \det(p \mid \mathfrak{g})^{-1}\det(p \mid \mathfrak{p})$. We thus define:

$$\delta(p) := |\det(p \mid (\mathfrak{p}\backslash\mathfrak{g})^*)|_F = |\det(p \mid \mathfrak{g})|_F^{-1} \cdot |\det(p \mid \mathfrak{p})|_F = \frac{\delta_P(p)}{\delta_G(p)} \in \mathbb{R}_{>0}$$

Remark. *Let (G, μ) be any topological group with a left invariant Haar measure. Note that $\mathrm{Ad}(g)^*\mu$ is still left invariant. Therefore we can define:*

$$\delta_G(g) \coloneqq \frac{\mathrm{Ad}(g)^*\mu}{\mu} \in \mathbb{R}^{>0}$$

This in facts give a homomorphism:

$$\delta_G : G \to \mathbb{R}^{>0}$$

This is the modulus character. From above we see we can in fact compute modulus character by computing determinant of adjoint action.

Remark. *If G has no homomorphism to $\mathbb{G}_m$, then δ_G is trivial. For example, this is true for unipotent groups and semisimple groups. On ther other hand, δ_G is obviously trivial for abelian groups. Therefore the modulus character is trivial for any reductive group.*

In particular, in our setting G is reductive, hence $\delta_G(p) = 1$. So in fact:

$$\delta(p) = \delta_P(p)$$

For example, if we take $B_2 \subset \mathrm{GL}_2$, then $\mathrm{Ad}\begin{pmatrix} a & b \\ & d \end{pmatrix}$ acts on $\mathfrak{b} = \mathrm{Lie}B$ by: $\begin{pmatrix} 1 & & \\ & 1 & \\ & & \frac{a}{d} \end{pmatrix}$.

Therefore: $\delta(b) = |\frac{a}{d}|$ in this case.

If θ is a smooth section of ω_X, then $|\theta|$ is a section of $|\omega_X|$-a real locally constant vector bundle over X. This is because $F^\times \xrightarrow{|\cdot|} \mathbb{R}_{>0} \subset \mathbb{R}^\times$. We can thus integrate sections of $|\omega_X|$. Concretely, we have:

$$|\omega_{P\backslash G}| = \mathbb{R} \times^P G$$

for P acts on $\mathbb{R}$ by $\delta_P = |\det(\mathrm{Ad}(\overline{p}) \mid \mathfrak{n}_P)|$. This is nothing but functions on G transformed under δ:

$$f(pg) = \delta_P(p) f(g)$$

If τ is a representation of P, and $\tau^\vee$ its dual representation. We claim that there will be a pairing:

$$\mathrm{Ind}_P^G(\tau) \times \mathrm{Ind}_P^G(\tau^\vee) \to \mathbb{C}$$

$$(f : G \to V, f' : G \to V^\vee) \mapsto \int_{P\backslash G} \langle f(g), f'(g) \rangle$$

Note that $\phi(g) = \langle f(g), f'(g) \rangle$ is a function on G with transformation law:

$$\phi(pg) = \delta(p)\phi(g)$$

So it is a section of $|\omega_{P\backslash G}|$, hence the integration makes sense. In other words, *normalized induction commutes with duality.*

Example. *We have:*

$$\mathbb{C} \subset \mathrm{Ind}_P^G(\delta_P^{-\frac{1}{2}}) = \{f : P \setminus G \to \mathbb{C}\}$$

$$\mathrm{Ind}_P^G(\delta_P^{\frac{1}{2}}) = \{\text{smooth sections of } |\omega_{P\setminus G}|\} \xrightarrow{\int_{P\setminus G}} \mathbb{C}$$

This realizes $\mathbb{C}$ *as a quotient of* $\mathrm{Ind}_P^G(\delta_P^{\frac{1}{2}})$*. This is an instance of the duality as we described above.*

Facts: For $G = \mathrm{GL}_2(F)$, our only choice for parabolic induction is $P = B, L = T = F^\times \times F^\times \xrightarrow{\chi_1 \times \chi_2} \mathbb{C}^\times$. Then:

$$\mathrm{Ind}_B^G(\chi_1, \chi_2)$$

is irreducible, unless:

- $\chi_1 \cdot \chi_2^{-1} = |-|$, then Ind has a 1-dimensional quotient, with irreducible kernel;
- $\chi_1^{-1} \cdot \chi_2 = |-|$, then Ind has a 1-dimensional sub, with irreducible quotient.

In our previous example,

$$\delta_B = (|-|, |-|^{-1})$$

So:

$$\mathbb{C} \subset \mathrm{Ind}_B^G(|-|^{-\frac{1}{2}}, |-|^{\frac{1}{2}}) = \mathrm{Ind}_B^G(\delta_P^{-\frac{1}{2}})$$

$$\mathrm{Ind}_B^G(\delta_P^{\frac{1}{2}}) \twoheadrightarrow \mathbb{C}$$

as we've seen before.

Fact: $\mathrm{Ind}_B^G(\chi_1, \chi_2) \simeq \mathrm{Ind}_B^G(\chi_2, \chi_1)$ unless in reducible cases. This isomorphism can be given by explicit operators called *interwining integrals*.

We thus get the *principal series*: a class of irreducible representations parametrized by:

$$\{X^*(F^\times \times F^\times)\}/S_2$$

Other irreducible representations of $\mathrm{GL}_2(F)$ are parametrized by E/F quadratic extensions (if $p \neq 2$ there will be 3 of such), and pick a character of $E^\times$. The constructions come from Weil representations, or c-$\mathrm{Ind}_K^G(-)$ for certain $K \subset G$ compact opens.

2.3 Spherical Hecke Algebras and Their Representations

For the purpose of studying automorphic forms, we're interested in so called **spherical** (unramified) representations of G. Let $G = \mathrm{GL}_n(F), K = G(\mathcal{O}) = \mathrm{GL}_n(\mathcal{O})$ (can generalize to split groups).

Definition 2.9. An irreducible smooth representation of G is called **spherical** if $V^K \neq 0$.

Theorem 2.10. *Spherical representations of* $\mathrm{GL}_n(F)$ *are in bijection with* $(\mathbb{C}^\times)^n/S_n$

Construction by induction. Consider:

$$B \twoheadrightarrow (F^\times)^{\times n} \xrightarrow{(\chi_1,\cdots,\chi_n)} \mathbb{C}^\times$$

Assume each χ_i is unramified, namely it factors through the valuation map and is therefore given by $\chi_i(\varpi)$. Therefore given each element in $(\mathbb{C}^\times)^n$, we can form:

$$\mathrm{Ind}_B^G(\chi_1, \cdots, \chi_n)$$

This will always have a K-fixed vector. To see this, we need to solve for:

$$f(bgk) = \delta(b)^{\frac{1}{2}} \prod_{i=1}^{n} \chi_i(b_i) f(g)$$

We need:

Theorem 2.11 (Iwasawa Decomposition)**.** *We have decomposition:*

$$\mathrm{GL}_n(F) = N_n(F) \cdot \varpi^\Lambda \cdot K$$

for $\Lambda = \mathbb{Z}^n, \varpi^\Lambda = \{\mathrm{diag}(\varpi^{\lambda_1}, \cdots, \varpi^{\lambda_n})\}$

Remark. *The content of the theorem is the following. Right multiplication by K corresponds column operations that are invertible over $\mathcal{O}$; left multiplication by $N_n(F)$ corresponds to row operations preserving ordering of rows and cannot scale. Any matrix can be transformed into a unique diagonal matrix of the desired form ϖ^Λ in this way.*

Remark. *Compare with the real setting, we have Iwasawa decomposition:*

$$\mathrm{GL}_n(\mathbb{R}) = N_n(\mathbb{R}) \cdot (\mathbb{R}_{>0})^n \cdot \mathrm{O}(n)$$

From Iwasawa decomposition, we're actually seeking functions:

$$f : N_n(F) \setminus G/K \to \mathbb{C}$$

under the desired transformation law. Therefore we only need to specify its value at 1. But in particular we need:

$$f(1) = f(b \cdot b^{-1}), \quad b \in B \cap K = \begin{pmatrix} \mathcal{O}^\times & \cdots & \mathcal{O} \\ & \cdots & \cdots \\ & & \mathcal{O}^\times \end{pmatrix}$$

Therefore, the formula reads:

$$f(1) = f(b \cdot b^{-1}) = \prod_{i}^{n} \chi_i(b_i) f(1)$$

Hence to have a K-fixed vector, we need all χ_i to be unramified. In fact in this case, we have a one-dimensional family of K-fixed vectors.

Fact: $\mathrm{Ind}_B^G(\chi)$ has a finite length. It follows that it has a unique spherical subquotients.
Recall I have:

$$\mathbb{C} \subset \mathrm{Ind}_B^G(\delta^{-\frac{1}{2}})$$

$$\mathrm{Ind}_B^G(\delta^{\frac{1}{2}}) \twoheadrightarrow \mathbb{C}$$

Fact: $\mathrm{Ind}_B^G(\chi)$ and $\mathrm{Ind}_B^G(w\chi)$ for $w \in S_n$ have same irreducible subquotients. In partcular, they give isomorphism spherical representations.

Hence we have a well-defined map:

$$\{\chi = (\chi_1, \cdots, \chi_n) \mid \chi_i \text{ unramified}\}/S_n \to \{\text{spherical reps of } G\}$$

This is a bijection.

Suppose I have any compact open subgroup $K \subset G$, I can consider:

$$\mathrm{inv} : \mathrm{Rep}_{\mathrm{sm}}(G) \to \mathcal{H}_K\text{-mod}$$

$$V \mapsto V^K = e_K V$$

where $\mathcal{H}_K = C_c(K \setminus G/K)dg = e_K \mathcal{H} e_K$. This is a unital algebra.

The functor inv has a left adjoint:

$$\mathrm{ind}_! : M \mapsto C_c(G/K) \otimes_{\mathcal{H}_k} M = \mathcal{H} e_K \otimes_{\mathcal{H}_K} M \in \mathrm{Rep}_{\mathrm{sm}}(G)$$

The functor inv also has a right adjoint:

$$\mathrm{ind}_* : M \mapsto \mathrm{Hom}_{\mathcal{H}_K}(C_c(K \setminus G), M) = \mathrm{Hom}_{\mathcal{H}_K}(e_K \mathcal{H}, M) \in \mathrm{Rep}_{\mathrm{sm}}(G)$$

Remark. *The above construction has nothing special to do with p-adic group representation. In fact, given any ring R and idempotent $e \in R$, we have:*

$$\mathrm{inv} : R\text{-mod} \to eRe\text{-mod}, V \mapsto eV$$

$$\mathrm{ind}_! : eRe\text{-mod} \to R\text{-mod}, \quad M \mapsto Re \otimes_{eRe} M$$

$$\mathrm{ind}_* : eRe\text{-mod} \to R\text{-mod}, \quad M \mapsto \mathrm{Hom}_{eRe}(eR, M)$$

with the desired adjunctions. Moreover, note that:

$$\mathrm{inv} \circ \mathrm{ind}_*(M) = e\mathrm{Hom}_{eRe}(eR, M) \simeq \mathrm{Hom}_{eRe}(eRe, M) \simeq M$$

So that $\mathrm{inv} \circ \mathrm{ind}_*$ *is identity. Thus the identiy map:*

$$M \to \mathrm{inv} \circ \mathrm{ind}_* M$$

induces

$$\mathrm{ind}_! M \to \mathrm{inv}_* M$$

Let $\mathrm{Rep}_{\mathrm{sm}}(G,K)$ to be the collection of smooth representations of G generated by V^K (every quotient of V has K-fixed vectors), and that every nonzero subrepresentation of V has nonzero K-fixed vectors.

Fact:

$$\mathrm{Rep}_{\mathrm{sm}}(G,K) \xrightarrow{\mathrm{inv}} \mathcal{H}_K\text{-mod}$$

is an equivalence.

In fact, there will always be a natural transformation:

$$\mathrm{ind}_! \to \mathrm{ind}_*$$

the inverse of the above inv will be given by:

$$\overline{\mathrm{ind}_!} = \overline{\mathrm{ind}_*}$$

be the image of $\mathrm{ind}_! \to \mathrm{ind}_*$.

In particular, the irreducible objects inside $\mathrm{Rep}_{\mathrm{sm}}(G,K)$ are those irreducible smooth (hence admissible) ones such that $V^K \neq 0$. Hence we have:

Corollary 2.12. *We have a bijection between irreducible smooth representations of G with $V^K \neq 0$, with simple modules of $\mathcal{H}_K$.*

Also note tht

$$\mathrm{Irr}_{\mathrm{sm}}(G) = \bigcup_K \mathrm{Irr}(G,K)$$

Therefore to understand ireducible smooth representations of G, we only need to understand all simple modules for $\mathcal{H}_K$ for each compact K.

Now let $G = \mathrm{GL}_n(F), K = \mathrm{GL}_n(\mathcal{O})$. Then $\mathcal{H}_K$ is called the **spherical Hecke algebra**. We have:

$$\mathcal{H}_K = C_c(\mathrm{GL}_n(\mathcal{O}) \setminus \mathrm{GL}_n(F)/\mathrm{GL}_n(\mathcal{O}))$$

with $\int_{\mathrm{GL}_n(\mathcal{O})} dg = 1$ be the normalization of the measure.

Theorem 2.13 (Satake isomorphism)**.** *Let $G = \mathrm{GL}_n(F)$, then:*

$$\mathcal{H}_K \simeq \mathbb{C}[t_1^{\pm 1}, \cdots, t_n^{\pm 1}]^{S_n}$$

In particular $\mathcal{H}_K$ is commutative (this is true for any split reductive group). Therefore we have an isomorphism:

$$(\mathbb{C}^\times)^n/S_n \simeq \mathrm{Max}(\mathcal{H}_K) \simeq \{\text{spherical reps of } G\}$$

But we've already constructed a map:

$$(\mathbb{C}^\times)^n/S_n \to \{\text{spherical reps of } G\}$$

The remaining thing to check is that these two constructions are the same (this boils down to understanding the Satake isomorphism).

Note that the geometric object

$$\mathrm{GL}_2(F)/\mathrm{GL}_2(\mathcal{O})$$

that appears in the construction of the left adjoint for the inv functor parametrizes the $\mathcal{O}$-lattices inside F^2, given by isomorphism:

$$\mathrm{GL}_2(F)/\mathrm{GL}_2(\mathcal{O}) \to \{\mathcal{O}\text{-lattices} \subset F^2\}$$

$$\overline{g} \mapsto g \cdot \mathcal{O}^2$$

In fact if we consider:

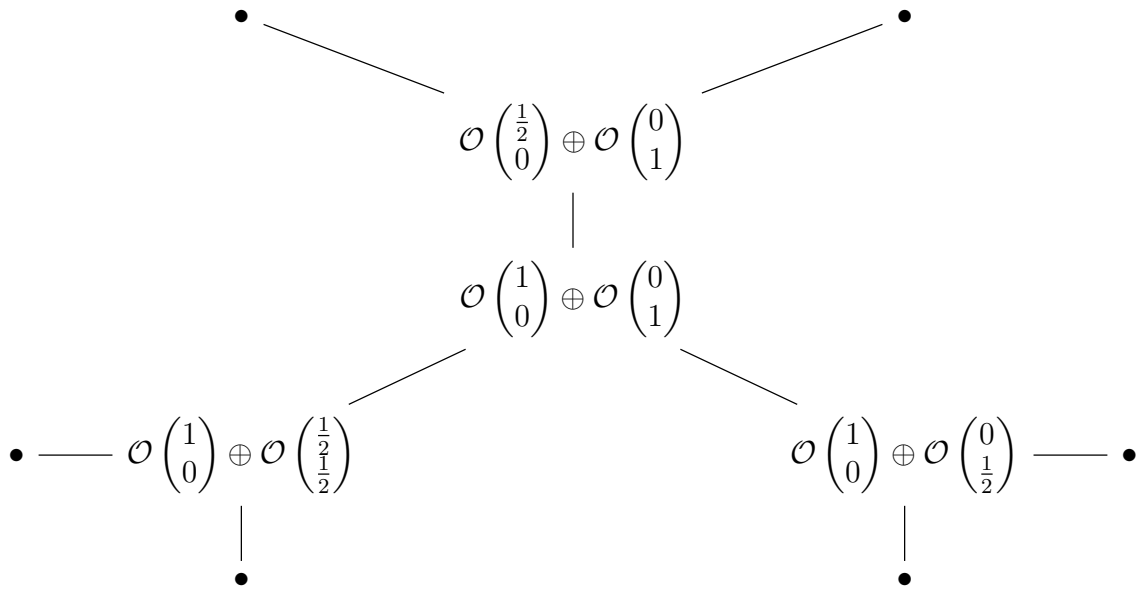
$$\mathrm{PGL}_2(F)/\mathrm{PGL}_2(\mathcal{O}_F)$$

it is in bijection with $\mathcal{O}$-lattices inside F^2 up to scaling by $F^\times$. I can understand these collection of lattices by certain combinatorial object called $q+1$-regular trees T_{q+1}. The origin of the tree will be the lattice $\mathcal{O}\begin{pmatrix}1\\0\end{pmatrix} + \mathcal{O}\begin{pmatrix}0\\1\end{pmatrix}$. Now if given a lattice $\mathcal{O}\cdot a + \mathcal{O}\cdot b$, we can specify a superlattice by specifying an element in $\mathbb{P}^1(\mathbb{F}_q)$. For $[x:y] \in \mathbb{P}^1(\mathbb{F}_q)$, then the corresponding superlattice is:

$$\mathcal{O}\cdot\langle a, b, x\varpi^{-1}a + y\varpi^{-1}b\rangle \supset \mathcal{O}\cdot\langle a, b\rangle$$

Note this is well-defined.

For example let's work for $F = \mathbb{Q}_2$ so that $q = 2$.



Note we can go back to $\mathcal{O}\begin{pmatrix}1\\0\end{pmatrix} \oplus \mathcal{O}\begin{pmatrix}0\\1\end{pmatrix}$ from $\mathcal{O}\begin{pmatrix}\frac{1}{2}\\0\end{pmatrix} \oplus \mathcal{O}\begin{pmatrix}0\\1\end{pmatrix}$ since we're allowed to rescale by $F^\times$. In general, the combinatorial object that will govern such lattices are certain simplicial complexes called **buildings**.

Let's now study the structure of $\mathcal{H}_K = C_c(K \setminus G/K)$. I can informally think of the spherical Hecke algebra as functions on:

$$\mathrm{GL}_n(F)/\mathrm{GL}_n(\mathcal{O}) \times \mathrm{GL}_n(F)/\mathrm{GL}_n(\mathcal{O})$$

which are invariant under the diagaonal $\mathrm{GL}_n(F)$ action with appropriate support conditions. I have a bijection:

$$K \setminus G/K \xrightarrow{\sim} G\text{-orbits of } (\Lambda_1, \Lambda_2)$$
$$g \mapsto (\mathcal{O}^n, g\mathcal{O}^n)$$

where Λ_1, Λ_2 are pairs of full rank $\mathcal{O}$-lattices inside F^n. One can check that this is well-defined (g,kg,gk has the same image), and is indeed a bijection.

Fact: (Smith normal form) Given two lattices $\Lambda_1, \Lambda_2 \subset F^n$, I can find a basis $e_1, \cdots, e_n \in F^n$ such that:

$$\Lambda_1 = \mathcal{O} \cdot e_1 \oplus \cdots \oplus \mathcal{O}e^n$$
$$\Lambda_2 : \varpi^{d_1}\mathcal{O} \cdot e_1 \oplus \cdots \oplus \varpi^{d_n}\mathcal{O} \cdot e_n$$

for $d_1 \geq \cdots \geq d_n$ intrinsic to Λ_1, Λ_2. This is saying that:

$$G\text{-orbits of } (\Lambda_1, \Lambda_2) \simeq \{d_1 \geq \cdots \geq d_n\}$$

We can reformulate this fact by **Cartan decomposition**:

$$G = \coprod_{d_1 \geq \cdots \geq d_n} K\mathrm{diag}(\varpi^{d_1}, \cdots, \varpi^{d_n})K$$

Proof. (of the fact) We can assume $\Lambda_1 \subset \Lambda_2$, by enlarging Λ_2 appropriately. In fact we can arrange that $\Lambda_1 \not\subset \varpi\Lambda_2$. Then I have:

$$\Lambda_1/\varpi\Lambda_1 \to \Lambda_2/\varpi\Lambda_2$$

Denote V the image of this map. Choose a basis:

$$V = \mathrm{Span}(\overline{e_1}, \cdots, \overline{e_r})$$

and lift to $e_1, \cdots, e_r \in \Lambda_1 \subset \Lambda_2$. So that:

$$\mathcal{O}e_1 \oplus \cdots \oplus \mathcal{O}e_r \subset \Lambda_1$$

Let $\overline{\Lambda_1} = \Lambda_1/\mathcal{O}e_1 \oplus \cdots \oplus \mathcal{O}e_r$, $\overline{\Lambda_2} = \Lambda_2/\mathcal{O}e_1 \oplus \cdots \oplus \mathcal{O}e_r$. Now that $\overline{\Lambda_1} \subset \overline{\Lambda_2}$ lattices in $F^n/\mathrm{Span}(e_1, \cdots, e_r)$. And then I do induction to get $\varepsilon_{r+1}, \cdots, \varepsilon_n$ for $(\overline{\Lambda_1}, \overline{\Lambda_2})$, and lift to $e_{r+1}, \cdots, e_n \in \Lambda_1$. Then $e_1, \cdots, e_n$ will be $\mathcal{O}$-basis for Λ_1. We can then adjust $e_{r+1}, \cdots, e_n$ by some elements in $e_1, \cdots, e_r$ to make sure $\varpi^{d_{r+i}}e_{r+i} \in \Lambda_2$. □

Now Cartan decomposition for $\mathrm{PGL}_2(F)$ reads:

$$\mathrm{PGL}_2(F) = \coprod_{d \geq 0} K \begin{pmatrix} \varpi^d & \\ & 1 \end{pmatrix} K$$

or equivalently:

$$\{d \geq 0\} \xrightarrow{\sim} \mathrm{PGL}_2(F)\text{-orbits of } ([\Lambda_1], [\Lambda_2])$$

Now the Bruhat-Tits tree for $\mathrm{PGL}_2(F)$, the vertices correspond to a scaling class of lattice, and $[\Lambda], [\Lambda']$ shares an edge iff $([\Lambda], [\Lambda'])$ correspond to $d = 1 \iff \Lambda = \mathcal{O}e_1 \oplus \mathcal{O}e_2, \Lambda' = \varpi\mathcal{O} \cdot e_1 \oplus \mathcal{O} \cdot e_2$ for some basis $e_1, e_2 \in F^2$. This is a regular tree of degree $q+1$. And we will see in homework that PGL_2-orbits of $[\Lambda], [\Lambda']$ by the distance between their distance.

In general for PGL_n, the Bruhat-Tits tree will be replaced by BT-buildings: certain $(n-1)$-dimensional simplicial complex.

The Cartan decomposition tells me $\mathcal{H}_K$ has basis:

$$\mathbf{1}_{K\mathrm{diag}(\varpi^{d_1}, \cdots, \varpi^{d_n})K}$$

Lemma 2.14. *$\mathcal{H}_K$ is commutative.*

Proof. We use the Gelfand trick. Consider $g \mapsto {}^t g$. This induces a map:

$$\tau : \mathcal{H}_K \to \mathcal{H}_K$$

Note that by construction τ is an anti-involution. But τ fixes the distinguished basis we chose. So in fact $\tau = \mathrm{id}$, we're done. □

Remark. *For general split reductive group G, we have Cartan decomposition:*

$$G = \coprod_{\lambda \in X_*(T) \text{ dominant}} K\varpi^\lambda K$$

$$\varpi^\lambda \in T(F), \lambda \in X_*(T)$$

here we mean $\varpi^\lambda = \lambda(\varpi) \in T(F)$. Again we can deduce $\mathcal{H}_K$ is commutative by introducing Cartan involution:

$$\sigma : G \to G$$

defined over $\mathbb{Z}$, such that $\sigma(t) = t^{-1}$ for $t \in T(F)$. If we choose $G \supset B, B^-$ opposite Borels, we will have $\sigma(B) = B^-$. And consider $\tau : g \mapsto \sigma(g^{-1})$, this is an anti-involution, which induces $\tau : \mathcal{H}_K \to \mathcal{H}_K$ an anti-involution. But τ acts identity on T, we're therefore done as before.

Claim:

$$\mathbf{1}_{K\mathrm{diag}(\varpi^{d_1}, \cdots, \varpi^{d_n})K} \cdot \mathbf{1}_{K\mathrm{diag}(\varpi^{d'_1}, \cdots, \varpi^{d'_n})K} = \mathbf{1}_{K\mathrm{diag}(\varpi^{d_1+d'_1}, \cdots, \varpi^{d_n+d'_n})K} + \text{lower terms}$$

Let's try to compute this in the case of PGL_2. Note that:

$$\mathrm{LHS}(\mathrm{diag}(\varpi^{\ell_1}, \cdots, \varpi^{\ell_n})) = \#\{\Lambda \mid \mathcal{O}^n \xrightarrow{(d_1, \cdots, d_n)} \Lambda \xrightarrow{(d'_1, \cdots, d'_n)} \varpi^{\ell_1}\mathcal{O} \oplus \cdots \oplus \varpi^{\ell_n}\mathcal{O}\}$$

In the case of PGL_2, then very concretely (for $d \geq d'$):

$$\mathrm{LHS}(\begin{pmatrix} \varpi^\ell & \\ & 1 \end{pmatrix}) = \begin{cases} 0, & \text{if } d + d' < \ell, \text{ or } d + d' \not\equiv \ell (\mathrm{mod} 2); \\ 1, & \text{if } d + d' = \ell; \\ (q-1)q^{k-1}, & \text{if } d + d' - 2k = \ell, 1 \leq k \leq d' - 1; \\ c_{d,d'}, & \text{if } \ell = d - d'. \end{cases}$$

where $c_{d,d'} = \begin{cases} q^{d'}, & \text{if } d > d'; \\ q^{d'-1}(q+1), & \text{if } d = d'. \end{cases}$. In other words, if T_d is the characteristic function on $K\mathrm{diag}(\varpi^d, 1)K$, then we have:

$$T_d * T_{d'} = T_{d+d'} + (q-1)\sum_{k=1}^{d'-1} q^{k-1} T_{d+d'-2k} + c_{d,d'} T_{d-d'}$$

In general give the two partitions, I can compare them using the dominant ordering (this is in fact equivalent to the closure ordering on the level of K-double cosets). From this description, I see the multiplication of indicator functions indeed have our desired form.

In other words, we can give a filtration on $\mathcal{H}_K$ using partial ordering, and then:

$$\mathrm{gr}\mathcal{H}_K \simeq \mathbb{C}[X_*(T)^{\mathrm{dom}}]$$

In the special case PGL_2, this is just isomorphic to $\mathbb{C}[t]$.

Abstractly, for G semisimple, $\mathcal{H}_K$ is a polybomial ring, with the number of variables equal the rank of the group.

Theorem 2.15 (Satake)**.** *Let G be split reductive. Then:*

$$\mathcal{H}_K \simeq \mathbb{C}[X_*(T)]^W$$

Recall abstractly, $\mathcal{H}_K$ is a polynomial ring in $t = \mathbf{1}_{K\begin{pmatrix} \varpi & \\ & 1 \end{pmatrix}K}$. On the other hand Satake says we should view:

$$\mathcal{H}_K \simeq \mathbb{C}[\mathbb{Z}]^{S_2} = \mathbb{C}[x, x^{-1}]^{S_2} = \mathbb{C}[x + x^{-1}]$$

We should view $t \mapsto x + x^{-1}$. Let's describe Satake transform in general. Start from $\phi \in \mathcal{H}_K$, we need to output $S\phi \in \mathbb{C}[X_*(T)]^W \subset \mathbb{C}[X_*(T)]$. So the data I need to specify is:

$$S\phi(\underline{d})$$

for each $\underline{d} \in X_*(T)$. Our formula will be:

$$S\phi(\underline{d}) = \delta_B^{\frac{1}{2}} \begin{pmatrix} \varpi^{d_1} & & \\ & \cdots & \\ & & \varpi^{d_n} \end{pmatrix} \int_{N(F)} \phi(\begin{pmatrix} \varpi^{d_1} & & \\ & \cdots & \\ & & \varpi^{d_n} \end{pmatrix} x) dx$$

To check this makes sense we need to check $S\phi(\underline{d})$ unchanged after permuting $\underline{d}$.

The idea for the formula of Satake isomorphism is the following. Recall we first constructed spherical representations using $\mathrm{Ind}_B^G(\chi)$. Now let χ vary. Note this parabolic induction is roughly the collection of χ-eigenfunctions under T of $N \backslash G$ Since we're looking at unramified characters, we're looking at $X_*(T) = T(F)/T(\mathcal{O})$-eigenfunctions on $T(\mathcal{O})N \backslash G$. Therefore we should look at the universal principal series:

$$\mathrm{Ind}_{T(\mathcal{O})N}^G(\mathbb{C})$$

This has a left G-action and a right $X_*(T)$-action, which commute. Let's look at

$$(\mathrm{Ind}_{T(\mathcal{O})N}^{G}(\mathbb{C}))^K$$

this has a left $\mathcal{H}_K$-action and a right $X_*(T)$-action. From Iwasawa decomposition we know:

$$(\mathrm{Ind}_{T(\mathcal{O})N}^{G}(\mathbb{C}))^K \simeq C(T(\mathcal{O})N \setminus G/K) \simeq \mathrm{Fun}(X_*(T))$$

where $X_*(T)$ acts by tanslation. If I pass to

$$M_{\mathrm{sph}} = C_c(T(\mathcal{O})N \setminus G/K)$$

it admits a $\mathbb{C}[X_*(T)]$-action, this is actually a free rank one module. Therefore I have a map:

$$\mathcal{H}_K \to \mathrm{End}_{\mathbb{C}[X_*(T)]}(M_{\mathrm{sph}}) \simeq \mathbb{C}[X_*(T)]$$

It turns out that this is precisely the Satake transform we hoped. The reason for $\mathcal{H}_K$ acts land in W-invariant part is because actually from the fact that:

$$\mathrm{Ind}_B^G(\chi) \simeq \mathrm{Ind}_B^G(w\chi)$$

for generic χ, from this we see actually

$$C_c(T(\mathcal{O})N \setminus G/K)$$

actually admits an additional right W-action.

Let's summarize what we get so far. We identify spherical representations of $\mathrm{Max}(\mathcal{H}_K)$. But note:

$$\mathrm{Max}\ \mathbb{C}[X_*(T)] = \mathrm{Hom}(X_*(T), \mathbb{C}^\times) = \widehat{T}$$

is defined to be the Langlands dual of T (notably this is a *complex* torus). Therefore: spherical representations are classified by $\widehat{T}/W$.

2.4 Langlands Dual Group

Let $\underline{G}/F$ be a split reductive group, with maximal torus T. Then we can associate certain combinatorial data:

$$(\Lambda = X^*(T), R = \{\text{roots}\}, \Lambda^\vee = X_*(T), R^\vee = \{\text{coroots}\})$$

There is an obvious symmetry. From the data:

$$(\Lambda^\vee, R^\vee, \Lambda, R)$$

we get $\widehat{G}/\mathbb{C}$ a complex reductive group (up to isomorphism), in which we have a maximal torus $\widehat{T}$, with:

$$X_*(\widehat{T}) = X^*(T), X^*(\widehat{T}) = X_*(T)$$

The roots of $(\widehat{G}, \widehat{T}) = R^\vee$.

Example. *Let $G = \mathrm{GL}_n$, then $\widehat{G} = \mathrm{GL}_n$. Since the root data is:*

$$(\mathbb{Z}^n, \{e_i - e_j \mid i \neq j\}, \mathbb{Z}^n, \{e_i^* - e_j^*\})$$

Therefore $\widehat{\mathrm{GL}_n} = \mathrm{GL}_n$.

Example. *Let $G = \mathrm{SL}_n$, then the root data is:*

$$(\Lambda = \mathbb{Z}^n/\Delta\mathbb{Z}, \{\overline{e_i - e_j}\}, \Lambda^\vee = (\mathbb{Z}^n)^{\Sigma=0}, \{e_i^* - e_j^*\})$$

It follows:

$$\widehat{G} = \mathrm{PGL}_n = \mathrm{GL}_n/\mathbb{G}_m$$

since Λ precisely gives the cocharacters in $\widehat{T} = (\mathbb{G}_m)^n/\mathbb{G}_m$

Example. *Let $G = \mathrm{Sp}_{2n}$, so $\widehat{G} = \mathrm{SO}_{2n+1}$. For G, the root data is:*

$$(\mathbb{Z}^n, \{e_i - e_j, e_i + e_j, 2e_i\}, \mathbb{Z}^n, \{e_i^* - e_j^*, e_i^* + e_j^*, e_i^*\})$$

The maximal torus consists of

$$\begin{pmatrix} x_1 & & & & & & \\ & \cdots & & & & & \\ & & x_n & & & & \\ & & & x_n^{-1} & & & \\ & & & & \cdots & & \\ & & & & & x_1^{-1} & \end{pmatrix}$$

The reason Langlands introduce the notion of Langlands dual group is that we can then view:

$$\{\text{spherical reps of } G(F)\} \longleftrightarrow \{\text{semisimple conjugacy classes of } \widehat{G}\}$$
$$\longleftrightarrow \{\mathbb{Z} \to \widehat{G} \text{ semisimple image}\}/\widehat{G} \simeq \{\text{unramified } W_F \to \widehat{G} \text{ s.s}\}/\widehat{G}$$

for the last identification, we view $\mathbb{Z} \simeq \mathrm{Fr}^{\mathbb{Z}}$.

2.5 Representations of Real Lie Groups

Recall for p-adic groups, we studied:

1. $\mathrm{Rep}_{\mathrm{sm}}(G) \longleftrightarrow (\mathcal{H}\text{-mod})_{\mathrm{sm}}$
2. Spherical representations.

We will do the same for real Lie groups. The basic examples of Lie groups are:

$$\mathrm{GL}_n(\mathbb{R}), \mathrm{GL}_n(\mathbb{C}), \mathrm{GL}_n(\mathbb{H}), U(p,q), O(p,q), \mathrm{Sp}(p,q), \mathrm{Sp}_{2n}(\mathbb{R})$$

Note that: $O(n), U(n), \mathrm{Sp}(n)$ are compact. In fact all groups we're considering have algberaic nature, and takes form $G(\mathbb{R})$ for algebraic group G. Now I can complexify and get $G(\mathbb{C})$, which are classified by the root data.

For $\mathrm{GL}_n(\mathbb{R}), U(p,q), \mathrm{GL}_{\frac{n}{2}}(\mathbb{H})$ for $p+q=n$, they all complexify to $\mathrm{GL}_n(\mathbb{C})$. Additional invariants are given by the maximal compact subgroups.

Theorem 2.16 (Cartan). *All maximal compact subgroups of $G(\mathbb{R})$ are conjugate.*

Example. *The maximal compact in $\mathrm{GL}_n(\mathbb{R})$ is $O(n)$; the maximal compact in $U(p,q)$ is $U(p) \times U(q)$; the maximal compact in $\mathrm{Sp}_{2n}(\mathbb{R})$ is $U(n)$.*

In the story of p-adic groups, we denote $K = \underline{G}(\mathcal{O})$; in the real case, we deonte K as a maximal compact subgroup $\subset G(\mathbb{R})$. Now let's classify real reductive groups.

Theorem 2.17 (Cartan). *Let $G/\mathbb{C}$ be a reductive group. There is a bijeciton:*

$$\{\text{real forms of } G\}/\sim \longleftrightarrow \{\theta : G \to G \mid \theta^2 = 1\}/\text{conj inside } \mathrm{Aut}(G)$$

Example. *Let $G(\mathbb{C}) = \mathrm{GL}_n(\mathbb{C})$. We have:*

1. *θ inner involution. It takes form:*

$$\begin{pmatrix} 1 & & & & & \\ & \cdots & & & & \\ & & 1 & & & \\ & & & -1 & & \\ & & & & \cdots & \\ & & & & & -1 \end{pmatrix}$$

   of blocks of size (p,q). This gives $U(p,q)$.

2. *There are outer involutions. We wish to get an automorphism:*

$$\mathrm{GL}_n(\mathbb{C}) \to \mathrm{GL}_n(\mathbb{C})$$

   Abstractly write $\mathbb{C}^n = V$, then we wish to define:

$$\mathrm{GL}(V) \to \mathrm{GL}(V^*)$$

   For $\phi : V \simeq V^*$, we can define:

$$g \mapsto \phi \circ g \circ \phi^{-1}$$

   From this perspective, the outer autormorphisms of $\mathrm{GL}(V)$ are classified by:

$$\mathrm{Isom}(V, V^*)/\mathbb{C}^\times$$

   If we're seeking for involution, then we must either get a symmetric pairing or an alternating pairing. This will give to:

$$\mathrm{GL}_n(\mathbb{R}), \mathrm{GL}_{\frac{n}{2}}(\mathbb{H})$$

Let's now discuss representations of real groups. Let $V/\mathbb{C}$ be a Hilbert space. A continuous representation of G is:

$$G \times V \to V$$

notably G may not preserve the inner product. If G does, we call this representation **unitary**.

Example. *Let G act on a manifold X. Then let's look at:*

$$L^2(X, |\omega_X|^{\frac{1}{2}})$$

This is a Hilbert space with a group action.

Definition 2.18. Let V be a continuous G-representation. We call $v \in V$ is smooth if: for every $v' \in V$, the map:

$$m_{v,v'} : G \to \mathbb{C}, \quad g \mapsto \langle v', gv \rangle \in \mathbb{C}$$

is a smooth function. Call the subspace of smooth vectors V^{sm}

In fact $V^{\mathrm{sm}} \subset V$ is dense. To see this, given any $v \in V$, for any $\phi \in \mathrm{Meas}_c^\infty(G)$ we have:

$$\phi * v = \int_G \phi(g) gv$$

is smooth. But now using ϕ to approximate the delta function we see the density. However, V^{sm} is no longer a Hilbert space, but it remains to be a Fréchet space (which is actually the right category to work with).

Now we wish to pass to:

$$V^{\mathrm{sm}}_{K\text{-fin}} \subset V^{\mathrm{sm}}$$

the collection of K-finite vectors. This is again a dense subspace. So we're not losing information when studying this space. However, we're losing the global G-action: if v is K-finite, then $g \cdot v$ need not be K-finite. But we still have:

1. K-action.
2. $\mathfrak{g}$-action. Recall for $v \in V^{\mathrm{sm}}$, $X \in \mathfrak{g}$, we define:

$$X \cdot v = \frac{d}{dt}|_{t=0} e^{tX} \cdot v$$

We must check that if v is K-finite, then so is $X \cdot v$. But note that:

$$k \cdot X \cdot v = (k \cdot X \cdot k^{-1}) \cdot k \cdot v = (\mathrm{Ad}(k) \cdot X) \cdot k \cdot v$$

Therefore as sets:

$$K \cdot \mathfrak{g} \cdot v = \mathfrak{g} \cdot K \cdot v$$

is finite-dimensional. In other words:

$$\mathfrak{g} \cdot v \subset V^{\mathrm{sm}}_{K\text{-fin}}$$

This motivates the following definition.

Definition 2.19. A $(\mathfrak{g}, K)$-module M is a $\mathbb{C}$-vector space M (countable dimensional) with a $\mathfrak{g}$-action, and a continuous K-action such that:

- M is locally finite under K.

- The action of $\mathrm{Lie}K = \mathfrak{k}$ on M is the same as the restriction of the $\mathfrak{g}$-action to $\mathfrak{k}$.
- For any $v \in M$, $X \in \mathfrak{g}$, $k \in K$, we have:

$$k \cdot X \cdot v = (\mathrm{Ad}(k)X) \cdot k \cdot v$$

Remark. *In homework, we will check the third condition is redundant when K is connected.*

We thus have constructed a functor:

$$\{\text{continuous Hilbert space representations of } G\} \to (\mathfrak{g}, K)\text{-mod}$$

This not an equivalence in general. To fix this, we restrict to "representations of G with moderate growth". This condition will play a role in our future discussion of automorphic representations. Let's now concentrate on $(\mathfrak{g}, K)$-modules.

In practice, we want to complexify both $\mathfrak{g}$ and K:

$$(\mathfrak{g}_{\mathbb{C}}, K_{\mathbb{C}}) = (\mathfrak{g}, K)\text{-mod}$$

This is because the finite-dimensional representations of K are equivalent to finite dimensional algebraic representations of $K_{\mathbb{C}}$.

Now let's look at $(\mathfrak{sl}_2(\mathbb{C}), \mathrm{SO}(2, \mathbb{C}) = \mathbb{C}^\times)$-modules. Now we can decompose:

$$M = \bigoplus_{n \in \mathbb{Z}} M_n$$

where $\mathbb{C}^\times$ acts on M_n by n-th power. Let's choose a basis e, h, f of $\mathfrak{sl}_2(\mathbb{C})$. We may arrange h such that:

$$\mathbb{C} \cdot h = \mathrm{Lie}\mathrm{SO}(2, \mathbb{C}) \subset \mathfrak{sl}_2(\mathbb{C})$$

Very concretely, we have:

$$\mathrm{SO}(2) = \{\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}\}$$

And my choice are:

$$h = \frac{1}{i}\begin{pmatrix} & 1 \\ -1 & \end{pmatrix}$$

$$e = \frac{1}{2}\begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}$$

$$f = \frac{1}{2}\begin{pmatrix} 1 & -i \\ -i & -1 \end{pmatrix}$$

Note that M be a $\mathfrak{g}, K$-module requires that h acts on M_n by n. And since h, e, f form a $\mathfrak{sl}_2$-triple:

$$e : M_n \to M_{n+2}, \quad f : M_n \to M_{n-2}$$

The last commutation relation reads:

$$ef - fe : M_n \to M_n, \quad v \mapsto nv$$

We have the following class of representations:

1. Most simple modules are principal series representations (induced from $B(\mathbb{R})$):

$$\cdots \rightleftarrows \mathbb{C} \rightleftarrows \cdots \rightleftarrows \mathbb{C} \rightleftarrows \cdots$$

2. The algebraic representations of $\mathrm{SL}_2(\mathbb{C})$.

3. Discrete series representations. For eample we have D_5^+:

$$5 \rightleftarrows 7 \rightleftarrows 9 \rightleftarrows \cdots$$

and D_5^-:

$$\cdots \rightleftarrows -9 \rightleftarrows -7 \rightleftarrows -5$$

All these are called discrete series except for D_1^+ and D_1^-. The later two are called limit of discrete series. Namely, discrete series are $D_n^{\pm 2}$ for $n \geq 2$.

2.6 Harish-Chandra Isomorphism and Infinitesimal Character

To classify all principal series representations, we need to look at certain invariants of $(\mathfrak{g}, K)$-modules. Now we write $\mathfrak{g}$ to replace $\mathfrak{g}_{\mathbb{C}}$. Then we have a $U\mathfrak{g}$ action on M.

Theorem 2.20 (Harish-Chandra)**.** *We have an isomorphism:*

$$Z(U\mathfrak{g}) \simeq \mathrm{Sym}(\mathfrak{h})^W$$

To understand this theorem, we need to construct elements in $Z(U\mathfrak{g})$. To seek for $P \in Z(U\mathfrak{g})$, we need $XP = PX$ for all $X \in \mathfrak{g}$. Now we look at the action of G on $U\mathfrak{g}$ and take derivative, we get:

$$\mathfrak{g} \times U(\mathfrak{g}) \to U(\mathfrak{g}), \quad (X, P) \mapsto XP - PX$$

But my group G is connected, so we have:

$$Z(U\mathfrak{g}) = (U\mathfrak{g})^G$$

Note that:

$$\mathrm{Sym}(\mathfrak{g})^G \hookrightarrow \mathbb{C}\langle\mathfrak{g}\rangle^G \to (U\mathfrak{g})^G = Z(U\mathfrak{g})$$

Now for $\mathfrak{g}$ semisimple, we have an distinguished element in $\Omega \in \mathrm{Sym}^2(\mathfrak{g})^G$ Casimir. This is the same data as the killing form: choose a basis x_i and a dual basis x^i under the Killing form, then:

$$\Omega = \sum x_i x^i$$

Now if we take $\{e, f, h\}$ to be one basis for $\mathfrak{sl}_2$, then the dual basis is $\{f, e, \frac{1}{2}h\}$ under the trace pairing (which is just Killing form rescaled). Hence my choice:

$$\Omega = ef + fe + \frac{1}{2}h^2$$

This can be viewed as an element in $(U\mathfrak{g})^G = Z(U\mathfrak{g})$. In fact, Harish-Chandra's theorem is claiming that:

$$\mathrm{Sym}(\mathfrak{h})^W \simeq \mathrm{Sym}(\mathfrak{g})^G \hookrightarrow \mathbb{C}\langle\mathfrak{g}\rangle^G \to (U\mathfrak{g})^G = Z(U\mathfrak{g})$$

is an isomorphism, where the first isomorphism is Chevalley's restriction theorem. This is an isomorphism of vector spaces by passing to associated graded, which commutes with taking G-invariant (everything is algebraic and G-reps are semisimple). To establish compatible ring structure, we need to give $U\mathfrak{g}$ a grading:

$$U(\mathfrak{g}) = \bigoplus_{\lambda \in X^*(H)} (U\mathfrak{g})_\lambda$$

In particular:

$$U(\mathfrak{g})_0 = U(\mathfrak{g})^H \supset U(\mathfrak{g})^G$$

The goal is to construct some two-sided ideal I such that:

$$\mathrm{HC} : Z(U\mathfrak{g}) = (U\mathfrak{g})^G \subset (U\mathfrak{g})^H \to (U\mathfrak{g})^H/I \simeq \mathrm{Sym}(\mathfrak{h})^W$$

Our choice is:

$$I = (\mathfrak{n}_- U(\mathfrak{g}))_0 = (U(\mathfrak{g})\mathfrak{n}_+)_0$$

this is a two-sided ideal by the above two identifications. This construction ensures that HC is ring homomorphism. In terms of our previous example $\mathfrak{sl}_2$, we have:

$$\mathrm{HC} : \Omega = \frac{1}{2}h^2 + ef + fe = \frac{1}{2}h^2 + h + 2fe \mapsto \frac{1}{2}h^2 + h$$

This shows that in the last bit $\mathrm{Sym}(\mathfrak{h})^W$ should be interpreted as the W-invariant under the **dot action**: the $-\rho$-shifted action, where $\rho = \frac{1}{2}\sum_{\alpha>0} \alpha$ so that $d\rho \in \mathfrak{h}^*$.

For a $(\mathfrak{g}, K)$-module M, we can canonically decompose into K-isotypical components:

$$M = \bigoplus_{\sigma \in \mathrm{Irr}(K)} M(\sigma)$$

where $M(\sigma) = \sigma \otimes \mathrm{Hom}_K(\sigma, M)$. This already incorporates the K-finiteness of the action. Moreover, I have an additional compatibility condition on two $\mathrm{Lie}(K)$-actions.

Definition 2.21. An **admissible** $(\mathfrak{g}, K)$-module is one such that each above $M(\sigma)$ is finite dimensional. A G-representation V is admissible if its corresponding $(\mathfrak{g}, K)$-module is admissible.

Definition 2.22. Let $(V, \|\|)$ be a Hilbert space representation of G. We call V **unitary** if G-action preserves the norm.

If $(V, \|\|)$ is unitary, then $M = V^\infty_{K\text{-fin}}$ inherits the norm (though not necessarily complete). Therefore, for the induced bilinear form $\langle -, - \rangle$, we have:

$$\langle Xu, v\rangle + \langle u, Xv\rangle = 0, \quad X \in \mathfrak{g}$$

$$\langle ku, kv\rangle = \langle u, v\rangle, \quad k \in K$$

The second condition is only needed for K is not connected.

Now $\mathfrak{g} = \mathfrak{g}_{\mathbb{C}}, G = G_{\mathbb{C}}$, recall we have Harish-Chandra isomorphsim:

$$Z(U\mathfrak{g}) \simeq (U\mathfrak{g})^G \simeq \mathrm{Sym}(\mathfrak{h})^W \simeq \mathbb{C}[\mathfrak{h}^*]^W$$

where W-action will fix the point $-\rho \in \mathfrak{h}^*$. Very explicitly, given any $z \in Z(U\mathfrak{g})$, we arrange every monomial appearing in the expression with negative root on the left, Cartan elements in the middle, and positive roots on the right. Therefore our expression of z will have some part inside $\mathrm{Sym}(\mathfrak{h})$, and the remaining involves positive and negative roots. Then the image of the Harish-Chandra map is just by discarding the later. There are two things to theck:

1. Harish-Chandra map is ring homomorphism (quotienting out by a two-sided ideal).
2. Image of Harish-Chandra map are W-invariant under the shifted action.
3. The morphism:
$$S^\bullet(\mathfrak{g})^G \to U(\mathfrak{g})^G \to (U\mathfrak{h})^{(W,\bullet)}$$
is an isomorphism, by passing to associated graded.

We need to check the second step. To check W-invariance, we need to study the action on Verma modules. For simplicity, let's assume $\mathfrak{g} = \mathfrak{sl}_2$, $W = \langle s \rangle$. For a weight λ, we denote M_λ for the highest weight Verma M_λ. This is irreducible except for $\lambda = k \in \mathbb{Z}_{\geq 0}$. Obviously in the later case:

$$0 \to M_{-k-2} \to M_k \to V_k \to 0$$

On any Verma M_λ, any $z \in Z(U\mathfrak{g})$ acts as scalar. In particular $\Omega = \frac{1}{2}h^2 + ef + fe$ will act by the same scalar on M_{-k-2} and M_k. Now that the image of Ω under the Harish-Chandra map is $\frac{1}{2}h^2 + h$, and Ω acts on M_λ by $\frac{1}{2}\lambda^2 + \lambda$. This tells us that $\mathrm{HC}(\Omega)$, viewed as a polynomial in $\mathfrak{h}^*$, will take the same value at k and $-k-2$.

Now fix a simple $(\mathfrak{g}, K)$-module M, any $z \in Z(U\mathfrak{g})$ acts on M via a scalar. This gives a homomorphism:

$$Z(U\mathfrak{g}) \to \mathbb{C}$$

which is a point in $\mathrm{Max}\, Z(U\mathfrak{g}) \simeq \mathfrak{h}^*/_{-\rho}W$. Therefore, starting from any simple $(\mathfrak{g}, K)$-module M, we can the a W-orbit on $\mathfrak{h}^*$. This is called the **infinitesimal character** for M.

Fact: For a fixed infinitesimal character $\lambda \in \mathfrak{h}^*/_{-\rho}W$, there are only finitely many isomorphism classes of simple $(\mathfrak{g}, K)$-modules. This finite number has an upper bound independent of λ (only depends on the group).

Example. *Consider $(\mathfrak{g}, K) = (\mathfrak{sl}_2, \mathrm{SO}(2))$. Let's compute infinitesimal characters on each principal series $(\mathfrak{g}, K)$-modules. Consider the scenario:*

$$n-2 \leftrightarrows n \leftrightarrows n+2$$

Now let C_{n-1} be the scalar that ef or fe acts between $n-2$ and n. From the commutation relation $[e, f] = h$ one deduce that $C_{n-1} - C_{n+1} = n$. Now we can explicitly check that $\Omega = \frac{1}{2}h^2 = ef + fe$ acts on the n-th piece by $\frac{1}{2}n^2 + C_{n-1} + C_{n+1}$ (this is independent of n). Now if we fix λ be the eigenvalue that Ω acts, we can then solve for all C_i.

Now for discrete series D_k^+ , let k be the lowest weight in the $(\mathfrak{g}, K)$-module, then:

$$C_{k-1} = 0, C_{k+1} = -k$$

So that Ω acts by $\frac{1}{2}k^2 + 0 - k = \frac{k(k-2)}{2}$. Similarly we can compute the action on D_k^-.

2.7 Principal Series and Discrete Series

Let's now discuss principal series. Let $B \subset \mathrm{SL}_2(\mathbb{R})$ be Borel. We need to induce a character:

$$\chi : \mathbb{R}^\times \to \mathbb{C}^\times$$

This is given by a pair $(s, \varepsilon) \in \mathbb{C} \times \{0, 1\}$

$$y \mapsto \mathsf{sgn}(y)^\varepsilon |y|^s$$

Let's consider the normalized induction:

$$\mathrm{Ind}_B^G(\mathsf{sgn}^\varepsilon \cdot |-|^s) = \{f : G \to \mathbb{C} \mid f(bg) = \chi(b)\delta_B(b)^{\frac{1}{2}} f(g)\}$$

Recall that:

$$\delta_B(\begin{pmatrix} y & \\ & \frac{1}{y} \end{pmatrix}) = |\frac{y}{y^{-1}}| = |y|^2$$

If $\varepsilon = 0$, then $\begin{pmatrix} -1 & \\ & -1 \end{pmatrix}$ will act trivially; and for $\varepsilon = -1$, the above matrix acts as -1. This precisely tells us about the parity of weights in the corresponding $(\mathfrak{g}, K)$-module. In homework, we showed that Ω acts by $\frac{1-s^2}{2}$. Also notably unitary induction of a unitary representation is unitary. In particular, for $s \in i\mathbb{R}$, we have:

$$\mathrm{Ind}_B^G(\mathsf{sgn}^\varepsilon \cdot |-|^s)$$

is unitary. These are **unitary principal series**. However, these do not exhaust all unitary representations. We still have **complementary series** $\mathrm{Ind}_B^G(\mathsf{sgn}^\varepsilon \cdot |-|^s)$ for $-1 < s < 1$.

To understand Discrete series for G, let's compare it with G_c the compact Lie group with the same complexification with G. For G_c, we have Peter-Weyl:

$$L^2(G_c) = \widehat{\bigoplus} V \boxtimes V^*$$

For G, the discrete series representations are precisely those appearing as subrepresentations of $L^2(G)$. Equivalently, we can require the matrix coefficient to be L^2-functions. Following our comparison in the compact group case, we have a canonical copy:

$$V \boxtimes V^* \subset L^2(G)$$

In the case of $\mathrm{SL}_2(\mathbb{R})$, suppose we want to construct D_k^-. The idea is that we should have a distinguished copy:

$$D_k^- \otimes (\text{weight } k \text{ line in } D_k^+) \subset L^2(G)$$

The weight k line is characterized by the property that it has weight k under the $\mathrm{SO}(2)$-action, and f acts by 0. It will translate to:

$$\phi(g \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}) = e^{ik\theta}\phi(g)$$

with vanishing f-action. Hence we should view ϕ as a section of certain line bundle on G/K. In our case:

$$G/K \simeq \mathcal{H} = \{z \in \mathbb{C} \mid \text{im } (z) > 0\}$$

$$gK \mapsto g \cdot i$$

Under this identification, f will act by $\overline{\partial}$. Therefore, we need to consider:

$$V = \{\phi : \mathcal{H} \to \mathbb{C} \mid \phi \text{ holomorphic }, \|\phi\|^2 = \int_{\mathcal{H}} |\phi(z)|^2 y^k \frac{dxdy}{y^2} < \infty\}$$

This is a Hilbert space. And we have a unitary action for $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$$(g^{-1}\phi)(z) = (cz+d)^{-k}\phi(\frac{az+b}{cz+d})$$

This preserves $\|-\|_k$. Our space $V = D_k^-$ is the **anti-holomorphic discrete series**.

For general G semisimple, the discrete series representations exist iff G has a maximal compact torus (equivalently, G is an inner form of G_c). For example for $\mathrm{SL}_n(\mathbb{R})$ with $n \geq 3$, there will be no maximal compact tori. So there will be no discrete series in this setting. On the contrary, $\mathrm{SU}(p,q)$ are inner forms of $\mathrm{SU}(p+q)$, so they have discrete series.

Theorem 2.23 (Classification of Discrete Series)**.** *Let G be simply connected. Start from $T \subset G$ maximal compact torus. Look at $\lambda \in X^*(T)$ regular (i.e. $\langle \lambda, \alpha^\vee \rangle \neq 0$). Then the discrete series are precisely parametrized by:*

$$X^*(T)^{\mathrm{reg}}/W(G,T)$$

For a sanity check, let's consider $G = G_c$, we know the representations should be parametrized by $X^*(T)/W$ (no regularity condition). The secret is that we should send λ in the first description to $\lambda - \rho$ in the second description.

3 Automorphic Forms and Automorphic Representations

3.1 Modular Forms

Let $G = \mathrm{SL}_2$, and $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ a congruent subgroup, i.e. $\Gamma \supset \Gamma(N) = \ker(\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N))$ for some N. Let's fix an integer $k \geq 2$. Let $\mathbb{H}$ be the upper half plane.

Definition 3.1 (Cusp)**.** Note that $\partial\mathbb{H} = \mathbb{P}^1(\mathbb{R})$. The cusps of $\mathbb{H}$ are $\mathbb{P}^1(\mathbb{Q}) \subset \partial\mathbb{H}$. For a congruent subgroup Γ, the cusp for Γ is:

$$\Gamma \setminus \mathbb{P}^1(\mathbb{Q}) \subset \Gamma \setminus \partial\mathbb{H}$$

In particular, for $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, there is only one cusp ∞.

Definition 3.2. A weight k, level Γ **modular form** is a holomorphic function: $f : \mathbb{H} \to \mathbb{C}$ with the properties:

1. For any $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma, z \in \mathbb{H}$, we have:

$$f(\frac{az+b}{cz+d}) = (cz+d)^k f(z)$$

2. f is holomorphic at the cusps. Let's explicitly describe how to check the holomorphicity at ∞ (any finite cusp can be translate to this scneario by applying some element in $\mathrm{SL}_2(\mathbb{Z})$). Note that $f(z) = f(z+N)$ for some large N, so f descend to a function on:

$$\mathbb{C}/N\mathbb{Z} \simeq \mathbb{C}^*$$

$$z \mapsto q = \exp(2\pi i z/N)$$

Then the holmorphicity at ∞ is nothing but the holomorphicity at $q = 0$.

A modular form is called a **cusp form** if it vanishes at all cusps.

We view $\mathbb{H} \subset \mathbb{P}^1_{\mathbb{C}}$ be an analytic open subset. In the case $k = 2$, we have $f(z)dz$ a 1-form on $\mathbb{H}$, we compute

$$f(\frac{az+b}{cz+d})d(\frac{az+b}{cz+d}) = f(\frac{az+b}{cz+d})(cz+d)^{-2}dz = f(z)dz$$

Therefore, weight 2 modular forms of level Γ are precisely the holomorphic 1-forms on $\Gamma \setminus \mathbb{H}$ with the cusp condition. Very explicitly, we let $q = \exp(2\pi i z)$, we can express:

$$f(z) = \phi(q) = a_0 + a_1 q + a_2 q^2 + \cdots$$

$$f(z)dz = \frac{1}{2\pi i}\phi(q)\frac{dq}{q}$$

The holomorphicity at cusp reads $\phi(q)$ has at most a simple pole at $q = 0$.

3.2 Automorphic Forms

Let's view $\mathbb{H} = \mathrm{SL}_2(\mathbb{R})/\mathrm{SO}_2(\mathbb{R})$. So we have:

$$\Gamma \setminus \mathrm{SL}_2(\mathbb{R})/\mathrm{SO}_2 \simeq \Gamma \setminus \mathbb{H}$$

We wish to interpret this as:

$$\mathrm{SL}_2(\mathbb{Q}) \setminus \mathrm{SL}_2(\mathbb{A})/K$$

In fact this is a general scenario. Let $G/\mathbb{Q}$ be a reductive group. Let's fix a maiximal compact $K_\infty \subset G(\mathbb{R})$. We also let $K_f \subset G(\mathbb{A}_f)$ be an open compact subgroup, such that there exists a finite set S of finite places with

$$K_f = \prod_{p \notin S} G(\mathbb{Z}_p) \times \prod_{p \in S} K_p$$

where K_p are open compact in $G(\mathbb{Q}_p)$. Now let:

$$K = K_\infty \cdot K_f \subset G(\mathbb{A})$$

$$\Gamma = G(\mathbb{Q}) \cap K_f$$

for example, if $K_f = \mathrm{SL}_2(\widehat{\mathbb{Z}})$, then $\Gamma = \mathrm{SL}_2(\mathbb{Z})$.

Fact: (Weil's theorem) For nice G, the canonical map

$$\Gamma \setminus G(\mathbb{R}) \to G(\mathbb{Q}) \setminus G(\mathbb{A})/K_f$$

is a bijection for all K_f. Here "nice" includes all semisimple, simply-connected, split groups (the split condition can be relaxed, for example it also works for $\mathrm{SU}(p,q)$ for $p, q \neq 0$).

This bijection means

$$G(\mathbb{Q})G(\mathbb{R}) \subset G(\mathbb{A})$$

is dense. This is called the **strong approximation** property for $(G, \mathbb{R})$.

Proposition 3.3. *The pair $(G, \mathbb{R})$ has strong approximation if:*

1. *G is semisimple and simply connected;*
2. *$G \sim \prod G_i$ for G_i $\mathbb{Q}$-simple, such that $G_i(\mathbb{R})$ non-compact for each i.*

Let's see the failure of strong approximation in the case of $G = \mathbb{G}_m$. Consider F to be any number field, then $K_f = \widehat{\mathcal{O}_F}$. We've seen:

$$1 \to \mathcal{O}_F^\times \setminus F_\infty^\times \to F^\times \setminus \mathbb{A}_F^\times / K_f \to \mathrm{Cl}(F) \to 1$$

This means the failure is precisely $\mathrm{Cl}(F)$.

In summary, we obtained an identification:

$$\Gamma \setminus \mathrm{SL}_2(\mathbb{R}) \xrightarrow{\sim} \mathrm{SL}_2(\mathbb{Q}) \setminus \mathrm{SL}_2(\mathbb{A})/K_f$$

for $\Gamma = \mathrm{SL}_2(\mathbb{Q}) \cap K_f$. Furhter quotienting out K_∞ on the right:

$$\Gamma \setminus \mathbb{H} \simeq \mathrm{SL}_2(\mathbb{Q}) \setminus \mathrm{SL}_2(\mathbb{A})/K$$

as desired.

Definition 3.4. Let $G/\mathbb{Q}$ be a reductive group. An **automorphic form** for G is a smooth function:

$$\phi : G(\mathbb{Q}) \setminus G(\mathbb{A}) \to \mathbb{C}$$

The smoothness means:

1. $g_\infty \mapsto \phi(x g_\infty)$ is smooth for $g_\infty \in G(\mathbb{R})$
2. $g_f \mapsto \phi(x g_f)$ is locally constant function for $g_f \in G(\mathbb{A}_f)$.

We have the following further constraints:

1. Now $\mathfrak{g}$ acts on such functions by derivatives of $G(\mathbb{R})$-action. We require ϕ to be $Z(U\mathfrak{g})$-finite (this is essentially imposing the condition that infinitesimal character is well-defined).

2. ϕ is K_∞-finite.

3. We also require there to exists open compact subgroup $K_f \subset G(\mathbb{A}_f)$ depending on ϕ such that ϕ is fixed by K_f. In this case we call K_f the level of ϕ.

4. We need f to have **moderate growth**. We choose a closed embedding:

$$G(\mathbb{A}) \hookrightarrow \mathbb{A}^N$$

Given $x = (x_1, \cdots, x_N) \in \mathbb{A}^N$, we can define the **height**:

$$\|x\| = \prod_v (\max_i(|x_i|_v, 1))$$

We can pullback this height function to $G(\mathbb{A})$ to get:

$$\|-\| : G(\mathbb{A}) \to \mathbb{R}_{>0}$$

We say ϕ has moderate growth if there exists C, r such that:

$$|\phi(g)| \leq C\|g\|^r$$

It is clear that the moderate growth condition is independent of the choice of the embedding.

Example. *Let $G = \mathrm{SL}_2/\mathbb{Q}$, and choose $K_f \subset G(\mathbb{A}_f)$, $k \geq 2$. We obtain $\Gamma = G(\mathbb{Q}) \cap K_f \subset G(\mathbb{R})$. Let $M_k(\Gamma)$ be the space of holomorphic modular forms of weight k, level Γ. We can define an embedding:*

$$M_k(\Gamma) \to \mathcal{A}^{K_f} \subset C^\infty(G(\mathbb{Q}) \setminus G(\mathbb{A})/K_f) = C^\infty(\Gamma \setminus G(\mathbb{R}))$$

$$f \mapsto (\phi_f : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto f(\frac{ai+b}{ci+d})(ci+d)^{-k})$$

One can check that $\phi_f(\gamma g) = \phi_f(g)$. Let's explain the dictionary between the properties of Modular forms with the properties with automorphic forms:

1. *f has weight $k \implies \phi_f(gs) = s^k \phi_f(g)$ for $s \in K_\infty = \mathrm{SO}_2$. In other words, ϕ_f is an eigenfunction under K_∞, with weight k.*

2. *f is holomorphic means that it is killed by the Cauchy-Rimeann operator. But the Cauchy Riemann operator is the lowering operator L (we normally call it f in $\mathfrak{sl}_2$), so that $Lf = 0$. This implies that:*

$$\Omega f = \frac{k(k-2)}{2} f$$

In particular, this tells us:

$$\dim Z(U\mathfrak{g}) \cdot f = 1$$

3. Holomorphicity at cusps correspond to moderate growth, as we will explain below.

Example. *Let us unravel the growth condition in the case $G = \mathrm{SL}_2$. Recall:*

$$\Gamma \setminus \mathrm{SL}_2(\mathbb{R}) \simeq \mathrm{SL}_2(\mathbb{Q}) \setminus \mathrm{SL}_2(\mathbb{A})/K_f$$

So to specify an automorphism we need to specify a function:

$$\phi : \Gamma \setminus \mathrm{SL}_2(\mathbb{R}) \to \mathbb{C}$$

Note that $\Gamma \setminus \mathrm{SL}_2(\mathbb{R})$ can be covered by:

$$\begin{pmatrix} 1 & x \\ & 1 \end{pmatrix} \begin{pmatrix} a & \\ & a^{-1} \end{pmatrix}$$

for x on some bounded interval and a^2 bounded below. In other words, ϕ becomes some function on a Siegel domain in $\mathbb{C}$. Moderate growth for ϕ translates to that $\phi(x+iy)$ has at most polynomial growth in $y \to \infty$ (for x fixed). For ϕ a modular form of weight k, one can express using the q-coordinates:

$$\phi(z) = a_0 + a_1 q + a_2 q^2 + \cdots$$

with $|q| = e^{-2\pi y}$. Note if ϕ contains negative powers of q then it will not be of polynomial growth: the is precisely the condition of holomorphicity at the cusps.

Now let $\mathcal{A}$ be the vector spaces of all automorphic forms for G. This is still a $(\mathfrak{g}, K_\infty)$-module. This is also a smooth $G(\mathbb{A}_f)$-module. Roughly speaking, this is a representation of $G(\mathbb{A})$. There is a notion of admissibility for $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$-mod: each irreducible representation of $K_\infty \times K_f$ appears with finite multipicities (this is independent of the choice of K_f). The space $\mathcal{A}$ will not be admissible, but it will be admissible after the following adjustment. Let's fix $J \subset Z(U\mathfrak{g})$ an ideal with finite codimension. Consider $\mathcal{A}(J) \subset \mathcal{A}$ be the annhilator of J. Then we have:

Theorem 3.5 (Harish-Chandra)**.** *$\mathcal{A}(J)$ is admissible $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$-module. In particular,*

$$\mathcal{A}(J, K_f) = \mathcal{A}(J)^{K_f}$$

is an admissible $(\mathfrak{g}, K_\infty)$-module.

Definition 3.6. An **automorphic representation** of $G(\mathbb{A})$ is an irreducible $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$-module that appears as a subquotient of $\mathcal{A}$. This will be a admissible module.

Given $J \subset Z(U\mathfrak{g})$ a finite codimensional ideal, σ an irreducible representation of K_∞, K_f some compact open in $G(\mathbb{A}_f)$, we can cut out:

$$\mathcal{A}(J, \sigma, K_f) \subset \mathcal{A}$$

Our previous computation for SL_2 tells us:

$$M_k(\Gamma) \simeq \mathcal{A}(J_{\frac{k(k-2)}{2}}, k, K_f)$$

More generally, we can change $J_{\frac{k(k-2)}{2}}$ to any other ideals, we get the more general notion of **Masse forms**. A very sepcial case for Harish-Chandra's theorem tells us:

$$\dim M_k(\Gamma) < \infty$$

To appreciate Harish-Chandra's theorem, let's try to prove the finiteness of modular forms by hand. Let's fix a weight $k = 2\ell$, we're seeking for Γ-invariant forms:

$$f(z)(dz)^\ell$$

Namely, every $f \in M_k(\Gamma)$ gives rise to a holomorphic section of $\omega^{\otimes \ell}$ on $\Gamma \setminus \mathbb{H}$, here $\omega = \mathcal{O}(-2)|_{\mathbb{H}}$, and fix $\mathbb{H} \hookrightarrow \mathbb{P}^1_{\mathbb{C}}$. By adding cusps, we get a compact Riemann surface $\overline{\Gamma \setminus \mathbb{H}}$. The holomorphicity at cusp implies that we should seek for $\overline{\sigma_f}$ holomorphic sections of $(\omega(\text{cusp}))^{\otimes \ell}$ on $\overline{\Gamma \setminus \mathbb{H}}$. We call $\omega(\text{cusp})$ the log-differentials, and $M_k(\Gamma)$ bijects with the above sections of holomorphic line bundles, in particular is finite dimensional.

In the case SL_3, we get:

$$\mathrm{SL}_3(\mathbb{Z}) \setminus \mathrm{SL}_3(\mathbb{R})/\mathrm{SO}_3$$

this is 5-dimensional, so there has no hope to be a complex manifold.

The generalization for $\mathbb{H}$ is the following. Let's consider $X = G(\mathbb{R})/K_\infty$, this is a symmetric space. For certain $G(\mathbb{R})$, X has a complex structure, in this case they're called **Hermitian symmetric domains**.

Example. Sp_{2n}, $K_\infty = U_n$. *Then* $\dim_{\mathbb{R}} X = n(n+1)$. *We can embed:*

$$X \hookrightarrow \mathrm{LagGr}(\mathbb{C}^{2n})$$

this is an open embedding. If $V = (\mathbb{R}^{2n}, \omega)$, $\mathrm{Sp}_{2n} = \mathrm{Sp}(V)$, *and* $V_{\mathbb{C}} = \mathbb{C}^{2n}$. *I have:*

$$G(\mathbb{R}) = \mathrm{Sp}(V) \subset \mathrm{Sp}(V_{\mathbb{C}})$$

acting on $\mathrm{LagGr}(V_{\mathbb{C}})$. *Now* $K_\infty = U(V, J, h)$. *Note that* h *is determined by* J, ω. *Since we start with a complex structure* J, *we can write:*

$$(V, J) = V_0 \otimes_{\mathbb{R}} \mathbb{C} \implies V_{\mathbb{C}} = V_{\mathbb{C}}^{1,0} \oplus V_{\mathbb{C}}^{0,1} \simeq V \oplus \overline{V}$$

The copy $V^{1,0}$ *is a Lagrangian, the stablizer under* $\mathrm{Sp}(V)$ *is precisely* $U(V, J, h)$.

For G simple, $G(\mathbb{R})/K_\infty$ is a Hermitian symmetric domain precisely when $Z(K_\infty) \supset S^1$. In the case $\mathrm{SL}_n(\mathbb{R})$, SO_n has finite center so it's not symmetric domain. In the case of $\mathrm{SU}(p, q)$, its maximal compact $S(U(p) \times U(q))$ will contain a copy of S^1. In this case

$$\mathrm{SU}(p, q)/S(U(p) \times U(q))$$

is open in $\mathrm{Gr}\binom{p+q}{p}$

3.3 Geometry of $G(\mathbb{Q}) \setminus G(\mathbb{A})$

Let's study the geometry of:

$$G(\mathbb{Q}) \setminus G(\mathbb{A})/K_f$$

Let's assume strong approximation holds so that

$$G(\mathbb{Q}) \setminus G(\mathbb{A})/K_f = \Gamma \setminus G(\mathbb{R})$$

Let's introduce the notion of Siegel domain. For simplicity let's work with split groups. Recall we have Iwasawa decomposition:

$$G(\mathbb{R}) = N(\mathbb{R})A(\mathbb{R})^\circ K_\infty$$

So that we have surjection:

$$\Gamma \cap N(\mathbb{R}) \setminus N(\mathbb{R}) \cdot A(\mathbb{R})^\circ \to \Gamma \setminus G(\mathbb{R})/K_\infty$$

In the case of $\mathrm{SL}_2(\mathbb{R})$, $\Gamma \cap N(\mathbb{R})$ just consist of translations. Now take a compact $\Omega_N \subset N(\mathbb{R})$ (essentially fundamental domain for $\Gamma \cap N(\mathbb{R}) \setminus N(\mathbb{R})$). We let $A_\varepsilon \subset A(\mathbb{R})^\circ$ equals:

$$\{a \in A(\mathbb{R})^\circ \mid \alpha_i(a) \geq \varepsilon\}$$

for each simple root α_i. We call the image of:

$$\Omega_N \cdot A_\varepsilon \to G(\mathbb{R})/K_\infty$$

a **Siegal domain**.

Fact: For G semisimple split, Γ given. There exists a Siegel domain $\Omega_N \cdot A_\varepsilon$ such that:

$$\Omega_N \cdot A_\varepsilon \to \Gamma \setminus G(\mathbb{R})/K_\infty$$

is surjective.

Fix a Haar measure dg on $G(\mathbb{A})$. So we have an induced dg on $G(\mathbb{Q}) \setminus G(\mathbb{A})$.

Theorem 3.7. *For G semisimple (split), then:*

$$\mathrm{vol}(G(\mathbb{Q}) \setminus G(\mathbb{A}), dg) < \infty$$

Proof. Choose K_f, it's enough to show

$$G(\mathbb{Q}) \setminus G(\mathbb{A})/K_f$$

has finite volume. So it suffices to show $\Gamma \setminus G(\mathbb{R})$ has finite volume. Now:

$$\Omega_N A_\varepsilon K_\infty \twoheadrightarrow \Gamma \setminus G(\mathbb{R})$$

Generally, if I have a unique decomposition:

$$H_1 \cdot H_2 \to G$$

with G unimodular, then:

$$d_\ell h_1 d_r h_2 = dg$$

Let's apply this to:

$$(NA) \cdot K \to G$$

we get:

$$d_\ell(na) \cdot dk = dg_\infty$$

The issue is that NA is not unimodular. Instead, let's consider:

$$B = AN$$

then

$$d_\ell b = da \cdot dn$$

$$d_\ell b = a^{-2\rho} dn \cdot da$$

the factor $a^{2\rho} = \det(\mathrm{Ad}(a^{-1}) \mid \mathrm{Lie}(N))$ To summarize $dg_\infty = a^{-2\rho} dn \cdot da \cdot dk$. So that:

$$\mathrm{vol}(\Omega_N A_\varepsilon K_\infty, dg_\infty) = \int_{\Omega_N} dn \cdot \int_{A_\varepsilon} a^{-2\rho} d^\times a \cdot \int_{K_\infty} dk$$

The middle term will becoem:

$$\int_\varepsilon^\infty a^{-1} \frac{da}{a} < \infty$$

where we interpret:

$$a = \begin{pmatrix} \sqrt{a} & \\ & \sqrt{a}^{-1} \end{pmatrix}$$

□

Remark. *In the case of* SL_2, $dn = dx$, $da = dy$, *and that:*

$$dg_\infty = \frac{dxdy}{y^2} \cdot dk$$

which is precisely the hyperbolic metric.

Theorem 3.8 (Langlands)**.** *We have:*

$$\mathrm{vol}(\mathrm{SL}_n(\mathbb{Z}) \setminus \mathrm{SL}_n(\mathbb{R})) = \zeta(2)\zeta(3)\cdots\zeta(n)$$

where we fix a top form on $\mathfrak{sl}_n(\mathbb{R})$ *such that* $\mathfrak{sl}_n(\mathbb{Z})$ *has covolume* 1.

3.4 Cusp Forms

Recall that:

$$\mathcal{A} \subset C^\infty(G(\mathbb{Q}) \setminus G(\mathbb{A}))$$

such that: $Z(U\mathfrak{g})$-finite, $K = K_f \times K_\infty$-finite, and satisifies the growth condition.

Let's recall the definition of The cuspidal modular form $f(z)$ of weight k and level Γ. Let's write: $\Gamma \cap \begin{pmatrix} 1 & * \\ & 1 \end{pmatrix} = \begin{pmatrix} 1 & n\mathbb{Z} \\ & 1 \end{pmatrix}$, so that $f(z+n)$ and therefore we can write:

$$f(z) = \sum_{m \geq 0} a_m q^m, \quad q = e^{2\pi i z/n}$$

A cusp form is requiring that $a_0 = 0$. Let's now define the general notion of cusp form.

Definition 3.9. An automorphic form $f \in \mathcal{A}$ is **cuspidal** if:

$$\int_{N_P(\mathbb{Q}) \setminus N_P(\mathbb{A})} f(ng)dn = 0$$

for all $g \in G(\mathbb{A})$ and all proper parabolic $P \subsetneq G$ (defined over $\mathbb{Q}$). Note the integral is well-defined since $N_P(\mathbb{Q}) \setminus N_P(\mathbb{A})$ is compact.

Remark. *It is enough to check for a finite number of parabolics. For* GL_n*, we only need to test for two block (maximal) parabolic subgroups. For* $P \supset Q$*, then* $N_P \subset N_Q$ *so that the vanishing of the integral in the case of* P *implies the vanishing of the integral in the case of* Q*.*

For $G = \mathrm{SL}_2$, we only need to check for $P = B$. Recall we have a map:

$$M_k(\Gamma) \to \mathcal{A}, \quad f \mapsto \phi_f$$

$$\phi_f(\begin{pmatrix} a & b \\ c & d \end{pmatrix}) = f(\frac{ai+b}{ci+d})(ci+d)^{-k}$$

Now note that:

$$N(\mathbb{Q}) \setminus N(\mathbb{A}) = \mathbb{Q} \setminus \mathbb{A}$$

Let's also assume $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, then we need to compute:

$$\int_{\mathbb{Q} \setminus \mathbb{A}} \phi_f(ng)dn = \int_0^1 f(z+x)dx$$

where we identified:

$$\mathbb{Q} \setminus \mathbb{A}/\widehat{\mathbb{Z}} = \mathbb{Z} \setminus \mathbb{R}, \quad z = g \cdot i$$

Convert to the q-coordinate, this converts to a countour integral along the q-disc:

$$\frac{1}{2\pi i} \oint f(q)\frac{dq}{q} = a_0 = \int_0^1 f(z+x)dx$$

Let G be semisimple. Let's denote $\mathcal{A}_{\text{cusp}} \subset \mathcal{A}$ be the subspace of cusp forms. Let's consider:

$$\mathcal{A} \subset L^2(G(\mathbb{Q}) \setminus G(\mathbb{A}))$$

by picking a Haar measure on $\mathbb{A}$. This is a dense subspace. We thus have:

$$\mathcal{A}_{\text{cusp}} \subset L^2_{\text{cusp}}(G(\mathbb{Q}) \setminus G(\mathbb{A})) \subset L^2(G(\mathbb{Q}) \setminus G(\mathbb{A}))$$

It turns out $\mathcal{A}_{\text{cusp}} \subset L^2_{\text{cusp}}(G(\mathbb{Q}) \setminus G(\mathbb{A}))$ is again dense.

Consider $G(\mathbb{A})$ acts on $L^2(G(\mathbb{Q}) \setminus G(\mathbb{A}))$ unitarily:

$$(g \cdot \phi)(x) = \phi(xg)$$

When we restricting to $L^2_{\text{cusp}}(G(\mathbb{Q}) \setminus G(\mathbb{A}))$, we have:

$$\widehat{\bigoplus}(\text{irreps of } G(\mathbb{A}))$$

with each irreducible piece appears with finite multiplicities (for GL_n, the multiplicity is at most one). The key point here is that we get a discrete family. Now we write:

$$L^2 = L^2_{\text{cusp}} \oplus (L^2_{\text{cusp}})^{\perp}$$

The second component also contains discrete part (irreducible sub). For example, constant function $1 \in L^2$. Actually $1 \in (L^2_{\text{cusp}})^{\perp}$. There will be a discrete spectrum interpolating from cusp forms and constant function 1. For SL_2, we have:

$$L^2_{\text{disc}} = L^2_{\text{cusp}} \oplus \mathbb{C} \cdot 1$$

Now we can write:

$$L^2 = L^2_{\text{disc}} \oplus^{\perp} L^2_{\text{cont}}$$

The continuous part is constructed by parabolic induction (Eisenstein series). Concretely, given $M \subset G$ Levi, and:

$$\pi \subset L^2_{\text{disc}}(M(\mathbb{Q}) \setminus M(\mathbb{A}), \text{fixed central character})$$

an irreducible representation of M. For example if $M = T$, we are considering:

$$T(\mathbb{Q}) \setminus T(\mathbb{A}) = (\mathbb{Q}^{\times} \setminus \mathbb{A}^{\times})^n \to S^1$$

This depends on a n continuous parameters. In general continuous parameter comes from the choice of the central character, and there will be $\dim Z(M)$ of them.

3.5 Automorphic Forms for Function Fields

Let $G/\mathbb{F}_q$ split semisimple group. Let $X/\mathbb{F}_q$ be a smooth projective curve, $F = \mathbb{F}_q(X)$. We still want to consider:

$$C^{\infty}(G(F) \setminus G(\mathbb{A}))$$

This has a $G(\mathbb{A})$-action by right translation R.

Definition 3.10. A function $f \in C^\infty(G(F) \setminus G(\mathbb{A}))$ is an automorphic form if:

$$\text{Span } R(G(\mathbb{A})) \cdot f$$

is an admissible $G(\mathbb{A})$-module. Let $\mathcal{A}$ be the subspace of automorphic forms.

Definition 3.11. We call $f \in \mathcal{A}$ is cuspidal if for all $P \subset G$ parabolic defined over $\mathbb{F}_q$ (any G defined over F comes from a group defined over $\mathbb{F}_q$), we have: for all $g \in G(\mathbb{A})$:

$$\int_{N_P(F)\setminus N_P(\mathbb{A})} f(ng)dn = 0$$

Let's now explain the geometric interpretation of automorphic forms. Fix $K \subset G(\mathbb{A})$ compact open, for example we can take:

$$K_0 = \prod_{x \in |X|} G(\mathcal{O}_x)$$

Inside of it, we have the congruent subgroups:

$$K_D = \prod_{x \in |X|} \ker(G(\mathcal{O}_x) \to G(\mathcal{O}_x/\mathfrak{m}_x^{N_x})), \quad D = \sum N_x \cdot x$$

Let's look at:

$$C(G(F) \setminus G(\mathbb{A})/K_0)$$

Recall for $\mathrm{GL}_n(\mathbb{Q}) \setminus \mathrm{GL}_n(\mathbb{A})/O_n \times \mathrm{GL}_n(\widehat{\mathbb{Z}})$ can be interpreted as the groupoid of metrized lattices of rank n (equiv. a vector bundle over Spec $\mathbb{Z}$). Similarly, we have:

$$\mathrm{GL}_n(F) \setminus \mathrm{GL}_n(\mathbb{A})/K_0$$

is canonically identified with vector bundles of rank n over X. Just as in the homework, we can identify:

$$\prod{}' \mathrm{GL}_n(F_x)/\mathrm{GL}_n(\mathcal{O}_x) = \mathrm{GL}_n(\mathbb{A})/K_0 \simeq \{(\mathcal{E}, \tau) \mid \mathcal{E} \text{ v.b. rk } n/X, \tau : \mathcal{E}|_{\text{Spec } F} \simeq F^n\}$$

and we will be done we can quotient by $\mathrm{GL}_n(F)$ on both sides. Now LHS is just a collection of $\mathcal{O}_x$-lattices in F_x^n such that almost all lattices are standard; so we are sending:

$$(\mathcal{E}, \tau) \mapsto (\Lambda_x \subset F_x^n)_{x \in |X|}$$

for $\Lambda_x = \mathcal{E}|_{\text{Spec } \mathcal{O}_x}$, $F_x^n = F^n|_{\text{Spec } F_x}$, and:

$$\tau|_{\text{Spec } F_x} \colon \mathcal{E}|_{\text{Spec } F_x} \simeq F^n|_{\text{Spec } F_x}$$

Note most Λ_x will be standard for almost all place since τ extends to a trivialization on some open subset $U \subset X$.

In general, the double quotient:

$$\mathrm{GL}_n(F) \setminus \mathrm{GL}_n(\mathbb{A})/K_D$$

can be interpreted as:

$$\{(\mathcal{E}, \tau_D) \mid \mathcal{E} \text{ v.b. rk } n/X, \tau_D : \mathcal{E}|_{\text{Spec } \mathcal{O}_D} \simeq \mathcal{O}_D^{\oplus n}\}$$

For classical groups G, we can give similar interpretations. For example, for $G = \mathrm{Sp}_{2n}$, we are replacing vector bundle with a symplectic vector bundle (a vector bundle with a symplectic pairing). Similar for orthogonal groups.

Let's now discuss the geometric interpretation of cusp forms. For $G = \mathrm{GL}_2$, we need to give a geometric meaning of the integral:

$$\int_{N(F)\backslash N(\mathbb{A})} f(ng)dn$$

Consider:

$$B(F) \setminus B(\mathbb{A})/K_0 \cap B(\mathbb{A}) \to G(F) \setminus G(\mathbb{A})/K_0$$

Concretely, LHS is a pair:

$$\{(\mathcal{L} \subset \mathcal{E}) \mid \text{rk } \mathcal{E} = 2, \mathcal{L} \in \mathrm{Pic}(X), \mathcal{E}/\mathcal{L} \in \mathrm{Pic}(X)\}$$

and the above map is just forgetting $\mathcal{L}$. Now we have:

$$N(F) \setminus N(\mathbb{A})/K_0 \cap N(\mathbb{A}) \to B(F) \setminus B(\mathbb{A})/K_0 \cap B(\mathbb{A})$$

The data for LHS is the groupoid of short exact sequences:

$$0 \to \mathcal{O}_X \to \mathcal{E} \to \mathcal{O}_X \to 0$$

Therefore:

$$\int_{N(F)\backslash N(\mathbb{A})} f(n)dn = \sum_{0\to\mathcal{O}_X\to\mathcal{E}\to\mathcal{O}_X\to 0} f(\mathcal{E})$$

Recall: $f : G(F) \setminus G(\mathbb{A})/K_0 \to \mathbb{C}$, so f will take in a vector bundle and spit out a complex number.

Let's discuss the meaning of

$$\int_{N(F)\backslash N(\mathbb{A})} f(ng)dn$$

in general. At each place x, we have Iwasawa decomposition:

$$N(F_x)T(F_x)K_x = G(F_x)$$

and gather all places:

$$N(\mathbb{A})T(\mathbb{A})K_0 = G(\mathbb{A})$$

We may assume $g \in T(\mathbb{A})$. Therefore, to compute the integral, we only need to sum over:

$$[g] \in T(F) \setminus T(\mathbb{A})/K_0 \cap T(\mathbb{A})$$

In the case of GL_2, we know:

$$T(F) \setminus T(\mathbb{A})/K_0 \cap T(\mathbb{A}) = \mathrm{Pic}(X)^2$$

So $[g]$ is the data of two vector bundles $\mathcal{L}_1, \mathcal{L}_2$. Hence:

$$\int_{N(F)\backslash N(\mathbb{A})} f(ng)dn = \sum_{0\to\mathcal{L}_1\to\mathcal{E}\to\mathcal{L}_2\to 0} f(\mathcal{E})$$

This is a finite sum since $\mathrm{Ext}^1(\mathcal{L}_2, \mathcal{L}_1)$ is finite. And f is cuspidal if the above sum is 0 for all $\mathcal{L}_1, \mathcal{L}_2$. More generally, for GL_n, f is cuspidal if:

$$\sum_{0\to\mathcal{E}_1\to\mathcal{E}\to\mathcal{E}_2\to 0} f(\mathcal{E}) = 0$$

where $\mathcal{E}_1, \mathcal{E}_2$ are vector bundles of rank $m, n-m$.

Theorem 3.12. *Let G be semisimple, $K \subset G(\mathbb{A})$ compact open. Then*

$$C_{\mathrm{cusp}}(G(F) \setminus G(\mathbb{A})/K)$$

is finite dimensional. Moreover, there is a finite subset $U_K \subset G(F) \setminus G(\mathbb{A})/K$ such that all cusp forms $f \in \mathcal{A}_{\mathrm{cusp}}(K)$ vanish outside U_K.

Corollary 3.13. *Let G be semisimple. Then any cuspidal form for G has compact support $\subset G(F) \setminus G(\mathbb{A})$.*

Over number fields, we have a similar result.

Fact: Any $f \in \mathcal{A}_{\mathrm{cusp}}$ has rapid decay at infinity. This is a strengthening of the moderate growth condition. For example, in the case of SL_2, the rapid decay condition is saying that $f(x+iy) \in o(y^{-N})$ for all N.

Indeed, for holomorphic cusp forms of weight k and $\Gamma \supset \begin{pmatrix} 1 & 1 \\ & 1 \end{pmatrix}$, we have expansion:

$$f(z) = a_1 e^{2\pi i z} + a_2 e^{4\pi i z} + \cdots$$

has exponential decay.

Let's discuss the notion of automorphic forms for reductive groups (e.g. $\mathrm{GL}_n(\mathbb{A})$). Consider:

$$\mathrm{GL}_n(F) \setminus \mathrm{GL}_n(\mathbb{A}) \xrightarrow{\det} F^\times \setminus \mathbb{A}^\times \xrightarrow{|\cdot|} \mathbb{R}^{>0}$$

Note now $\mathrm{GL}_n(F) \setminus \mathrm{GL}_n(\mathbb{A})$ has infinite volume. Let's fix $\omega : F^\times \setminus \mathbb{A}^\times \to \mathbb{C}^\times$ continuous. Now we identify:

$$F^\times \setminus \mathbb{A}^\times \simeq Z(F) \setminus Z(\mathbb{A})$$

where Z is the center of GL_n.We denote:

$$\mathcal{A}(\omega) \subset C^\infty(\mathrm{GL}_n(F) \setminus \mathrm{GL}_n(\mathbb{A}))^{(Z(F)\setminus Z(\mathbb{A}),\omega)}$$

this is saying that we fix the action of the center. Now observe that:

$$Z(F) \setminus Z(\mathbb{A})/K_Z \simeq \mathrm{Pic}(X)$$

acts on $G(F)\backslash G(\mathbb{A})/K_0$, or equivalently on $\mathrm{Vect}_n(X)$-the groupoid of rank n vector bundles on X. This action is just given by tensor product. So $f \in \mathcal{A}(\omega)$ is a function

$$\mathrm{Vect}_n(X) \to \mathbb{C}$$

such that $f(\mathcal{E}\otimes\mathcal{L}) = \omega(\mathcal{L})f(\mathcal{E})$, for all $\mathcal{L}\in \mathrm{Pic}(X), \mathcal{E}\in \mathrm{Vect}_n(X)$. As before we need to insist that $R(G(\mathbb{A}))\cdot f$ is an admissible $G(\mathbb{A})$-module. Let's decompose according to degrees:

$$\mathrm{Vect}_n(X) = \coprod_{d\in\mathbb{Z}} \mathrm{Vect}_n^d(X)$$

Now tensor with a line bundle of degree one increases the degree by n. So we only need to determine the value on:

$$\mathrm{Vect}_n^d(X), \quad 0\le d\le n-1$$

Proof. (of theorem 3.12, sketched). Let's assume $G=\mathrm{GL}_n, K=K_0$. Let $f\in\mathcal{A}_{\mathrm{cusp}}(\omega, K_0)$. Our goal is to show tht there exists:

$$U\subset \coprod_{i=0}^{n-1}\mathrm{Vect}_n^i(X)$$

finite such that f vanishes outside $\mathrm{Pic}(X)\cdot U\subset \mathrm{Vect}_n(X)$, where U is independent of f (dependent on K). Recall the geometric meaning of cuspidality is that for every maximal parabolic P, the function

$$f:\mathrm{Vect}_n(X)\to\mathbb{C}$$

need to satisfy $\phi(g)=\int_{N_P(F)\backslash N_P(\mathbb{A})} f(ng)dn = 0$. Now we have generalized Iwasawa decomposition (since $P=MN$):

$$N(F_x)M(F_x)K_x = G(F_x) \implies N(\mathbb{A})M(\mathbb{A})K_0 = G(\mathbb{A})$$

Hence $\phi(g)=0 \iff \phi|_{M(\mathbb{A})}=0$. In fact $\phi|_{M(\mathbb{A})}$ descends to a function on:

$$M(F)\setminus M(\mathbb{A})/K_{M,0}, \quad K_{M,0}=\prod M(\mathcal{O}_X)$$

The left invariance under $M(F)$ is from the fact that M normalizes N. But now:

$$\phi|_{M(\mathbb{A})}\colon \mathrm{Vect}_m(X)\times\mathrm{Vect}_{n-m}(X)\to\mathbb{C}$$

$$(\mathcal{E},\mathcal{E}')\mapsto \sum_{0\to\mathcal{E}'\to\mathcal{E}\to\mathcal{E}''\to 0}\frac{1}{|\mathrm{Aut}(\mathcal{E}'\to\mathcal{E}\to\mathcal{E}'')|}f(\mathcal{E})$$

where we view all extension $0\to\mathcal{E}'\to\mathcal{E}\to\mathcal{E}''\to 0$ as a groupoid. Now it's easy to see the automorphism group for every $\mathcal{E}$ is parametrized by $\mathrm{Hom}(\mathcal{E}'',\mathcal{E}')$, so the weighted coefficient is a constant (can effectively drop this term).

Now to proceed, we need to develop some reduction theory for $\mathrm{Vect}_n(X)$. Recall in homework, we identified $\mathrm{GL}_n(\mathbb{Q})\setminus\mathrm{GL}_n(\mathbb{A}_{\mathbb{Q}})/\mathrm{GL}_n(\widehat{\mathbb{Z}})\cdot O_n$ with metrized rank n $\mathbb{Z}$-lattices. To understand these objects, we used the shortest vector algorithem. In the case of function

fields, we need an anlogy of "shortest vector". Note that the shortest vector is determined by the line it spans, or equivalently a rank one sublattice. So we need to look at a sub-line-bundle:

$$\mathcal{L} \subset \mathcal{E}$$

The larger the degree is, the shorter the vector is. Here is a formalism due to Harder-Narosimhan:

1. We can define the *slope* of a vector bundle:

$$\mu(\mathcal{E}) \coloneqq \frac{\deg(\mathcal{E})}{\mathrm{rk}(\mathcal{E})} \in \mathbb{Q}$$

2. We call $\mathcal{E}$ *semistable*, if for all subbundle $\mathcal{F} \subset \mathcal{E}$, we have $\mu(\mathcal{F}) \leq \mu(\mathcal{E})$.

   Example. *If $\mathcal{E} = \mathcal{L}_1 \oplus \mathcal{L}_2$ with $\deg \mathcal{L}_1 > \deg \mathcal{L}_2$, we have a subbundle $\mathcal{L}_1$ with larger slope. So that $\mathcal{E}$ is not semistable. On the other hand, if $\deg \mathcal{L}_1 = \deg \mathcal{L}_2$, one sees that $\mathcal{E}$ is semistable.*

3. Every $\mathcal{E} \in \mathrm{Vect}_n(X)$ has a unique filtration by subbundles:

$$0 \subsetneq \mathcal{F}_1 \subsetneq \cdots \subsetneq \mathcal{F}_r = \mathcal{E}$$

   such that $\mathcal{F}_i/\mathcal{F}_{i-1}$ is semistable, and $\mu(\mathcal{F}_1/\mathcal{F}_0) > \mu(\mathcal{F}_2/\mathcal{F}_1) > \cdots$.

In light of the previous formalism, we can decompose:

$$\mathrm{Vect}_n^d(X) = \prod_{\nu \in \mathrm{HN}_d} \mathrm{Vect}_n^\nu(X)$$

where we record all slopes and ranks of each quotient in the unique filtration. One can arrange these data using the Harder Narosimhan polygon: it is a convex polygon with vertices:

$$(0,0), (\mathrm{rk}\ \mathcal{F}_1, \deg \mathcal{F}_1), (\mathrm{rk}\ \mathcal{F}_2, \deg \mathcal{F}_2), \cdots, (n,d)$$

Let's try to impose a bound on the HN polygons that can appear. Once we do this, combined with $\mathrm{Vect}_n^\nu(X)$ is finite, we're therefore done. More precisely, there must be $\nu_0 \in \mathrm{HN}_d$ such that all cusp forms on $\mathrm{Vect}_n^d(X)$ must vanish on $\mathrm{Vect}_n^\nu(X)$ where $\nu \not\leq \nu_0$.

Let's illustrate the idea in the case of GL_2 and $d = 0$. So I need to give a vertex $(1, e)$. Now assume I have $(1, e')$ with $e' > e$, then vector bundles with HN shape $(1, e')$ is given by $\mathcal{E}$ fitting into:

$$0 \to \mathcal{E}' \to \mathcal{E} \to \mathcal{E}'' \to 0$$

for $\deg \mathcal{E}' = e', \deg \mathcal{E}'' = -e'$. Now fix $\mathcal{E}' \in \mathrm{Pic}^{e'}(X), \mathcal{E}'' \in \mathrm{Pic}^{-e'} \in \mathrm{Pic}^{-e'}(X)$, recall we need:

$$\sum_{0 \to \mathcal{E}' \to \mathcal{E} \to \mathcal{E}'' \to 0} f(\mathcal{E}) = 0$$

Note for $e' \gg 0$, the extension must split. Therefore we must have:

$$f(\mathcal{E}' \oplus \mathcal{E}'') = 0$$

□

Now let F be a number field, G a semisimple group. Recall we have:

$$L^2_{\text{cusp}} \subset L^2(G(F) \setminus G(\mathbb{A}))$$

Fact: $L^2 = \widehat{\bigoplus}_{\pi \in \text{Irr}(G(\mathbb{A}))\pi}$, for each piece appears with finite multiplicity. Let's briefly discuss the proof of this fact in the case when $G(F) \setminus G(\mathbb{A})$ is compact (equiv. $\Gamma \setminus G(\mathbb{R})$ is compact). In this case any L^2 function is automatically cuspidal. Now I have $G(\mathbb{A})$ acting on $L^2(G(F) \setminus G(\mathbb{A}))$. Let's choose a sequence $\{f_n\} \subset C_c^\infty(G(\mathbb{A}), \mathbb{R})$ such that:

1. $f_n \to \delta_1$
2. $f_n(g) = f_n(g^{-1})$

Now $R(f_n)$ is self-adjoint, Hilbert-Schmidt (because $R(f_n)$ is represented by an integral kernel on $\Gamma \setminus G(\mathbb{R}) \times \Gamma \setminus G(\mathbb{R})$). In particular $R(f_n)$ are compact operators, so that the spectrum of $R(f_n)$ is discrete except at 0.

3.6 Tensor Product Theorem

Let $\mathcal{A}$ be the space of automorphic forms. We have a smooth action of $G(\mathbb{A}_f)$ on $\mathcal{A}$. Let's introduce:

$$\mathcal{H}_f = \bigotimes_{v \nmid \infty}{}' \mathcal{H}_v$$

where $\mathcal{H}_v = C_c^\infty(G(F_v))dg$. We have distinguished idempotent $e_v = e_{G(\mathcal{O}_v)} \in \mathcal{H}_v$. The restricted product means fro all but finitely many v, the v-vector must be e_v. There is naturally a $\mathbb{C}$-algebra structure on $\mathcal{H}_f$ (but again it is not unital). Now $\mathcal{H}_f$ has a family of idempotents taking form: for $K \subset G(\mathbb{A}_f)$ compact open, then we take

$$e_K = \bigotimes_v e_{K_v} \in \mathcal{H}_f$$

In fact, we can view:

$$\mathcal{H}_f = C_c^\infty(G(\mathbb{A}_f))dg$$

and we can construct idempotent e_K as in the case of p-adic groups. Therefore $(\mathcal{H}_f, \{e_K\})$ behaves very similar to local Hecke algebra $(\mathcal{H}_v, \{e_{K_v}\})$. For example, I have an equivalence of categories:

$$\text{Rep}^{\text{sm}}(G(\mathbb{A}_f)) \simeq \mathcal{H}_f\text{-mod}_{\text{sm}}$$

preserving the underling vector space.

Let's now discuss the infinite places. Assume $F = \mathbb{Q}$. Then $\mathcal{A}$ is a $(\mathfrak{g} = \text{Lie}(G(\mathbb{R})), K_\infty)$-module. In homework, we constructed a non-unital $\mathbb{C}$-algebra $\mathcal{H}_{\mathfrak{g},K_\infty}$, with the property that we have an equivalence:

$$(\mathfrak{g}, K_\infty)\text{-mod} \to \mathcal{H}_{\mathfrak{g},K_\infty}\text{-mod}_{\text{sm}}$$

where: $\mathcal{H}_{\mathfrak{g},K_\infty} = \text{Alg}(K_\infty) \otimes_{U(\mathfrak{k}_\mathbb{C})} U(\mathfrak{g}_\mathbb{C})$, for $\mathfrak{k} = \text{Lie}(K_\infty)$. The algebra:

$$\text{Alg}(K_\infty) = \bigoplus_{V \in \text{Irr}(K_\infty)} \text{End}_\mathbb{C} V$$

is non-unital, but again we have a family of idempotents:

$$e_\Sigma = \sum_{\sigma \in \Sigma} 1_\sigma$$

for Σ a finite set of irreducible representations of K_∞.

Example. *If $K_\infty = S^1$, then $\mathrm{Alg}(K_\infty) \simeq \mathbb{C}[z, z^{-1}] = \bigoplus_{n\in\mathbb{Z}} \mathbb{C}z^n$. This is an identification of vector spaces. Under this identification, we have z^n are actually idempotents.*

Let's discuss the multiplication on $\mathcal{H}_{\mathfrak{g},K_\infty}$. Obviously we should impose:

$$\phi \cdot X = \phi \otimes X$$

Suppose we allow δ functions, then we must have:

$$X \cdot \delta_y = \delta_y \cdot \mathrm{Ad}(y^{-1})X = \delta_y \otimes \mathrm{Ad}(y^{-1})X$$

More conceptually, $\mathcal{H}_{\mathfrak{g},K_\infty}$ is the distribution on $G(\mathbb{R})$ supported on K_∞ and finite under left/right multiplications by K . In other words, we can evaluate functions on K_∞, but can take derivative on any direction on $G(\mathbb{R})$.

In this setting, we have a family of idempotents $e_\Sigma \otimes 1 \in \mathcal{H}_{\mathfrak{g},K_\infty}$. Smooth module means it must be fixed by one of $e_\Sigma \otimes 1$.

Now we can define the global Hecke algebra:

$$\mathcal{H} = \mathcal{H}_{\mathfrak{g},\infty} \otimes \mathcal{H}_f = \otimes' \mathcal{H}_v$$

and $\mathcal{A}$ will have a $\mathcal{H}$ action. Therefore, an automorphic representation π is an irreducible admissible subquotientes of $\mathcal{A}$ on $\mathcal{H}$-mod. Therefore, we need to study irreducible admissible $\mathcal{H}$-mod.

Theorem 3.14 (Tensor Product Theorem)**.** *1. Let M_v be a simple $\mathcal{H}_v$-module, such that for almost all v, M_v is spherical ($M_v^{G(\mathcal{O}_v)} \neq 0$, in fact one dimensional). Choose a basis $m_v \in M_v^{G(\mathcal{O}_v)}$. We can form:*

$$M = \bigotimes{}' M_v$$

meaning that almost all factors equal m_v. This is an irreducible admissible $\mathcal{H}$-module.

2. For an irreducible admissible $\mathcal{H}$-module M, then M takes the above form.

Sketched. For the first part, we need to see for each $e \in \mathcal{H}$ idempotent, we must see eM is finite dimensional. Now we can write:

$$e_S \otimes (\otimes_{v\notin S} e_v)$$

for some finite collection of places S. Then:

$$eM = e_S M_S \otimes (\otimes_{v\notin S} m_v)$$

has finite dimension. To show simplicity, we first look at the toy case. Let $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$, and $M_i \in \mathcal{H}_i$-mod are admissible irreducible (admissible means that $M = \bigcup eM$, and each eM is finite dimensional). We will show $M = M_1 \otimes M_2$ is admissible irreducible $\mathcal{H}$-mod. In fact, all admissible irreducible $\mathcal{H}$-module takes this form. We will use the following:

Lemma. *Let $(\mathcal{H}, \{e\})$ be a non-unital algebra with a family of idempotents. Then M is an irreducible admissible $\mathcal{H}$-*mod *or 0, is equivalent to: for any e, eM is an irreducible $e\mathcal{H}e$-module of finite dimension or 0.*

Now if $e_i \in \mathcal{H}_i$ is idempotent, $M = M_1 \otimes M_2$, then:

$$(e_1 \otimes e_2) = e_1 M \otimes e_2 M$$

This is an irreducible module for $e_1\mathcal{H}_1 e_2 \otimes e_2 \mathcal{H} e_2$. This justifies M is irreducible. Conversely, if M is an irreducible admissible $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$-module, then we can choose e_1, e_2 such that:

$$(e_1 \otimes e_2)M$$

is an irreducible $e_1\mathcal{H}e_1 \otimes e_2\mathcal{H}_2 e_2$-module. We again use the structure theory of simple modules of tensor product of finite dimensional algebras, and deduce that:

$$(e_1 \otimes e_2)M \simeq N_1(e_1) \otimes N_2(e_2)$$

We can comaptibly choose these isomorphisms so that it makes sense to talk about:

$$\bigcup N_1(e_1^i) \otimes N_2(e_2^i) = N_1 \otimes N_2$$

for e_1^i, e_2^i approximating the identity, and $N_\ell = \bigcup N_\ell(e_{\ell^i})$. This is because the obstruction of the lifting process is given by H^1 of the filtered sysmtem (which is contractible). Now there exists a unique map:

$$N_1 \otimes N_2 \to M$$

compatible with the tensor product decompoisition of $(e_1^{(i)} \otimes e_2^{(j)})M$. We therefore proved out toy case.

In general, given M irreducible admissible $\mathcal{H}$-module, then eM is an irreducible finite-dimensional $e\mathcal{H}e$-module. We may assume $e = \otimes e_v^{(0)}$, where $e_v^{(0)} = e_v$ for $v \notin S$. Therefore:

$$eM = M_S \otimes (\otimes_{v \notin S} m_v)$$

is a module of: $e_S \mathcal{H}_S e_S \otimes (\otimes' e_v \mathcal{H}_v e_v)$. The second bit is restricted tensor product of spherical Hecke algebras, so it is actually a commutative unital $\mathbb{C}$-algebra. We can deal with the first factor $M_S(e_S)$ by inductively applying the toy case argument. □

3.7 Hecke Operators

Fix a level $K \subset G(\mathbb{A}_f)$, and consider $\mathcal{A}_G(K) \subset C^\infty(G(F) \setminus G(\mathbb{A})/K)$. Given any set X with a right $G(\mathbb{A}_f)$-action (e.g. $X = G(F) \setminus G(\mathbb{A})$), we obtain $C^\infty(X)$ as a left $\mathcal{H}_f = C_c^\infty(G(\mathbb{A}_f))dg$-module via:

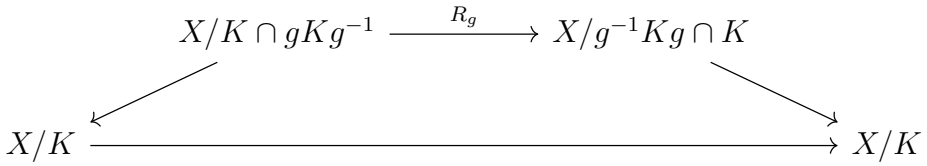
$$(h * f)(x) = \int_{G(\mathbb{A}_f)} f(xg)h(g)$$

After imposing K-invariance, we can realize $C^\infty(X/K)$ as a $\mathcal{H}_{f,K} = C_c^\infty(K \setminus G(\mathbb{A}_f)/K)$-module, given by the same formula:

$$(h * f)(x) = \sum_{g \in G(\mathbb{A}_f)/K} f(xg)h(g)$$

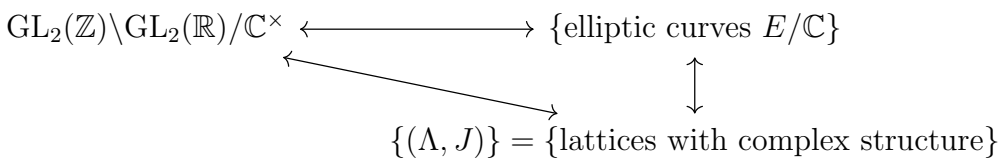
This gives a $\mathcal{H}_{f,K}$-module structure on $\mathcal{A}_G(K)$.

Recall that $\mathcal{H}_{f,K}$ has a basis: $\mathbf{1}_{KgK}$. We have a diagram:



The module structure is given by: $\mathbf{1}_{KgK} * f$ equals pull-pull-push in the above diagram. Here pushforward is just summing over the fibre.

Example. *Let's make the Hecke operator in the case of modular forms explicit. We view modular forms as functions on the space of ellitpic curves. Given $\tau \in \mathbb{H}$, one can form $E_\tau = \mathbb{C}/\mathbb{Z}\tau \oplus \mathbb{Z}$. We claim that we have bijections:*



The vertical map is given by:

$$E \mapsto (H_1(E, \mathbb{Z}), H_1(E, \mathbb{R}) \simeq \mathrm{Lie}(E/\mathbb{C}))$$

The converse map is given by:

$$(\Lambda, J) \mapsto (\Lambda \otimes \mathbb{R}, J)/\Lambda$$

Note that $\mathrm{GL}_2(\mathbb{C})/\mathbb{C}^\times$ is identified with the set of complex structures on $\mathbb{R}^2$. They have two connected components, corresponding to two orientations on $\mathbb{R}^2$. But now:

$$\mathrm{GL}_2(\mathbb{R})/\mathbb{C}^\times \simeq \mathbb{C} \setminus \mathbb{R} = \mathbb{H}^+ \cup \mathbb{H}^-$$

It follows that:

$$\mathrm{SL}_2(\mathbb{Z}) \setminus \mathrm{SL}_2(\mathbb{R})/\mathrm{SO}_2 \simeq \mathrm{GL}_2(\mathbb{Z}) \setminus \mathrm{GL}_2(\mathbb{R})/\mathbb{C}^\times$$

Forget about the $\mathbb{C}^\times$ factor, we have:

$$\mathrm{GL}_2(\mathbb{Z}) \setminus \mathrm{GL}_2(\mathbb{R}) = \{(\Lambda, J, v) \mid 0 \neq v \in \Lambda \otimes_{\mathbb{Z}} \mathbb{R}\}$$

which is equivalent as a groupoid to:

$$\{(E, v) \mid E : \text{elliptic curve}/\mathbb{C}, 0 \neq v \in \mathrm{Lie}(E/\mathbb{C})\}$$

Therefore we can interpret a holomorphic modular form as a function on elliptic curve with a tangent direction at origin:

$$f : \{(E, v) \mid 0 \neq v \in E\} \to \mathbb{C}$$

with condition

$$f(E, cv) = c^k f(E, v)$$

and certain holomorphicity and growth rate condition. Note k must be even, since $(E, v) \simeq (E, -v)$ by inversion on elliptic curves.

Let's now begin describing the Hecke operator. Let $K = \mathrm{GL}_2(\hat{\mathbb{Z}}) \subset \mathrm{GL}_2(\mathbb{A}_f)$. Let's choose:

$$T_n = \mathbf{1}_{\{g \in \mathrm{Mat}_2(\hat{\mathbb{Z}}) | (\det(g)) = n\hat{\mathbb{Z}}\}}$$

More explicitly:

$$T_p = \mathbf{1}_{\mathrm{GL}_2(\widehat{\mathbb{Z}}^{(p')})} \otimes \mathbf{1}_{\mathrm{GL}_2(\mathbb{Z}_p)\begin{pmatrix} p & \\ & 1 \end{pmatrix}\mathrm{GL}_2(\mathbb{Z}_p)}$$

$$T_{p^e} = \mathbf{1}_{\mathrm{GL}_2(\widehat{\mathbb{Z}}^{(p')})} \otimes \sum_{0 \leq a \leq b, a+b=e} \mathbf{1}_{\mathrm{GL}_2(\mathbb{Z}_p)\begin{pmatrix} p^a & \\ & p^b \end{pmatrix}\mathrm{GL}_2(\mathbb{Z}_p)}$$

$$T_n = (\bigotimes_{p|n} T_{p^{v_p(n)}}) \otimes \mathbf{1}_{\mathrm{GL}_2(\hat{\mathbb{Z}}^{(n')})}$$

These almost generate $\mathcal{H}_{f,K}$. The only things we're missing are:

$$\delta_m = \mathbf{1}_{m \cdot \mathrm{GL}_2(\hat{\mathbb{Z}})}, \quad m \in \mathbb{Q}_{>0}$$

In other words:

$$\mathcal{H}_{f,K} = \mathbb{C}[\delta_m, T_n \mid m \in \mathbb{Q}_{>0}, T_n]$$

Let's see how these operators act on modular forms $M_k(\mathrm{GL}_2(\mathbb{Z}))$. The action of δ_m is easy:

$$\delta_m * f = m^k f$$

The more interesting part is the action of T_n. One should put a Hermitian inner product (the hyperbolic metric) on the vector space of modular forms, and T_n will act by self-adjoint operators. Therefore, $M_k(\mathrm{GL}_2(\mathbb{Z}))$ has an eigen-decomposition. Classically, one can write:

$$M_k(\mathrm{GL}_2(\mathbb{Z})) = \mathbb{C} \cdot E_k \oplus S_k(\mathrm{GL}_2(\mathbb{Z}))$$

where the first term is Eisenstein series (recording the leading coefficient in the q-expansion, hence one dimensional), and the second term is cusp forms. This is in fact an eigenspace decomposition, and T_n acts by $1 + n^{k-1}$ on Eisenstein series. For the cusp forms, we can further decompose:

$$S_k(\mathrm{GL}_2(\mathbb{Z})) = \bigoplus(\text{1-dim eigenspaces})$$

This is a special case of the multiplicity one theorem. For $k = 2$, we have a canonical isomorphism:

$$S_2(\Gamma) \oplus \overline{S_2(\Gamma)} \simeq H^1(\overline{\Gamma \setminus \mathbb{H}}, \mathbb{C})$$

On this $\mathbb{C}[T_n]$ has two-dimensional eigenspaces. An **eigenform** *is one of the eigenvector in the above multiplicity one decomposition. Fix an eigenform f, let's record:*

$$T_n * f = a_n f, \quad (a_1 = 1)$$

Fact: *The q-expansion of f is up to a scalar:*

$$\sum_{n \neq 1} a_n q^n$$

Let's interpret the Hecke operator geometrically.

$$(T_n * f)(E, v) = \sum_{E' \to E, \deg n \text{ isogeny}} f(E', v)$$

How to enumerate all E'? Given $E = \mathbb{C}/\Lambda$, $E' \to E$ bijects with: $\mathbb{C}/\Lambda' \to \mathbb{C}/\Lambda$ such that $\Lambda' \subset \Lambda$ has index n. By fixing an isomorphism $\Lambda \simeq \mathbb{Z}^2$, then $\Lambda' \hookrightarrow \Lambda$ bijects with:

$$\{g \in \mathrm{Mat}_2(\mathbb{Z}) \mid \det(g) = \pm n\}/\mathrm{GL}_2(\mathbb{Z})$$

Finally, given an eigenform f, we can define its L-function:

$$L(s, f) = \sum_{n \geq 1} \frac{a_n}{n^s}$$

Fact:

$$\int_0^\infty f(iy) y^s \frac{dy}{y} = (2\pi)^{-s} \Gamma(s) L(s, f)$$

This is an entire function in s.

Example. *Let's now discuss Hecke operators in function field case. Given $X/\mathbb{F}_q$, $K = \mathrm{GL}_2(\mathcal{O})$. Now an automorphic form is a function:*

$$f : \{\text{v.b. of rank 2 on } X\} \to \mathbb{C}$$

Now $T_D \in \mathcal{H}_K$ for $D \subset X$ an effective divisor. For $D = x$ a single point, we get:

$$T_x = \mathbf{1}_{\mathrm{GL}_2(\mathcal{O}_x)} \begin{pmatrix} \varpi_x & \\ & 1 \end{pmatrix} \mathrm{GL}_2(\mathcal{O}_x)$$

In addition I have δ_D for $D \in \mathrm{Div}(X)$ (not necessarily effective, corresponding to allowing rational numbers in the case of number field). We have formula:

$$(\delta_{D'} * f)(\mathcal{E}) = f(\mathcal{E}(D'))$$

$$(T_D * f)(\mathcal{E}) = \sum_{\mathcal{E} \subset \mathcal{E}', \mathrm{div}(\mathcal{E}'/\mathcal{E}) = D} f(\mathcal{E}')$$

4 Whittaker Models and L-functions

Suppose for now $G = \mathrm{GL}_2$. Given an automorphic representation π, and $\phi \in \pi$ and view ϕ as a function on $\mathrm{GL}_2(F) \setminus \mathrm{GL}_2(\mathbb{A})$. Let's consider:

$$s \mapsto \int_{F^\times \setminus \mathbb{A}^\times} \phi \begin{pmatrix} a & \\ & 1 \end{pmatrix} |a|^{s-\frac{1}{2}} d^\times a$$

For a nice choice of ϕ, we will get the L-function $L(s, \pi)$. A very nice case is when F is a function field, π is unramified, and choose ϕ to lie in the spherical vector at all places.

We expect:

1. Holomorphic modular forms should be certain L-functions.
2. For pure tensor automorphic representation $\pi = \otimes' \pi_v$, we should have a Euler product decomposition:
$$L(s, \pi) = \prod_v L_v(s, \pi_v)$$

This factorization needs tools from Whittaker functional on π. Another important application of Whittaker functional is the **multiplicity one theorem** as we stated below.

Let $G = \mathrm{GL}_n$, F a global field. Let $\mathcal{A}_{G,\mathrm{cusp}}$ be the space of cuspidal automorphic forms. Our first goal is to prove:

Theorem 4.1 (Multiplicity One)**.** *Every abstract irreducible admissible representation of $G(\mathbb{A})$ appears at most once in $\mathcal{A}_{G,\mathrm{cusp}}$.*

Theorem 4.2 (Strong Multiplicity One)**.** *Suppose π, π' are both cuspidal representations of $G(\mathbb{A})$. Suppose there exists a finite set of places of F such that $\pi_v \simeq \pi'_v$ as $G(F_v)$-representations for all $v \notin S$, then in fact: $\pi \simeq \pi'$.*

4.1 Whittaker Model for p-adic Groups

The main tool for attack these theorems are local Whittaker models. Let's focus on the case for p-adic groups. Let G be a split reductive group over a local non-archimidean field F. Let $B \subset G$ be a Borel.

Let's fix $\psi : F \to \mathbb{C}^\times$ a nontrivial character. We will get a map:

$$\psi_N : N(F) \to \prod_{\alpha_i \text{ simple roots}} N_{\alpha_i}(F) \xrightarrow{\sim} \prod_{\alpha_i} F \xrightarrow{\sum} F \xrightarrow{\psi} \mathbb{C}^\times$$

Note we need to choose isomorphisms $N_{\alpha_i}(F) \simeq F$ for each α_i.

Definition 4.3. Let (π, V) be a smooth irreducible representation of $G(F)$. A Whittaker functional on (π, V) is a nonzero linear map:

$$\Lambda : V \to \mathbb{C}$$

such that:

$$\Lambda(\pi(u)\cdot v)=\psi_N(u)\Lambda(v)$$

for all $v\in V, u\in N(F)$. In other words, viewing $\Lambda\in V^*$ is an eigenvector:

$$\pi^*(u)\cdot\Lambda=\psi_N(u)^{-1}\Lambda$$

Remark. *Λ is not required to be fixed by some open subgroup of $G(F)$.*

Theorem 4.4. *1. For any irreducible smooth $G(F)$-representation (π,V), if Whittaker functional exists, it is unique up to a scalar.*

2. "Most" π admits Whittaker functionals.

Remark. *Let's appreciate the second statement in the case of $\mathrm{GL}_2(F)$. For finite dimensional representations, they must be one dimensional:*

$$\mathrm{GL}_2(F)\xrightarrow{\det}F^\times\xrightarrow{\chi}\mathbb{C}^\times$$

There is no Whittaker functional ($N(F)$-actions are trivial). For all other infinite dimensional cases, there is always a Whittaker functional. In general, the existence relies on the "sizes" of the representations. For GL_n, the "largest" representations always have Whittaker functionals. For example $\mathrm{Ind}_B^G(\chi)$ is "largest", and it has size comparable to $C^\infty(B(F)\setminus G(F))$. On the other hand, $\mathrm{Ind}_P^G(\chi)$ for P larger parabolic, it will not have Whittaker functionals.

Let $\Lambda: V\to\mathbb{C}$ be a Whittaker functional. It can be interpreted as an $N(F)$-equivariant map. So by Frobenius reciprocity, this corresponds to a $G(F)$-equivariant map:

$$\rho: V\to\mathrm{Ind}_{N(F)}^{G(F)}(\psi_N)$$

They're related by:

$$\Lambda(v)=\rho(v)(1)$$

Therefore, Whittaker functionals on π biject with embeddings of $\rho:\pi\hookrightarrow\mathrm{Ind}_N^G(\psi_N)$. So the previous theorem translates to:

Theorem 4.5. *Every irreducible π appears at most once as a subrepresentation of $\mathrm{Ind}_N^G(\psi_N)$. Most irreducible π can be embedded in $\mathrm{Ind}_N^G(\psi_N)$.*

Example. *Let $G=\mathrm{GL}_2$, $\pi=\mathrm{Ind}_B^G(\chi_1,\chi_2)$, for χ_i unramified characters. Concretely:*

$$\{f:G(F)\to\mathbb{C}\mid f(\begin{pmatrix}a&b\\0&d\end{pmatrix}g)=|\frac{a}{d}|^{\frac{1}{2}}\chi_1(a)\chi_2(d)f(g)\}$$

A Whittaker functional has property:

$$\Lambda(\pi(u)f)=\psi(u)\Lambda(f)$$

$$\iff\Lambda(g\mapsto f\left(g\begin{pmatrix}1&u\\&1\end{pmatrix}\right))=u\Lambda(f)$$

Our candidate for the Whittaker functional should be some sorts of averaging. Let's first consider:

$$\Lambda_0 : C^\infty((B(F), \psi) \setminus G(F)) \to \mathbb{C}$$

$$\phi \mapsto \int_{N(F)} \phi(u)\psi_N(u^{-1})du$$

Then

$$\Lambda_0(\pi(u)\phi) = \psi_N(u)\Lambda_0(\phi)$$

is by design. But the problem is that $\phi(u) = \phi(1)$ for all $u \in N(F)$, which makes $\Lambda_0 \equiv 0$.

More generally, we can pick pick:

$$\Lambda_g : \phi \mapsto \int_{N(F)} \phi(gu)\psi_N(u^{-1})du$$

This still satisfies the eigen property. We only need to pick g such that this is not constantly zero. Moreover, the choice of g essentially lies in $B(F)\backslash G(F)/N(F)$. So we're in the setting of Bruhat decomposition. Therefore, in the case of GL_2, our only choice is:

$$\phi \mapsto \int_F \phi(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & x \\ & 1 \end{pmatrix})\psi(-x)dx$$

This is not identically zero essentially becasue $B(F)sN(F) \subset G(F)$ is a free product.

In general, let $w_0 \in W$ be the longest element in W, we should consider:

$$\Lambda : \phi \mapsto \int_{N(F)} \phi(w_0 \cdot u)\psi_N(u^{-1})du$$

For similar reason, we know $\Lambda \not\equiv 0$. If $w_0 \neq w \in W$, then the similar formula will always give me zero. The reason for this is the following. There will exist some simple root α_i such that $w(\alpha_i) > 0$. Now suppose u_i lies in the root subgroup of α_i, then:

$$w \cdot u_i = (\mathrm{Ad}(w)u_i) \cdot w$$

with the first factor lies in $N(F)$. By some computation involving change of variable will show that the integral is constantly zero.

Moreover, this argument also shows for principal series there is always only one Whittaker functional up to rescaling. Consider:

$$\mathcal{H} = C_c^\infty(G(F)) \to \mathrm{Ind}_B^G(\chi) \xrightarrow{\Lambda} \mathbb{C}$$

Call this composition a distribution D. Then the distribution satisfies:

$$D(L(b)R(u)f) = (\chi\delta^{\frac{1}{2}})(b)\psi_N(u)D(f)$$

In other words:

$$D \in \mathcal{D}(G(F))^{B(F)\times N(F), \chi\boxtimes\psi_N}$$

But the space of bi-invariant distribution is one-dimensional by a similar argument as before. In conclusion:

$$\mathrm{Ind}_B^G(\chi)$$

always admits a nonzero Whittaker functional unique up to scalar. Similarly, one can prove that for all $B \subsetneq P$, $\mathrm{Ind}_P^G(\chi)$ has no Whittaker functional.

Let's go back to the case of GL_2. In the representation $\mathrm{Ind}_B^G(\chi_1, \chi_2)$, we have a specific spherical vector $f_0 \in V^K$ for $K = \mathrm{GL}_2(\mathcal{O})$:

$$f_0\left(\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} k\right) = |\frac{a}{d}|^{\frac{1}{2}} \chi_1(a)\chi_2(d)$$

Let's normalize $\Lambda(f_0) = 1$. Now:

$$\mathrm{Ind}_B^G(\chi_1, \chi_2) \hookrightarrow \mathrm{Ind}_N^G(\psi_N)$$

Hence the image of f_0 must lie in:

$$(\mathrm{Ind}_N^G(\psi_N))^K = \{\phi : G(F)/K \to \mathbb{C} \mid \phi(ug) = \psi_N(u)\phi(g), \forall u \in N(F), g \in G(F)\}$$

Now Iwasawa decomposition implies that ϕ is determined by its value on $T(F)/T(\mathcal{O})$.

Lemma. *Assume ψ has conductor $\mathcal{O}_F$. For any ϕ, $\phi_{a,b} = \phi\left(\begin{pmatrix} \varpi^a & \\ & \varpi^b \end{pmatrix}\right) = 0$ for all $a < b$. The map:*

$$\mathrm{Ind}_N^G(\psi_N)^K \to C(\{(a,b) \in \mathbb{Z}^2 \mid a \geq b\})$$
$$\phi \mapsto (\phi_{a,b})$$

is an isomorphism.

Let $g = \begin{pmatrix} \varpi^a & \\ & \varpi^b \end{pmatrix}, u \in N(F)$. Then:

$$\psi_N(u)\phi(g) = \phi(ug) = \phi(g(g^{-1}ug)) = \phi(g)$$

for $g^{-1}ug \in K$. Therefore, if $\phi(g) \neq 0$, then $\psi_N(u) = 1$ for all $u \in N(F) \cap gKg^{-1}$. One can compute:

$$N(F) \cap gKg^{-1} = \begin{pmatrix} 1 & \varpi^{a-b}\mathcal{O}_F \\ & 1 \end{pmatrix}$$

Therefore we must have $a \geq b$ since ψ has conductor $\mathcal{O}_F$.

Now let's consider $\mathcal{H}_G^{\mathrm{sph}}$-action on $(\mathrm{Ind}_B^G(\chi_1, \chi_2))^K$. Then $\rho(f_0)$ must be an eigen vector for $\mathcal{H}_G^{\mathrm{sph}}$ with some character ξ. This determines $\rho(f_0)$ up to a scalar.

In homework, we will see there is an isomorphism of $\mathcal{H}_G^{\mathrm{sph}}$-modules:

$$\mathrm{Ind}_N^G(\psi_N)^K \simeq C(\mathbb{Z}^2_{\mathrm{dom}})$$

We will get $\rho(f_0)_{a,b}$ honest numbers for a, b.

Before delving into the proof of existence and uniqueness of Whittaker functionals for p-adic groups, let's consider the analogue of finite groups of Lie type. Replace F by $\mathbb{F}_q$ and all constructions and statements before make sense.

Theorem 4.6 (Gelfand-Graev)**.** *Each irreducible representation of $G(\mathbb{F}_q)$ has at most one Whittaker functional up to scalar. Equivalently,* $\mathrm{Ind}_{N(\mathbb{F}_q)}^{G(\mathbb{F}_q)}(\psi_N)$*. Equivalently,*

$$\mathcal{E} = \mathrm{End}_{G(\mathbb{F}_q)}(\mathrm{Ind}_{N(\mathbb{F}_q)}^{G(\mathbb{F}_p)}(\psi_N)) = \{f : G(\mathbb{F}_q) \to \mathbb{C} \mid f(u_1 g u_2) = \psi_N(u_1) f(g) \psi_N(u_2)\}$$

is commutative.

Proof. Again we use Gelfand trick. We try to define an anti-involution ι on $G(\mathbb{F}_q)$ such that:

1. It preserves $(N(\mathbb{F}_q), \psi_N)$.
2. It fixes all double cosets:
$$N(\mathbb{F}_q) \setminus G(\mathbb{F}_q) / N(\mathbb{F}_q)$$

If one can find such ι, then it follows that our ring $\mathcal{E}$ is commutative. However, we need to relax the second condition a little bit. Let's consider:

$$\iota(g) = w_0 {}^t g w_0$$

for longest element w_0. It is clear that it preserves $(N(\mathbb{F}_q), \psi_N)$. However, different diagonal matrices belong to different double cosets, but:

$$\iota \cdot \mathrm{diag}(a_1, \cdots, a_n) = \mathrm{diag}(a_n, \cdots, a_1)$$

However, look at our functions on $\mathcal{E}$. If $f \in \mathcal{E}$ is supported on the diagonal, then f is necessarily supported on the scalar mtrices. In general, we can describe explicitly the relevant double cosets in $N(\mathbb{F}_q) \setminus G(\mathbb{F}_q) / N(\mathbb{F}_q)$. For example in the case of GL_2, the relevant double cosets are: $w_0 T(\mathbb{F})$ and $\begin{pmatrix} a & \\ & a \end{pmatrix}$. In general for GL_n, the relevant double cosets takes form:

$$w_J w_0 \cdot \begin{pmatrix} C_1 & & \\ & \cdots & \\ & & C_r \end{pmatrix}$$

where $J = J_1 \oplus J_2 \oplus \cdots \oplus J_r$, and C_i are scalar matrices. Then in fact ι will fix all *relevant* double cosets. □

To pass to p-adic groups, we need distribution analogue of the above story. Let's consider:

$$\mathcal{D}(G(F))^{(N(F)\times N(F), \psi_N \boxtimes \psi_N^{-1})}$$

These are distributions D such that:

$$D(L(u_1)R(u_2)f) = \psi(u_1)\psi(u_2^{-1})D(f)$$

If D comes from a function Φ, then:

$$\Phi(u_1 g u_2) = \psi(u_1^{-1})\psi(u_2^{-1})\Phi(g)$$

Now if I have anti-involution ι (as before) on $G(F)$, it acts on $\mathcal{D}(G(F))^{(N\times N, \psi\times\psi^{-1})}$ via:

$$({}^\iota D)(f) = D(f \circ \iota)$$

Theorem 4.7. *Te action of ι on $\mathcal{D}(G(F))^{N\times N,\psi\times\psi^{-1}}$ is the identity.*

Proof. The proof is similar as in the finite field case, where ι fixes all relevant double cosets. □

Proof. (of uniqueness of Whittaker functional). Let $\Lambda : V \to \mathbb{C}$ and $\Lambda' : V^\vee \to \mathbb{C}$ be both Whittaker functionals. Suppose $\Lambda \in V^\vee, \Lambda' \in V$ are both smooth, then I can consider the matrix coefficient:

$$\langle \pi(g)\Lambda', \Lambda\rangle \in C^\infty((N(F),\psi)\setminus G(F)/(N(F),\psi))$$

Hence this is invariant under ι. But almost never Λ, Λ' are smooth. In this case we're getting a distribution:

$$D : \mathcal{H}_G = C_c^\infty(G(F)) \to \mathbb{C}$$
$$\phi \mapsto \langle \Lambda * \phi, \Lambda'\rangle$$

where $(\Lambda * \phi)(v) = \Lambda(\pi(\phi)v)$. The smoothness of ϕ will implies $\Lambda * \phi \in V^\vee$ is an element in the smooth dual. One can check that D satisfies the desired eigen-properties.

Now let's start from two Whittaker functionals $\Lambda_1, \Lambda_2 : V \to \mathbb{C}$. To use the previous observation, we wish to turn Λ_2 into $\Lambda_2' : V^\vee \to \mathbb{C}$. But $V^\vee$ can be realized on V. Define for $v \in V$:

$$\pi^*(g)\cdot v = \pi({}^\iota g^{-1})v$$

Fact: $(\pi^*, V) \simeq (\pi^\vee, V^\vee)$. The proof of this fact is by comparing characters. If I have an admissible representation (π, V), then $\pi(\phi) : V \to V$ has finite-dimensional image $\phi \in C_c(K\setminus G(F)/K)$. Now let's just send:

$$\phi \mapsto \mathrm{Tr}(\pi(\phi), V^K)$$

This defines a distribution $\Theta(\phi)$. One should check the following general fact:

$$\Theta_{\pi^\vee} = \Theta_\pi \circ \text{inversion}$$

On the other hand:

$$\Theta_{\pi^*} = \Theta_\pi \circ \text{inversion} \circ \iota$$

But ι acts by identity on characters because ι fixes each conjugacy class, this proves the fact.

Let's go back to the proof. We view $\Lambda_2 : V \simeq V^\vee \to \mathbb{C}$ as a funcitonal:

$$\Lambda_2' : V^\vee \to \mathbb{C}$$

This will be a Whittaker functional with respect to (N, ψ^{-1}). Let's now define:

$$D(\phi) = (\Lambda_1 * \phi, \Lambda_2') \in \mathcal{D}(G(F))^{(N\times N,\psi,\psi^{-1})}$$

Similarly, I can form:

$$D'(\phi) = \langle \Lambda_2 * \phi, \Lambda_1'\rangle$$

A calculation shows $D' = {}^\iota D$. The previous theorem tells us: $D = {}^\iota D$, i.e.

$$\langle \Lambda_1 * \phi, \Lambda_2'\rangle = \langle \Lambda_2 * \phi, \Lambda_1'\rangle, \quad \forall \phi \in C_c^\infty(G(F))$$

We have:

$$\Lambda_1 * (-) : \mathcal{H}_G \to V^\vee$$

this is $G(F)$-equivariant map. Similarly we have $\Lambda_2 * (-) : \mathcal{H}_G \to V^\vee$.
Claim: $\Lambda_1 * \phi = 0 \iff \Lambda_2 * \phi = 0$ This claim implies the theorem. Now only need to prove the claim. Note that if:

$$\Lambda_1 * \phi = 0 \implies \Lambda_1 * \phi * \phi' = 0 \implies \langle \Lambda_1 * \phi * \phi', \Lambda_2' \rangle = 0$$
$$\implies \langle \Lambda_2 * \phi * \phi', \Lambda_1' \rangle = 0 \implies \Lambda_2 * \phi = 0$$

□

4.2 Whittaker Model for Real Groups

Suppose I have a $G(\mathbb{R})$-representation V. Fix $\psi_N : N(\mathbb{R}) \to S^1$. We still can talk about the notion of Whittaker functional:

$$\Lambda : V \to \mathbb{C}$$

with property: $\Lambda(u \cdot v) = \psi(u)\Lambda(v)$. We still want to have some uniqueness result.

Let's begin with the example an irreducible prinical series (π, V) for $\mathrm{SL}_2(\mathbb{R})$. If there's unique Whittaker functional, then there will be an embedding:

$$V \hookrightarrow \mathrm{Fun}((N(\mathbb{R}), \psi_N) \setminus G(\mathbb{R}))$$

As a $(\mathfrak{g}, K)$-module, we have an embedding:

$$V \hookrightarrow C^\infty((N(\mathbb{R}), \psi_N) \setminus G(\mathbb{R}))_{K\text{-fin}}$$

Suppose v_0 is a weight zero vector in V. Let's specify the image of v_0

$$i : v_0 \mapsto f \in C^\infty((N(\mathbb{R}), \psi_N) \setminus G(\mathbb{R})/\mathrm{SO}_2) \simeq C^\infty(\mathbb{R}^{>0})$$

So $i(v_0)$ is determined by its value on $\begin{pmatrix} a & \\ & a^{-1} \end{pmatrix}$ for $a > 0$. Also recall that V has a infinitesimal character λ (the scalar that Ω acts by). Then $i(v_0)$ is an eigenvector under the Casimir action on $C^\infty(\mathbb{R}^{>0})$ (the Casimir acts via a quadratic differential operator). Therefore we're looking at a 2nd order ODE:

$$\Omega f = \lambda f$$

This has 2 linearly independent solutions.

There are two solutions to this non-uniqueness.

1. Moderate growth. For all $v \in V$, let's look at:

$$a \mapsto \Lambda\left(\pi \begin{pmatrix} a & \\ & 1 \end{pmatrix} v\right)$$

this must have moderate growth as $|a| \to \infty$ (i.e. this function is controlled by some $C \cdot |a|^N$). The conclusion is that any irreducible $(\mathfrak{g}, K)$-module has up to scalar at most one embedding into $C^{\infty,\mathrm{mod}}((N(\mathbb{R}), \psi_N), G(\mathbb{R}))_{K\text{-fin}}$. For GL_n, we need to choose a height function:

$$\|g\| = \max(|a_{ij}|, |a_{ij}^*|)$$

2. Require Λ to be continuous with respect to the Frechet topology on V.

They will give the same Whittaker functional.

4.3 Global Whittaker Model

We start with an automorphic representation π of $G(\mathbb{A}_f) \times (\mathfrak{g}_\infty, K_\infty)$. A Whittaker model for π is an $G(\mathbb{A})$-embedding:

$$\pi \hookrightarrow \mathcal{A}((N(\mathbb{A}), \psi_N) \setminus G(\mathbb{A}))$$

The symbol $\mathcal{A}$ means that we impose conditions as in the definition of automorphic forms:

1. $Z(U\mathfrak{g}_\infty)$-finite.
2. $K = K_\infty$-finite.
3. Moderate growth.

Theorem 4.8. *There is at most one Whittaker model for π (i.e. as a subspace of $\mathcal{A}(\cdots)$ is unique).*

To be more precise, we need to start from a unitary character:

$$\psi : F \setminus \mathbb{A} \to S^1$$

and extend to:

$$\psi_N : N(F) \setminus N(\mathbb{A}) \to S^1$$

Proof. (for function fields). Let F be a function field. It is enough to show uniqueness (up to scalar) of:

$$\Lambda : V = \otimes'_{v \in |X|} V_v \to \mathbb{C}$$

Let S be finitely many places such that π_v is spherical for $v \notin S$, and ψ_v has conductor $\mathcal{O}_v$ for all $v \notin S$. Therefore, for all $v \notin S$, there exists χ_v such that π_v is a subrepresentation of $\mathrm{Ind}_B^G(\chi_v)$. Let $\xi_v \in \mathrm{Ind}_B^G(\chi_v)$ be the unique spherical vector such that $\xi_v(1) = 1$. Then there exists unique Whittaker functional $\Lambda_v^0 : \mathrm{Ind}_B^G(\chi_v) \to \mathbb{C}$ such that $\Lambda_v^0(\xi_v) = 1$.

Now if $\Lambda \neq 0$, then we can choose a pure tensor $\eta = \xi^S \otimes \eta_S \in V$ such that $\Lambda(\eta) \neq 0$ (enlarging S), where:

$$\xi^S = \otimes_{v \notin S} \xi_v, \quad \eta_S = \otimes_{v \in S} \eta_v$$

Now we can look at:

$$\Lambda_S = \Lambda|_{\xi^S \otimes (-)} \colon V_S \coloneqq \otimes_{v \in S} V_v \to \mathbb{C}$$

a nonzero Whittaker functional for the irreducible $G(\mathbb{A}_S)$-module V_S. Let $\Lambda_v : V_v \to \mathbb{C}$ be a nonzero Whittaker model for all $v \in S$. The local uniqueness tells me:

$$\Lambda_S = c \cdot (\otimes_v \Lambda_v) : \otimes_{v \in S} V_v \to \mathbb{C}$$

We claim that:

$$\Lambda = c \cdot (\otimes_{v \notin S} \Lambda_v^0) \otimes (\otimes_{v \in S} \Lambda_v) : \otimes'_{v \in |X|} (V_v, \xi_v) \to \mathbb{C}$$

We can check this on pure tensors. One should first show that they're proportional on $\xi^{S'} \otimes (\otimes_{v \in S'} V_v)$ for $S' \supset S$, and show the constant is one by evaluating at $\eta \in \xi^S \otimes V_S$. $\square$

4.4 Proof of Multiplicity One Theorem

We focus in the case for GL_2. The goal is to show that $\pi \subset \mathcal{A}_{\text{cusp}}$ at most once. The idea is to show:

1. π always has a Whittaker model.
2. Recover π uniquely from its Whittaker model.

We claim that for any fixed ψ we have an embedding:

$$\mathcal{A}_{\text{cusp}} \hookrightarrow \mathcal{A}((N(\mathbb{A}), \psi_N) \setminus G(\mathbb{A}))$$

$$\phi \mapsto \left(W_\phi^\psi : g \mapsto \int_{F\setminus\mathbb{A}} \phi\left(\begin{pmatrix} 1 & x \\ & 1 \end{pmatrix} g\right) \psi(-x)dx \right)$$

as $G(\mathbb{A})$-representations. Note this will immediately imply that π admits a Whittaker model.

The key point is that this is injective. The trick is that we should veriy ψ and therefore allow the use of Fourier inversion formula. Note that we can change ψ by $a \in F^\times$:

$$\psi_a = \psi \circ a$$

Once can check that:

$$\begin{aligned} W_\phi^{\psi_a}(g) &= \int \phi\left(\begin{pmatrix} 1 & a^{-1}x \\ & 1 \end{pmatrix} g\right) \psi(-x)dx \\ &= \int \phi\left(\begin{pmatrix} a^{-1} & \\ & 1 \end{pmatrix} \begin{pmatrix} 1 & x \\ & 1 \end{pmatrix} \begin{pmatrix} a & \\ & 1 \end{pmatrix} g\right) \psi(-x)dx \\ &= \int \phi\left(\begin{pmatrix} 1 & x \\ & 1 \end{pmatrix} \begin{pmatrix} a & \\ & 1 \end{pmatrix} g\right) \psi(-x)dx \\ &= W_\phi^\psi\left(\begin{pmatrix} a & \\ & 1 \end{pmatrix} g\right) \end{aligned}$$

Consider the function:

$$f : F \setminus \mathbb{A} \to \mathbb{C}$$

$$x \mapsto \phi\left(\begin{pmatrix} 1 & x \\ & 1 \end{pmatrix} g\right)$$

is a function on $F \setminus \mathbb{A}$. We can view $\psi \in (F \setminus \mathbb{A})^*$, then:

$$\hat{f}_{\phi,g}(\psi) = W_\phi^\psi(g)$$

Therefore $\hat{f}_{\phi,g}$ is a function on $(F \setminus \mathbb{A})^*$. Therefore by Fourier inversion formula:

$$\phi(g) = f_{\phi,g}(0) = \sum_{\psi' \in (F\setminus\mathbb{A})^*} \hat{f}_{\phi,g}(\psi') = \sum_{\psi' \in (F\setminus\mathbb{A})^*} W_\phi^{\psi'}(g)$$

Recall:

$$(F \setminus \mathbb{A})^*$$

is a one dimensional F-vector space. If we choose a basis ψ, then:

$$(F \setminus \mathbb{A})^* = \{\psi_a \mid a \in F^\times\} \cup \{\mathsf{triv}\}$$

Therefore we can rewrite the above summation as:

$$W_\phi^{\mathsf{triv}}(g) + \sum_{a \in F^\times} W_\phi^{\psi_a}(g) = W_\phi^{\mathsf{triv}}(g) + \sum_{a \in F^\times} W_\phi^{\psi}\left(\begin{pmatrix} a & \\ & 1 \end{pmatrix} g\right)$$

One can now check the injectivity. Indeed if $W_\phi^\psi \equiv 0$, then

$$\phi(g) = W_\phi^{\mathsf{triv}}(g) = \int_{F \setminus \mathbb{A}} \phi\left(\begin{pmatrix} 1 & x \\ & 1 \end{pmatrix} g\right) dx = 0$$

since ϕ is cuspidal. Now to show multiplicity one, we note that:

$$\mathrm{Hom}_{G(\mathbb{A})}(\pi, \mathcal{A}_{\mathrm{cusp}}) \hookrightarrow \mathrm{Hom}_{G(\mathbb{A})}(\pi, \mathcal{A}((N(\mathbb{A}), \psi_N) \setminus G(\mathbb{A}))) \xrightarrow{\sim} \mathrm{Hom}_{N(\mathbb{A})}(\pi, \mathbb{C}_{\psi_N})^{\text{mod growth}}$$

Now using the uniqueness of global Whittaker functional, we know the last space is one dimensional.

Let us briefly comment on how to prove multiplicity one for general GL_n. Let's start from GL_3 to see a pattern. Note that the whole unipotent subgroup is non-abelian so the Fourier theory won't apply. Instead, we start with:

$$U = \begin{pmatrix} 1 & & * \\ & 1 & * \\ & & 1 \end{pmatrix} \simeq \mathbb{G}_a \times \mathbb{G}_a$$

We consider:

$$\phi \mapsto \left(\phi_1 : g \mapsto \int_{(F \setminus \mathbb{A})^2} \phi\left(\begin{pmatrix} 1 & & y \\ & 1 & x \\ & & 1 \end{pmatrix} g\right) \psi(-x) dx dy\right)$$

This is Fourier transform of:

$$\begin{pmatrix} y \\ x \end{pmatrix} \mapsto \phi\left(\begin{pmatrix} 1 & & y \\ & 1 & x \\ & & 1 \end{pmatrix} g\right)$$

evaluated at $(0, \psi) \in ((F \setminus \mathbb{A})^2)^* = ((F \setminus \mathbb{A})^*)^2$. Note that $\begin{pmatrix} \mathrm{GL}_2 & \\ & 1 \end{pmatrix}$ normalizes my group U. Therefore the value of the Fourier transform at $(\psi_1, \psi_2) \neq (0, 0)$ can be recovered as $\phi_1\left(\begin{pmatrix} \gamma & \\ & 1 \end{pmatrix} g\right)$ for $\gamma \in \mathrm{GL}_2(F)$. Using ϕ is cuspidal, Fourier inversion enables us to recover ϕ from ϕ_1.

In summary, we get:

$$\mathcal{A}_{\mathrm{cusp}}(Q_3(F) \setminus \mathrm{GL}_3(\mathbb{A})) \xrightarrow{\sim} \mathcal{A}_{\mathrm{cusp}}\left(\left(\begin{pmatrix} F^\times & * & \\ & 1 & \\ & & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & & * \\ & 1 & * \\ & & 1 \end{pmatrix}, \psi_1\right) \setminus \mathrm{GL}_3(\mathbb{A})\right)$$

$$\phi \mapsto \phi_1$$

One can do another step further to get:

$$\phi_2 \in \mathcal{A}((N(\mathbb{A}), \psi_N) \setminus \mathrm{GL}_3(\mathbb{A}))$$

For a complete proof, see Exercise B.29.

4.5 Global Zeta Integral

Recall in Tate's Thesis, we introduced:

$$Z : \mathcal{S}(\mathbb{A}) \times (F^\times \setminus \mathbb{A}^\times)^* \to \mathbb{C}$$

$$Z(f, \chi) = \int_{\mathbb{A}^\times} f(a)\chi(a)d^\times a$$

We also wrote:

$$Z(s, f, \chi) = Z(f, \chi \cdot |-|^s)$$

for $s \in \mathbb{C}$. We have functional equation:

$$Z(f, \chi) = Z(\hat{f}, \chi^{-1} \cdot |-|)$$

Now let (π, V) be a cuspidal automorphic representation of GL_2. Given $\phi \in V \subset \mathcal{A}_{\text{cusp}}$ and $\chi : F^\times \setminus \mathbb{A}^\times \to \mathbb{C}^\times$, we can define the **Zeta integral** to be:

$$Z(\phi, \chi) = \int_{F^\times \setminus \mathbb{A}^\times} \phi\left(\begin{pmatrix} a & \\ & 1 \end{pmatrix}\right) \chi(a) d^\times a$$

Remark. *In the story for* GL_1*,* χ *plays the role of a 1-dimensional automorphic representation of* GL_1*, and* f *is purely auxillary. It's even not clear how to relate* $\mathbb{A}^\times$ *with* $\mathbb{A}$ *in general. This is a recent development by Braverman-Kazhdan.*

Let's also set:

$$Z(s, \phi, \chi) = \int_{F^\times \setminus \mathbb{A}^\times} \phi\left(\begin{pmatrix} a & \\ & 1 \end{pmatrix}\right) \chi(a) \cdot |a|^{s-\frac{1}{2}} d^\times a$$

Fact: This is holomorphic in $s \in \mathbb{C}$.

Let's now discuss the Euler product expansion. Recall we have Whittaker models:

$$\pi \hookrightarrow \mathcal{A}_{\text{cusp}} \hookrightarrow \mathcal{A}((N(\mathbb{A}), \psi_N) \setminus G(\mathbb{A}))$$

$$\phi \mapsto W_\phi$$

We can recover ϕ via:

$$\phi(g) = \sum_{b \in F^\times} W_\phi\left(\begin{pmatrix} b & \\ & 1 \end{pmatrix} g\right)$$

So that:

$$\phi\begin{pmatrix} a & \\ & 1 \end{pmatrix} = \sum_{b\in F^\times\cdot a} W_\phi \begin{pmatrix} b & \\ & 1 \end{pmatrix}$$

Then the formula for zeta integral reads:

$$Z(\phi,\chi) = \int_{a\in F^\times\backslash\mathbb{A}^\times} (\sum_{b\in F^\times a} W_\phi \begin{pmatrix} b & \\ & 1 \end{pmatrix})\chi(a)d^\times a = \int_{\mathbb{A}^\times} W_\phi \begin{pmatrix} b & \\ & 1 \end{pmatrix}\chi(b)d^\times b$$

This procedure is called *unfolding*. If $\phi = \otimes\phi_v$, then $W_\phi = \otimes W_{\phi_v} \in C^\infty((N(F_v),\psi_v)\backslash G(F_v))$. The factorization of W_ϕ is coming from the uniqueness of local Whittaker model (up to scalar). Also we have $\chi = \prod_v \chi_v$. Therefore:

$$Z(\phi,\chi) = \prod_v Z(W_{\phi_v},\chi_v)$$

where the local zeta integral is given by:

$$Z(W,\chi) = \int_{F_v^\times} W(b)\cdot\chi(b)d^\times b$$

for $W \in C^\infty(F_v^\times)$, $\chi : F_v^\times \to \mathbb{C}^\times$.

Now the analytic property of zeta integral comes from analytic property of cusp forms. Recall that if ϕ is a cusp form of central character ω (i.e. $\phi\left(\begin{pmatrix} a & \\ & a \end{pmatrix} g\right) = \omega(a)\phi(g)$) , then

$$\mathbb{A}^\times \to \mathbb{C}, \quad a \mapsto \phi\left(\begin{pmatrix} a & \\ & 1 \end{pmatrix} g\right)$$

has rapid decay (decay faster than inverse of any polynomial) as $|a|\to\infty$. On ther other hand, we can write:

$$\phi\left(\begin{pmatrix} a & \\ & 1 \end{pmatrix} g\right) = \phi\left(\begin{pmatrix} & 1 \\ 1 & \end{pmatrix}\begin{pmatrix} a & \\ & 1 \end{pmatrix} g\right) = \phi\left(\begin{pmatrix} 1 & \\ & a \end{pmatrix}\begin{pmatrix} & 1 \\ 1 & \end{pmatrix} g\right)$$

Now recall we impose a central character ω, we get:

$$\omega(a) = \phi\left(\begin{pmatrix} a^{-1} & \\ & 1 \end{pmatrix}\begin{pmatrix} & 1 \\ 1 & \end{pmatrix} g\right)$$

We may without loss assume ω is unitary, and therefore we see for $|a|\to 0$ we also have rapid decay. The consequence for all these is that zeta function is holomorphic.

Now I take χ to be trivial, then I get:

$$Z(s,\phi) = \int_{F^\times\backslash\mathbb{A}^\times} \phi\begin{pmatrix} a & \\ & 1 \end{pmatrix}|a|^{s-\frac{1}{2}}d^\times a = \prod_v Z(s,W_{\phi_v})$$

assuming $\phi = \otimes\phi_v$. Now for most v, π_v is spherical and ϕ_v is spherical, for those we set:

$$W_{\phi_v} : (N(F_v),\psi_v)\setminus G(F_v)/G(\mathcal{O}_v) \to \mathbb{C}$$

with normalization $1 \mapsto 1$.

Example. *Let $f \in S_k(\mathrm{SL}_2(\mathbb{Z}))$ be a modular form, this corresponds to $\phi : \mathrm{GL}_2(\mathbb{Q}) \setminus \mathrm{GL}_2(\mathbb{A}) \to \mathbb{C}$ given by:*

$$\phi(g_\infty) = (ci+d)^k \cdot f(\frac{ai+b}{ci+d}), \quad g_\infty = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Let me compute $Z(s,\phi)$ in this case.

$$Z(s,\phi) = \int_{\mathbb{Q}^\times \backslash \mathbb{A}^\times / \widehat{\mathbb{Z}}^\times} \phi \begin{pmatrix} a & \\ & 1 \end{pmatrix} |a|^{s-\frac{1}{2}} d^\times a = \int_0^\infty \phi \begin{pmatrix} a & \\ & 1 \end{pmatrix} a^{s-\frac{1}{2}} \frac{da}{a} = \int_0^\infty f(ai) a^{s-\frac{1}{2}} \frac{da}{a}$$

Now let's Fourier expand f:

$$f(z) = \sum_{n \geq 1} a_n e^{2\pi i n z}$$

Then:

$$\int_0^\infty f(ai) \cdot a^s \frac{da}{a} = (2\pi)^{-s} \Gamma(s) \cdot \sum_{n \geq 1} \frac{a_n}{n^s} \coloneqq \Lambda(s,f)$$

This is the **complete L-function** *of modular form f. This L-function enjoys functional equation:*

$$\Lambda(s,f) = (-1)^{\frac{k}{2}} \Lambda(k-s,f)$$

This can be proved using the invariance of f under the matrix $w_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

More generally, the zeta integral enjoys the functional equation:

$$Z(\phi,\chi) = Z(\pi(w_0)\phi, \omega^{-1}\chi^{-1})$$

where $\omega : F^\times \setminus \mathbb{A}^\times \to \mathbb{C}^\times$ is the central character of π. If ϕ corresponds to $f \in S_k(\mathrm{SL}_2(\mathbb{Z}))$, then its central character is $|-|^k$.

The natural generalization in the case of GL_n is the so called *Rankin-Selberg zeta integral*. The analogue of for the pair (π,χ) is:

$$(\pi,\pi') \subset \mathcal{A}_{\mathrm{cusp}}(\mathrm{GL}_n) \times \mathcal{A}(\mathrm{GL}_{n-1})$$

Now take $\phi \in V_\pi, \phi' \in V_{\pi'}$, we can define:

$$Z(s,\phi,\phi') = \int_{\mathrm{GL}_{n-1}(F)\backslash \mathrm{GL}_{n-1}(\mathbb{A})} \phi \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix} \phi'(A) |\det A|^{s-\frac{1}{2}} dA$$

where dA is the Haar measure on $\mathrm{GL}_{n-1}(\mathbb{A})$. It will have relation with $L(s,\pi\times\pi')$. Extra work is needed to produce $L(s,\pi)$.

4.6 Local Zeta Integral

Let π be an irreducible admissible representation of $\mathrm{GL}_2(F)$ for F nonarchimedean local field. Suppose I have local Whittaker functional:

$$\Lambda : V_\pi \to \mathbb{C}$$

which gives:

$$V_\pi \hookrightarrow C^\infty((N(F), \psi_N) \setminus G(F))$$
$$v \mapsto W_v$$

Denote $\mathcal{W}_\pi \subset C^\infty((N(F), \psi_N)\backslash G(F))$ be the image of the above map. Now for each $W \in \mathcal{W}_\pi$ and $\chi : F^\times \to \mathbb{C}^\times$, we have:

$$Z(W, \chi) = \int_{F^\times} W \begin{pmatrix} a & \\ & 1 \end{pmatrix} \chi(a) d^\times a$$

$$Z(s, W, \chi) = \int_{F^\times} W \begin{pmatrix} a & \\ & 1 \end{pmatrix} \chi(a)|a|^{s-\frac{1}{2}} d^\times a$$

Let's list a few properties:

1. Fix W, χ, we have $Z(s, W, \chi) \in \mathbb{C}(q^{-s})$ for q the cardinality of residue field of F.

2. As W varies in $\mathcal{W}_\pi$, then $\{Z(s, W, \chi)\}_{W \in \mathcal{W}_\pi}$ generates a fractional ideal for $\mathbb{C}[q^{-s}]$. Therefore it has common denominator in $\mathbb{C}[q^{-s}]$, normalized such that the constant term is 1. Then we define:

   $$L(s, \pi) = \frac{1}{\text{common denominator}}$$

   In fact the common denominator has degree ≤ 2 in q^{-s}.

   Example. *Let $\pi = \mathrm{Ind}_B^G(\chi_1, \chi_2)$ be irreducible spherical. Let $\chi_i(\varpi) = \alpha_i$. We can compute:*

   $$L(s, \pi) = \frac{1}{(1 - \alpha_1 q^{-s})(1 - \alpha_2 q^{-s})}$$

   Otherwise (except for π one dimensional, for which there is no Whittaker model so that L-function is not defined), we have:

   $$L(s, \pi) = \begin{cases} \dfrac{1}{1 - \alpha q^{-s}}, & \text{if } \pi \text{ is } \textit{Steinberg}; \\ 1, & \textit{otherwise.} \end{cases}$$

   The later case is called **supercuspidal**.

3. The functional equation:

   $$\frac{Z(1 - s, \pi(w_0)W, \omega^{-1}\chi^{-1})}{Z(s, W, \chi)} := \gamma(s, \pi, \chi, \psi)$$

   is independent of $W \in \mathcal{W}_\pi$, but depends on π, χ, s, ψ. This is the γ-factor. If everything is spherical: π is spherical, χ is unramified, ψ has conductor $\mathcal{O}$ then:

   $$\gamma(s, \pi, \chi, \psi) = 1$$

To prove these properties, we need extra tools of p-adic group representations. Let $\mathrm{GL}_2(F)$ act on V_π. Let's restrict to $N(F) = F$-action. Therefore we have $(C_c^\infty(F), *)$ acts on V_π. Now Fourier transform gives:

$$(C_c^\infty(F), *) \xrightarrow{\sim} (C_c^\infty(F), \cdot)$$

and the later acts on $C_c^\infty(F)$ by multiplication. More generally, if $\mathcal{E}$ is a sheaf on F, then: $\Gamma(F, \mathcal{E})$ will admit a $C^\infty(F)$-action. Via restriction we have $C_c^\infty(F)$ acts on $\Gamma_c(F, \mathcal{E})$.

We have the following generality. Let X be a totally disconnected, locally compact space. Look at:

$$\mathrm{Shv}_c(X, \mathbb{C}) \xrightarrow{\Gamma_c(X,-)} C_c^\infty(X)\text{-mod}$$

Note this functor has a right adjoint (and inverse):

$$M \mapsto \mathcal{E}_M$$

where $\mathcal{E}_M(U) = \mathbf{1}_U \cdot M$

Therefore the data of $(C_c^\infty(F), \cdot)$ action on M is the same as a sheaf $\mathcal{E}_\pi$ on F. The stalks of the sheaf $\mathcal{E}_\pi$ are easier to understand. In fact:

$$(\mathcal{E}_\pi)_0 = \varinjlim_{\varpi^n \mathcal{O}_F} \mathbf{1}_{\varpi^n \mathcal{O}_F} \cdot V_\pi = \varinjlim_{\varpi^n \mathcal{O}_F} \mathbf{1}_{\varpi^{-n} \mathcal{O}_F} * V_\pi$$

One can check that:

$$(\mathcal{E}_\pi)_0 \simeq (V_\pi)_{N(F)} = V_\pi / N(F) \cdot V_\pi$$

This is the **Jacquet module** of V_π.

More canonically we should write $(C_c^\infty(F), \cdot)$ by $(C_c^\infty(F^\vee), \cdot)$ where $F^\vee$ the Pontrygain dual of F. Now for $\psi \in F^\vee$, we have:

$$(\mathcal{E}_\pi)_\psi = (V_\pi)_{N(F), \psi}$$

The dual vector space of $(\mathcal{E}_\pi)_\psi$ is the space of ψ-Whittaker functionals on V_π. So:

1. Either π has no Whittaker funcitonal, then $\mathcal{E}_\pi$ is supported at $0 \in F^\vee$
2. Or π has Whittaker funcitonal, then $\dim(\mathcal{E}_\pi)_\psi = 1$ for all $\psi \neq 0$. Note $\mathcal{E}_\pi$ has $F^\times$-equivariant structure coming from:

$$\{\begin{pmatrix} a & \\ & 1 \end{pmatrix} \mid a \in F^\times\}$$

normalizing $N(F)$. This shows:

$$\mathcal{E}_\pi|_{F^\vee \setminus \{0\}} \simeq \mathbb{C}$$

Use this dictionary, we have

$$0 \to \Gamma_c(F^\vee \setminus \{0\}, \mathcal{E}_\pi) \to V_\pi = \Gamma_c(F^\vee, \mathcal{E}_\pi) \to (\mathcal{E}_\pi)_0 \to 0$$

which translates to:

$$0 \to \Gamma_c(F^\vee \setminus \{0\}, \mathcal{E}_\pi) \to V_\pi \to (V_\pi)_{N(F)} \to 0$$

Jacquet module are essential because by Frobenius reciprocity, a $T(F)$-equivariant map:

$$(V_\pi)_{N(F)} \to \mathbb{C}_\psi$$

is the same data as an embedding $G(F) \hookrightarrow \mathrm{Ind}_B^G(\chi)$. Note that such embedding exists for at most two $\chi = (\chi_1, \chi_2)$ via exchanging χ_1, χ_2. As a consequence, we know:

$$\dim(V_\pi)_{N(F)} \leq 2$$

In fact:

$$\dim(V_\pi)_{N(F)} = \begin{cases} 2, & \text{if } \pi \simeq \mathrm{Ind}_B^G(\chi_1, \chi_2) \text{ (hence is irreducible);} \\ 1, & \text{if } \pi \text{ is 1-dimensional, or Steinberg;} \\ 0, & \pi \text{ supercuspidal.} \end{cases}$$

The second case shows up in $\mathrm{Ind}_B^G(\chi_1, \chi_2)$ with $\chi_1 \cdot \chi_2^{-1} = |-|, |-|^{-1}$.

In summary, if we look at:

$$0 \to \Gamma_c(F^\vee \setminus \{0\}, \mathcal{E}_\pi) \to V_\pi \to (V_\pi)_{N(F)} \to 0$$

then V_π is slightly larger than $C_c^\infty(F^\vee \setminus \{0\})$, and the third term gives asymtotics of functions as $\psi \to 0$.

With Whittaker model $\mathcal{W}_{\pi,\psi}$ and $W \in \mathcal{W}_{\pi,\psi}$, we get a function:

$$(a \mapsto W \begin{pmatrix} a & \\ & 1 \end{pmatrix}) \in C^\infty(F^\times)$$

But more intrinsically, I should view:

$$(\psi \circ a) \mapsto W \begin{pmatrix} a & \\ & 1 \end{pmatrix} \in C^\infty(F^\vee \setminus \{0\})$$

This gives me a (canonical) embedding:

$$V_\pi \hookrightarrow C^\infty(F^\vee \setminus 0)$$

This is the **Kirillov model**. Using the sheafy observation, we actually have:

$$C_c^\infty(F^\vee \setminus 0) \hookrightarrow V_\pi \hookrightarrow C^\infty(F^\vee \setminus 0)$$

The precise location of V_π depends on the asymptotic behavior of $\psi \to 0$.

A Gloabl Class Field Theory

Let's give a very brief overview of global class field.

Let $\mathcal{I}_K$ be the ideal group of invertible ideals of $\mathcal{O}_K$, we have a canonical surjective homomorphism:

$$\phi : \mathbb{A}_K^\times \to \mathcal{I}_K$$
$$a \mapsto \prod \mathfrak{p}^{v_\mathfrak{p}(a_v)}$$

where we take product over finite places. Recall the idèle class group is defined to be $C_K = K^\times \backslash \mathbb{A}_K^\times$, then ϕ induces a diagram:

$$\begin{array}{ccccccccc} 1 & \longrightarrow & K^\times & \longrightarrow & \mathbb{A}_K^\times & \longrightarrow & C_K & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & \mathcal{P}_K & \longrightarrow & \mathcal{I}_K & \longrightarrow & \mathrm{Cl}(\mathcal{O}_K) & \longrightarrow & 1 \end{array}$$

Let L/K be a finite separable extension of global fields. Our setup ensures the all norms are compatible:

$$\begin{array}{ccccc} L^\times & \longrightarrow & \mathbb{A}_L^\times & \longrightarrow & \mathcal{I}_L \\ \downarrow{\scriptstyle N_{L/K}} & & \downarrow{\scriptstyle N_{L/K}} & & \downarrow{\scriptstyle N_{L/K}} \\ K^\times & \longrightarrow & \mathbb{A}_K^\times & \longrightarrow & \mathcal{I}_K \end{array}$$

The norm also descends to class groups:

$$\begin{array}{ccc} C_L & \twoheadrightarrow & \mathrm{Cl}_L \\ \downarrow{\scriptstyle N_{L/K}} & & \downarrow{\scriptstyle N_{L/K}} \\ C_K & \twoheadrightarrow & \mathrm{Cl}_K \end{array}$$

We now begin to construct global Artin homomorphism for global field K. Recall at each place v, we have a local Artin homomorphism:

$$\theta_{K_v} : K_v^\times \to \mathrm{Gal}(K_v^{\mathrm{ab}}/K_v)$$

With the following properties:

1. If L/K is finite abelian, and w a place of L aove K, we have a short exact sequence:

$$1 \to N_{L_w/K_v}(L_w^\times) \to K_v^\times \xrightarrow{\theta_{L/K}} \mathrm{Gal}(L_w/K_v) \to 1$$

2. If K_v is nonarchimedean and L_w/K_v is unramified, then $\theta_{L_w/K_v}(\varpi_v) = \mathrm{Frob}_{L_w/K_v}$ for all uniformizer ϖ_v of K_v.

We emphasize that L/K is an **abelian** extension of global fields. Let

$$\phi_w : \mathrm{Gal}(L_w/K_v) \hookrightarrow \mathrm{Gal}(L/K), \quad \sigma \mapsto \sigma|_L$$

be the restriction map (also note it is injective). If v is archimedean then the image is either trivial on generated by the involution corresponding to complex conjugation in $L_w \simeq \mathbb{C}$. If v is a finite place, and $\mathfrak{q}$ is the prime ideal corresponding to w, then the image is nothing but the decomposition group $D_{\mathfrak{q}}$. We can combine these cases to call $\phi_w(\mathrm{Gal}(L_w/K_v))$ the decomposition group of the place w.

We have a well-defined map

$$\phi_w \circ \theta_{L_w/K_v} : K_v^\times \to \mathrm{Gal}(L/K)$$

independent of the choice of w. Combine these and choose w arbitrarilty we get a well-defined map:

$$\theta_{L/K} : \mathbb{A}_K^\times \to \mathrm{Gal}(L/K)$$
$$(a_v) \mapsto \prod \phi_w(\theta_{L_w/K_v}(a_v))$$

This is well-defined since only finitely many places ramify (controlled by the discriminant), and for those unramified place, we have:

$$\phi_w(\theta_{L_w/K_v}(a_v)) = \mathrm{Frob}_v^{v(a_v)} = 1$$

One can also check $\theta_{L/K}$ is a continuous group homomorphism. These systems are compatible, in the sense that for $L_1 \supset L_2 \supset K$, I have: $\theta_{L_2/K} = \theta_{L_1/K}|_{L_1}$. Hence combine all these gadgets we get **global Artin homomorphism**

$$\theta_K : \mathbb{A}_K^\times \to \mathrm{Gal}(K^{\mathrm{ab}}/K)$$

The global Artin homomorphism is the "unique correct one" that extends local Artin homomorphism:

$$\begin{array}{ccc} K_v^\times & \xrightarrow{\theta_{L_w/K_v}} & \mathrm{Gal}(L_w/K_v) \\ \Big\downarrow{\scriptstyle \iota_w} & & \Big\downarrow{\scriptstyle \phi_w} \\ \mathbb{A}_K^\times & \xrightarrow[\theta_{L/K}]{} & \mathrm{Gal}(L/K) \end{array}$$

Theorem A.1 (Gloabl Artin Reciprocity)**.** *Let K be a global field. The above homorphism θ_K vanishes on $K^\times$, so that we have a continuous homomorphism:*

$$\theta_K : C_K \to \mathrm{Gal}(K^{\mathrm{ab}}/K)$$

with the following exact sequence always exact:

$$1 \to N_{L/K}(C_L) \to C_K \to \mathrm{Gal}(L/K)$$

Remark. *When K is number field θ_K is surjective but not injective, with kernel the connected component of the identity. When K is a global function field, θ_K is injective but not surjective; but its image is dense.*

We can summarize as:

Theorem A.2 (Main Theorem of GCFT)**.** *Let K be a global field. The global Artin homomorphism induces a canonical isomorphism:*

$$\overline{\theta}_K : \overline{C_K} \to \mathrm{Gal}(K^{\mathrm{ab}}/K)$$

as profinite groups

In addition, we have:

1. **Global Existence**. We have an inclusion reversing bijection:

$$\{\text{finite index open subgroups of } C_K\} \leftrightarrow \{\text{finite abelian extensions of } K\}$$

$$H \mapsto (K^{\mathrm{ab}})^{\theta_K(H)}$$

$$N_{L/K}(C_L) \leftarrow L$$

2. **Functoriality**. For L/K any finite separable extension (not necessarily abelian)

$$\begin{array}{ccc} C_L & \xrightarrow{\theta_L} & \mathrm{Gal}(L^{\mathrm{ab}}/L) \\ \downarrow{\scriptstyle N_{L/K}} & & \downarrow{\scriptstyle \mathrm{res}} \\ C_K & \xrightarrow{\theta_K} & \mathrm{Gal}(K^{\mathrm{ab}}/K) \end{array}$$

B Homworks

B.1 Homework 1

I consulted classification of algebraic tori for Exercise 1.2. I consulted the Hasse principle in one variable for Exercise 2.2.

Exercise B.1. *1. Let E/F be a finite extension of global fields. Show that $E^{\mathrm{Nm}=1}$ is a discrete subgroup of $\mathbb{A}_E^{\mathrm{Nm}=1}$ and the quotient $E^{\mathrm{Nm}=1}\backslash\mathbb{A}_E^{\mathrm{Nm}=1}$ is compact.*

2. Let T be an anisotropic torus over global field F. Show that $T(F)\backslash T(\mathbb{A}_F)$ is compact.

Proof. 1. Note that if $x \in E$ be such that $\mathrm{Nm}_{E/F}(x) = 1$. If we fix w a place of F, we know the Norm can be computed locally:

$$\prod_{v|w} \mathrm{Nm}_{E_v/F_w} = \mathrm{Nm}_{E/F} = 1$$

So that $E^{\mathrm{Nm}=1} \subset \mathbb{A}_E^{\mathrm{Nm}=1}$. Now let's show discreteness. Choose v be any place of E and consider:

$$B_{\mathcal{O}_v^\times}(1,\varepsilon) = U_v \subset \mathcal{O}_v^\times \subset E_v^\times$$

be a small neighborhood of 1 with $\varepsilon < 1$. Consider:

$$U_v \times \prod_{w\neq v} \mathcal{O}_w^\times$$

be an open neighborhood in $\mathbb{A}_E$. Now that $(U_v \times \prod_{w\neq v} \mathcal{O}_w^\times) \cap E^\times = \{1\}$ by product formula. So that $E^\times \hookrightarrow \mathbb{A}_E^\times$ is discrete. It follows that $E^{\mathrm{Nm}=1} \hookrightarrow \mathbb{A}_E^{\mathrm{Nm}=1}$ is discrete.

It remains to show compactness. We need to split into two cases:

(a) If E/F are function fields, say $E = \mathbb{F}_{q^r}(Y)$ for some curve Y. Recall we have a map:

$$\mathbb{A}_E^\times \to \mathrm{Div}(Y), \quad (x_v)_v \mapsto \sum \deg(v)\cdot v(x_v)\cdot v$$

If $\prod_{v|w} \mathrm{Nm}_{E_v/F_w}(x_v) = 1$, we compute:

$$\begin{aligned}\sum_v v(x_v) &= \sum_w \sum_{v|w} \deg(v)\cdot v(x_v) = \sum_w \sum_{v|w} w(\mathrm{Nm}_{E_v/F_w}(x_v)) \\ &= \sum_w w(\prod_{v|w} \mathrm{Nm}_{E_v/F_w}(x_v)) = 0\end{aligned}$$

This shows that $\mathbb{A}_E^{\mathrm{Nm}=1}$ has image lying in $\mathrm{Div}^0(Y)$. Hence the identification:

$$E^\times\backslash\mathbb{A}_E^\times/\prod_v \mathcal{O}_{E,v}^\times \simeq \mathrm{Pic}(Y)$$

restricts to an inclusion:

$$E^{\mathrm{Nm}=1}\backslash\mathbb{A}_E^{\mathrm{Nm}=1}/(\prod_v \mathcal{O}_{E,v}^\times)^{\mathrm{Nm}=1} \hookrightarrow \mathrm{Pic}^0(Y)$$

with RHS finite. Now that the norm map is a continuous map, it follows that $(\prod_v \mathcal{O}_{E,v}^\times)^{\mathrm{Nm}=1}$ is compact by the compactness of $(\prod_v \mathcal{O}_{E,v}^\times)$. This proves the desired statement for function fields.

(b) If E/F are number fields. Note that $\mathbb{A}_E^\times \subset \mathbb{A}_{E,1}^\times$, so that:

$$E^{\mathrm{Nm}=1}\backslash \mathbb{A}_E^{\mathrm{Nm}=1}/(\prod_{v \text{ finite}} \mathcal{O}_{E,v}^\times)^{\mathrm{Nm}=1} \hookrightarrow E^\times \backslash \mathbb{A}_{E,1}^\times / \prod_{v \text{ finite}} \mathcal{O}_v^\times$$

We've seen from lecture that RHS is compact. But again we have compactness of $(\prod_{v \text{ finite}} \mathcal{O}_{E,v}^\times)^{\mathrm{Nm}=1}$, this concludes the proof in this case.

2. Following the hint, we can present a homomorphism of F-tori:

$$\phi : T \to T' = \prod_i (R_{E_i/F}\mathbb{G}_m)^{\mathrm{Nm}=1}$$

with finite kernel. This induces a map:

$$T(F)\backslash T(\mathbb{A}_F) \to \prod_i (R_{E_i/F}\mathbb{G}_m(F))^{\mathrm{Nm}=1}\backslash (R_{E_i/F}\mathbb{G}_m(\mathbb{A}_F))^{\mathrm{Nm}=1}$$

with finite kernel. Also its image in RHS must be closed since it comes from an algebraic group homomorphism ($\phi(T)$ is a closed subgroup scheme). Therefore it suffices to prove RHS is compact. But each entry in RHS is nothing but:

$$E_i^{\mathrm{Nm}=1}\backslash (\mathbb{A}_E)^{\mathrm{Nm}=1}$$

Hence it remains to prove the claim. We fix a base global field F. We need to use the equivalence of categories:

$$\{\text{algebraic tori over } F \text{ up to isogeny}\} \simeq \{\mathrm{Gal}(\overline{F}/F)\text{-mod}_{\mathrm{fd}}(\mathbb{Q})\}$$

defined by:

$$T \mapsto X_*(T) \otimes_{\mathbb{Z}} \mathbb{Q}$$

In particular, irreducible torus corresponds to irreducible Galois modules. Without loss we may assume our anisotropic torus T is irreducible, so that it cooresponds to an irreducible Gal-module, therefore there must be an open subgroup $\mathrm{Gal}(\overline{F}/E)$ acting trivially. So:

$$T \otimes_F E \simeq \mathbb{G}_m^{\times \ell}$$

up to isogeny. This induces a natural natural map of F-group schemes:

$$\phi : T \to (R_{E/F}\mathbb{G}_m)^{\times \ell}$$

with finite kernel. It remains to show the image lies in the norm one part. But:

$$X_*(R_{E/F}\mathbb{G}_m^{\mathrm{Nm}=1}) \otimes_{\mathbb{Z}} \mathbb{Q} \simeq (\mathbb{Q}[\mathrm{Gal}(E/F)]/\mathrm{triv})^\vee$$

is the regular representation taking away the trivial representation. And by definition $X_*(T)$ doesn't have a Galois fixed point so that it doesn't contain the trivial representation as a component. It follows that:

$$X_*(T) \hookrightarrow X_*((R_{E/F}\mathbb{G}_m)^{\times \ell})$$

We're therefore done.

□

Exercise B.2. *Let F be a number field.*

1. *Let $\Gamma \subset \mathbb{A}_F$ be a discrete additive subgroup containing F show $F = \Gamma$.*
2. *Let $E \subset \mathbb{A}_F$ be a finite field extension of F. Show that $R = F$. In particular F is algebraically closed in $\mathbb{A}_F$.*

Proof. 1. We wish to use the approximation theorem:

$$F + \prod_{v \text{ finite}} \mathcal{O}_v + \prod_{v|\infty} F_v = \mathbb{A}_F$$

Let Γ be a group strictly contains F, then we may assume it contains an element

$$x \in (\prod_{v \text{ finite}} \mathcal{O}_v + \prod_{v|\infty} F_v) \backslash \mathcal{O}_F$$

To deduce a contradiction, we show for each basis element near 0:

$$\prod_{v \in S} U_v \times \prod_{v \notin S} \mathcal{O}_v$$

it intersects Γ at a nonzero point. Let

$$\{p_1, \cdots, p_k\} \subset \text{Spec } \mathbb{Z}$$

be the set of prime numbers below S. Then we can choose $N_1 \cdots N_k$ such that:

$$(p_1^{N_1} \cdots p_k^{N_k} x)_v \in U_v, \forall v \in S, v \nmid \infty$$

$$(p_1^{N_1} \cdots p_k^{N_k} x)_v \in \mathcal{O}_v, \forall v \notin S$$

Therefore we may assume the only failure is happening at infinity places. Let's truncate $x_\infty \in F \otimes_{\mathbb{Q}} \mathbb{R}$ at infinity places. Set $M = p_1^{N_1} \cdots p_k^{N_k}$. The goal is to find $a \in \mathcal{O}_F, b \in \mathbb{Z}$ such that $aM - bx_\infty \to 0$ in $F \otimes_{\mathbb{Q}} \mathbb{R} \simeq \mathbb{R}^d$. If $x_\infty \in F$ then we can directly make this equals 0. Otherwise, we view $x_\infty \in \mathbb{R}^d / \mathcal{O}_F \cdot M \simeq (S^1)^d$ a d-dimensional torus, and x_∞ is an irrational point on the torus. Thus it generates a dense subgroup of $(S^1)^d$ and we're done.

2. It suffices to show for each $x \in \mathbb{A}_F \backslash F$, x is not integral over F. Let's assume:

$$x = y + z \in \prod_{v \text{ finite}} \mathcal{O}_v + F \otimes_{\mathbb{Q}} \mathbb{R}$$

again by approximation. Suppose x has monic minimal polynomial f, then $f(y_v) = 0$ for all finite place v. This implies $f = 0$ has a solution in each $\mathcal{O}_v$. By Hasse principle in one variable, we know $f = 0$ actually has a solution in $\mathcal{O}_F$ and thus f has degree 1. So that $f = X - a$ for some $a \in \mathcal{O}_F$. Therefore we must have $z \in F \otimes_{\mathbb{Q}} \mathbb{R}$ equals the image of a under the diagonal embedding. It follows that $x = a \in \mathcal{O}_F \subset F$. We're done.

□

Exercise B.3. *Let F be a tamely ramified extension of $\mathbb{Q}_p$. Let $\psi : F \to \mathbb{C}^\times$ be the canonical additive character:*

$$\psi(x) = \psi_p(\mathrm{Tr}_{\mathbb{F}/\mathbb{Q}_p}(x))$$

Let dx be the Fourier self-dual Haar measure on F. Compute $\int_{\mathcal{O}_F} dx$.

Proof. Let e, f be the ramification index and residue class degree of $F/\mathbb{Q}_p$, so that $q = p^f$. We compute:

$$\begin{aligned}\hat{\hat{f}}(x) &= \int_{\xi \in F^*} \hat{f}(\xi)\xi(x)d\xi = \int_{\xi \in F^*} (\int_{y \in F} f(y)\xi(y)dy)\xi(x)d\xi \\ &= \int_{\xi \in F^*} \int_{y \in F} f(y)\xi(x+y)dydx\end{aligned}$$

Under the identification $F \to F^*$

$$z \mapsto (\xi_z : x \to \psi(zx))$$

The Fourier inversion formula reads:

$$f(-x) = \hat{\hat{f}}(x) = \int_{y,z \in F} f(y)\psi(z(x+y))dydz$$

Now set $f = \mathbf{1}_{\mathcal{O}_F}$ and $x = 0$:

$$1 = \int_{y \in \mathcal{O}_F, z \in F} \psi(zy)dydz$$

For fixed z, we look at

$$\int_{y \in \mathcal{O}_F} \psi(zy)dy$$

Note that $\psi(zy)$ is a character for $y \in \mathcal{O}_F$ a compact group, so the above integral will vanish for $\psi(zy)$ nonconstant. But recall:

$$\psi(zy) = \psi_p(\mathrm{Tr}_{F/\mathbb{Q}_p}(zy))$$

is constant precisely when $\mathrm{Tr}(zy) \in \mathbb{Z}_p, \forall y \in \mathcal{O}_F \iff z \in \mathfrak{D}^{-1}_{F/\mathbb{Q}_p} = (\varpi_F)^{1-e} \cdot \mathcal{O}_F$. In this case $\psi = 1$. Therefore:

$$1 = \int_{z \in (\varpi_F)^{1-e} \cdot \mathcal{O}_F} \int_{y \in \mathcal{O}_F} dydz = \mu(\varpi_F^{1-e} \cdot \mathcal{O}_F) \cdot \mu(\mathcal{O}_F)$$

For μ the measure associated to the form dx. But note that $\varpi_F^{1-e} \cdot \mathcal{O}_F \supset \mathcal{O}_F$ has index q^{e-1}, the invariance of the measure implies:

$$\mu(\varpi_F^{1-e} \cdot \mathcal{O}_F) = q^{e-1}\mu(\mathcal{O}_F)$$

It follows that:

$$1 = q^{e-1}\mu(\mathcal{O}_F) \cdot \mu(\mathcal{O}_F) \implies \int_{\mathcal{O}_F} dx = \mu(\mathcal{O}_F) = q^{\frac{1-e}{2}}$$

□

Exercise B.4. *Consider $F = \mathbb{F}_q(X)$ a function field. Recall we have a residue pairing:*

$$\mathrm{res} : \mathbb{A}_F \times \mathbb{A}_{\omega_F} \to \mathbb{F}_q$$

$$((f_v), (\theta_v)) \mapsto \sum_v \mathrm{Tr}_{\mathbb{F}_q(v)/\mathbb{F}_q}(\mathrm{res}_v(f_v\theta_v))$$

Prove that:

1. *$\omega_F = F^\perp$ under residue pairing.*
2. *Show that the residue pairing inducing an isomorphism:*

$$\omega_F \backslash \mathbb{A}_{\omega_F} / (\prod_v \omega_{\mathcal{O}_v}) \simeq \mathbb{F}_q$$

Proof. 1. Given $f \in F, \theta \in \omega_F$ the residue theorem tells us:

$$\sum_v \mathrm{Tr}_{\mathbb{F}_q(v)/\mathbb{F}_q}(\mathrm{res}_v(f\theta)) = 0$$

so that $\omega_F \subset F^\perp$. Now recall that:

$$F \backslash \mathbb{A} / \prod_v \mathcal{O}_v \simeq H^1(X, \mathcal{O}_X) \simeq \mathbb{F}_q^g$$

so in particular the group $F\backslash\mathbb{A}$ is compact, i.e.F is cocompact. So that its dual $F^\perp \hookrightarrow \mathbb{A}_{\omega_F}$ is discrete. Therefore the lemma will follow if we can prove ω_F is a maximal compact subgroup inside $\mathbb{A}_{\omega_F}$. But we saw from lecture that we can identify $\mathbb{A}_F$ with $\mathbb{A}_{\omega_F}$ by choosing some $\theta \in \omega_F$, and this will identify F with ω_F. So it suffices to show F is discrete inside $\mathbb{A}_F$, which we've already done in the first question.

2. Recall we have a fundamental exact sequence:

$$0 \to \mathcal{O}_X \to K_X \to \prod_v{}' i_{v,*}F_v/\mathcal{O}_v \to 0$$

where $i_v : \operatorname{Spec} \mathcal{O}_v \to X$ is the infinitesimal neightborhood at place v. This is a flasque resolution of the sheaf $\mathcal{O}_X$. Tensoring with canonical line bundle ω_X, we get a resolution of ω_X:

$$0 \to \omega_X \to \omega_X \otimes_{\mathcal{O}_X} K_X \to \prod_v{}' (i_{v,*}F_v/\mathcal{O}_v) \otimes_{\mathcal{O}_X} \omega_X \to 0$$

Using projection formula, the third term can be rewritten as:

$$\prod_v{}' i_{v,*}(F_v/\mathcal{O}_v \otimes_{\mathcal{O}_v} i_v^*\omega_X) = \prod_v{}' i_{v,*}(F_v/\mathcal{O}_v \otimes_{\mathcal{O}_v} \omega_{\mathcal{O}_v}) = \prod_v{}' i_{v,*}(\omega_{F_v}/\omega_{\mathcal{O}_v})$$

the second equality is because ω_v is rank one free module over $\mathcal{O}_v$, so in particular flat. Now let's look at the cohomology at first degree, this is nothing but:

$$\omega_F \backslash \prod_v{}' \omega_{F_v} / \prod_v \omega_{\mathcal{O}_v} = \omega_F \backslash \mathbb{A}_{\omega_F} / (\prod_v \omega_{\mathcal{O}_v}) \simeq H^1(X; \omega_X) \simeq \mathbb{F}_q$$

It remains to see this identification can be built from the residue Pairing. Indeed, consider:

$$\operatorname{res}(1, -) : \mathbb{A}_{\omega_F} \to \mathbb{F}_q, \quad (\theta_v) \mapsto \sum_v \operatorname{Tr}_{\mathbb{F}_q(v)/\mathbb{F}_q}(\operatorname{res}_v(\theta_v))$$

Note this map vanishes on both ω_F and $\prod_v \omega_{\mathcal{O}_v}$, so it induces a character

$$\omega_F \backslash \mathbb{A}_{\omega_F} / \prod_v \omega_{\mathcal{O}_v} \to \mathbb{F}_q$$

But the pairing is perfect so this is a nonzero map. But the target is a simple group so it must be surjective. However we've already seen the source is isomorphic to $\mathbb{F}_q$, this implies the above character is in fact an isomorphism.

□

Exercise B.5. *Let $F = \mathbb{F}_q(X)$ be a function field.*

1. *Using the duality of $\mathbb{A}_F$ and $\mathbb{A}_{\omega_F}$ to explain Serre duality:*

$$H^i(X, \mathcal{O}_X(D)) \times H^{1-i}(X, \omega_X(-D)) \to \mathbb{F}_q$$

for $i = 0, 1$.

2. *Using Poisson summation formula on $\mathbb{A}_F$ to recover the Riemann-Roch formula for $\chi(X, \mathcal{O}_X(D))$.*

Proof. 1. We always work in the derived category of sheaves over X. Let $p : X \to \{\mathrm{pt}\}$. Recall that we have

$$\mathcal{O}_X \simeq (K_X \to \prod{}' F_v/\mathcal{O}_v)$$

Tensoring with line bundle $\mathcal{O}_X(D)$, we get another two step complex:

$$\mathcal{O}_X(D) \simeq (K_X \to \prod{}' F_v/\varpi_v^{-n_v}\mathcal{O}_v)$$

where $D = \sum_v n_v \cdot v$, by projection formula. Note that $\mathcal{O}_X(D)$ is a line bundle (in particular flat), so that we can tensor termwise. Similarly from the previous question we can deduce that:

$$\omega_X(-D) \simeq (\omega_X \otimes K_X \to \prod \omega_{F_v}/\varpi^{n_v}\omega_{\mathcal{O}_v})$$

Now the goal is to show:

$$\mathsf{H}^i(Rp_*\mathcal{O}_X(D))^\vee \simeq \mathsf{H}^{1-i}(Rp_*\omega_X(-D))$$

But the left hand side is the same as:

$$\mathsf{H}^{1-i}((Rp_*\mathcal{O}_X(D))^\vee[1])$$

So it suffices to show:

$$(Rp_*\mathcal{O}_X(D))^\vee[1] \simeq Rp_*\omega_X(-D)$$

Indeed:

$$(Rp_*\mathcal{O}_X(D))^\vee[1] = (F \to \mathbb{A}_F/\prod_v \varpi^{-n_v}\mathcal{O}_v)^\vee[1] = ((\prod_v \varpi_v^{-n_v}\mathcal{O}_v)^\perp \to (\mathbb{A})^*/F^\perp)$$

But recall that under the residue pairing $\mathbb{A}^* \simeq \mathbb{A}_{\omega_F}$, we identify $F^\perp$ with ω_F, and $(\prod \varpi_v^{-n_v}\mathcal{O}_v)^\vee$ with $\varpi_v^{n_v}\omega_{\mathcal{O}_v}$ (this is the local residue pairing for DVR). So the dual complex is canonically identified with:

$$\prod_v \varpi_v^{n_v}\omega_{\mathcal{O}_v} \to \mathbb{A}_{\omega_F}/\omega_F$$

But then RHS reads:

$$\omega_F \to \prod \mathbb{A}_{\omega_F}/\varpi^{n_v}\omega_{\mathcal{O}_v}$$

and they represent the same element in the derived category.

2. Recall that Poisson summation formula reads:

$$\mathrm{vol}(F\backslash\mathbb{A}_F)\sum_F f(x) = \sum_{\theta\in\omega_F} \hat{f}(\theta)$$

Given any divisor $D = \sum_v n_v v$, we consider

$$f = \mathbf{1}_{\prod_v \varpi_v^{-n_v}\mathcal{O}_v}$$

Then the left hand side equals

$$\mathrm{vol}(F\backslash\mathbb{A}_F)\cdot q^{\dim H^0(X,\mathcal{O}_X(D))}$$

since $x\in F\cap\prod_v \varpi_v^{-n_v}\mathcal{O}_v \iff x+D\geq 0$ as a divisor. Now let's compute:

$$\hat{f}(\theta)=\int_{z\in\mathbb{A}_F} f(z)\langle z,\theta\rangle dz=\int_{z\in\prod_v \varpi_v^{-n_v}\mathcal{O}_v}\langle z,\theta\rangle dz$$

We're integrating a character over a compact group, so it will vanish unless its constant 1. Indeed, $\langle z,\theta\rangle=1$ iff $\theta\in H^0(X,\omega_X(-D))\subset F$. So that RHS equals:

$$\sum_{\theta\in H^0(X,\omega_X(-D))\subset F}\mathrm{vol}(\prod_v \varpi_v^{-n_v}\mathcal{O}_v)=q^{\dim H^0(\omega_X(-D))}\cdot\prod_v q^{n_v}=q^{\dim H^0(\omega_X(-D))+\deg D}$$

by the invariance of the measure. Now take $\log_q$ on both sides, we get:

$$\log_q(\mathrm{vol}(F\backslash\mathbb{A}_F))+\dim H^0(X,\mathcal{O}_X(D))=\dim H^0(\omega_X(-D))+\deg D$$

But by the first part, we have $\dim H^0(\omega_X(-D))=\dim H^1(\mathcal{O}_X(D))$, it follows that:

$$\chi(X,\mathcal{O}_X(D))=\deg D-\log_q(\mathrm{vol}(F\backslash\mathbb{A}_F))$$

Plug in $D=0$, we see:

$$\log_q(\mathrm{vol}(F\backslash\mathbb{A}_F))=1-g\implies\chi(X,\mathcal{O}_X(D))=\deg D-g+1$$

as required.

□

B.2 Homework 2

Let $F = \mathbb{F}_q(X)$ be a global function field. The goal of the next two questions is to deduce the functional equation for the zeta function $\zeta_X(s)$ from Tate's global functional equation for $Z(s, \chi, f)$.

Exercise B.6. *1. Choose a rational differential form θ on X. Explain how to use θ to obtain a Pontrayin self-duality for $\mathbb{A}$ and all its local fields F_v.*

2. Let $d_v x$ be the Fourier-self-dual Haar measure on F_v, and dx the Fourier self-dual Haar measure on $\mathbb{A}$ under the chose self-duality. Compute $\mathrm{vol}(\mathcal{O}_{F_v}, d_v x)$.

3. Give another proof that $F\backslash\mathbb{A}/\prod_v \mathcal{O}_v$ has q^g elements using the fact that

$$\mathrm{vol}(F\backslash\mathbb{A}_{\mathbb{F}}, dx) = 1$$

4. Let $d_v^\times x = \frac{1}{1-q_v^{-1}} \frac{d_v x}{|x|_v}$ be the normalized Haar measure on $F_v^\times$ and $d^\times x$ be the product. Derive a formula for $\mathrm{vol}(F^\times\backslash\mathbb{A}_F^{\times,1}, d^\times x)$ *in terms of point-counting on the Jacobian of X.*

Proof. 1. For each place v for F, we have $\theta_v \in \omega_{F_v}$. Thus we can view $\theta \in \mathbb{A}_{\omega_v}$. Explicitly, we have a map:

$$\theta : \mathbb{A} \to \mathbb{A}_\omega, \quad (a_v) \mapsto (a_v \theta_v)$$

This identifices $\mathbb{A}$ with $\mathbb{A}_\omega \simeq \mathbb{A}^*$. The local picture is just:

$$F_v \to \omega_{F_v}, \quad a_v \mapsto a_v \theta_v$$

With a selected character $\psi : \mathbb{F}_p \to S^1$, we get the pairing:

$$\langle -, - \rangle : F_v \times F_v \to S^1, \quad (a_v, b_v) \mapsto \psi \circ \mathrm{Tr}_{\mathbb{F}_{q(v)}/\mathbb{F}_p} \circ \mathrm{res}_v(a_v b_v \theta_v)$$

2. We still need to have Fourier inversion formula:

$$f(-x) = \int_{\xi \in F_v^*} \int_{y \in F_v} f(y)\xi(x+y) dy d\xi$$

Under the duality introduce in the first part, we have:

$$f(-x) = \int_{z \in F_v} \int_{y \in F_v} f(y)\langle z, x+y\rangle dy dz$$

Again choose $f = \mathbf{1}_{\mathcal{O}_{F_v}}$ and $x = 0$:

$$1 = \int_{z \in F_v} \int_{y \in \mathcal{O}_{F_v}} \langle z, y\rangle dy dz$$

We need to ask when is $\langle z, - \rangle : \mathcal{O}_{F_v} \to S^1$ a constant character 1. This happens when:

$$\mathrm{Tr}_{\mathbb{F}_{q(v)}/\mathbb{F}_p}(\mathrm{res}_v(zy\theta_v)) = 0 \iff z \in \varpi_v^{-\mathrm{ord}(\theta_v)} \mathcal{O}_{F_v}$$

Therefore we have:

$$1 = \int_{z \in \varpi_v^{-\mathrm{ord}(\theta_v)} \mathcal{O}_{F_v}} \int_{y \in \mathcal{O}_{F_v}} dydz = \mu(\mathcal{O}_{F_v}) \cdot q_v^{\mathrm{ord}(\theta_v)} \mu(\mathcal{O}_{F_v})$$

Hence:

$$\mathrm{vol}(\mathcal{O}_{F_v}, d_v x) = q_v^{-\frac{\mathrm{ord}(\theta_v)}{2}} = q^{-\frac{\deg(v) \cdot \mathrm{ord}(\theta_v)}{2}}$$

3. The previous part tells us:

$$\mathrm{vol}(\prod_v \mathcal{O}_{F_v}, dx) = \prod_v \mathrm{vol}(\mathcal{O}_{F_v}, d_v x) = q^{\sum_v - \frac{\deg(v) \cdot \mathrm{ord}(\theta_v)}{2}}$$

But we know that

$$\sum_v \deg(v) \cdot \mathrm{ord}(\theta_v) = \deg(K_X) = 2g - 2$$

It follows that $\mathrm{vol}(\prod_v \mathcal{O}_{F_v}, d_v x) = q^{1-g}$. Combining with the fact that $F \cap \prod_v \mathcal{O}_v = \mathcal{O}_F = \mathbb{F}_q$ has order q, and the fact $\mathrm{vol}(F \backslash \mathbb{A}_F, dx) = 1$, we deduce our result:

$$|F \backslash \mathbb{A} / \prod_v \mathcal{O}_v| = q^g$$

4. Recall we have a short exact sequence:

$$1 \to \mathcal{O}_F^\times \to \prod \mathcal{O}_v^\times \to F^\times \backslash \mathbb{A}_F^{\times,1} \to F^\times \backslash \mathbb{A}_F^{\times,1} / \prod \mathcal{O}_v^\times \simeq J_X(\mathbb{F}_q) \to 1$$

where J_X is the Jacobian for X. It follows that:

$$\mathrm{vol}(F^\times \backslash \mathbb{A}_F^\times, d^\times x) = \frac{1}{q-1} |J_X(\mathbb{F}_q)| \cdot \mathrm{vol}(\prod \mathcal{O}_v^\times, d^\times x)$$

But then:

$$\mathrm{vol}(\prod \mathcal{O}_v^\times, d^\times x) = \prod_v \mathrm{vol}(\mathcal{O}_v^\times, d_v^\times x) = \prod_v \mathrm{vol}(\mathcal{O}_v, d_v x) = q^{1-g}$$

from the previous part. It follows that:

$$\mathrm{vol}(F^\times \backslash \mathbb{A}_F^\times, d^\times x) = \frac{q^{1-g}}{q-1} |J_X(\mathbb{F}_q)|$$

□

Exercise B.7. *1. Let $f \in \mathcal{S}(\mathbb{A})$ be the characteristic function of $\prod_v \mathcal{O}_v$. Compute the Fourier transform $\hat{f}$ as a function on $\mathbb{A}$.*

2. Deduce the functional equation for $\zeta_X(s)$ using Tate's theorem $Z(s, 1, f) = Z(1 - s, 1, \hat{f})$.

Proof. 1. We compute:

$$\hat{f}(y) = \int_{x\in\mathbb{A}} f(x)\langle x,y\rangle dx = \int_{x\in\prod_v \mathcal{O}_v} \langle x,y\rangle dx = \prod_v \int_{x_v\in\mathcal{O}_v} \langle x_v,y_v\rangle d_v x$$

As we've seen in Q1, the inner integral:

$$\int_{x_v\in\mathcal{O}_v} \langle x_v,y_v\rangle d_v x = \begin{cases} \mu(\mathcal{O}_v) = q^{-\frac{\deg(v)\cdot \mathrm{ord}(\theta_v)}{2}}, & \text{if } y_v \in \varpi_v^{-\mathrm{ord}(\theta_v)}\mathcal{O}_{F_v}; \\ 0, & \text{otherwise.} \end{cases}$$

Therefore:

$$\hat{f} = q^{1-g}\cdot \mathbf{1}_{\prod \varpi_v^{-\mathrm{ord}(\theta_v)}\cdot\mathcal{O}_v}$$

2. We need to use Tate's theorem for $f = \mathbf{1}_{\prod_v \mathcal{O}_v}$:

$$Z(s,1,f) = Z(1-s,1,\hat{f})$$

This reads:

$$\int_{\mathbb{A}^\times} f(a)|a|^s d^\times a = \int_{\mathbb{A}^\times} \hat{f}(a)|a|^{1-s} d^\times a$$

$$\iff \prod_v \int_{\mathcal{O}_v\setminus\{0\}} |a_v|_v^s d_v^\times a = q^{1-g}\prod_v \int_{\varpi_v^{-\mathrm{ord}(\theta_v)}\cdot\mathcal{O}_v\setminus\{0\}} |a_v|_v^{1-s} d_v^\times a$$

Now that:

$$\begin{aligned} \int_{\mathcal{O}_v\setminus\{0\}} |a_v|_v^s d_v^\times a &= \sum_{n=0}^{\infty} \int_{\varpi_v^n\mathcal{O}_v^\times} q_v^{-ns} d_v^\times a \\ &= \mathrm{vol}(\mathcal{O}_v^\times, d_v^\times x)\sum_{n=0}^{\infty} q_v^{-ns} \\ &= \mathrm{vol}(\mathcal{O}_v^\times, d_v^\times x)\cdot \frac{1}{1-q_v^{-s}} \end{aligned}$$

Therefore LHS $= \mathrm{vol}(\prod \mathcal{O}_v^\times, d^\times x)\zeta_X(s) = q^{1-g}\zeta_X(s)$. Similarly, we compute:

$$\begin{aligned} \int_{\varpi_v^{-\mathrm{ord}(\theta_v)}\cdot\mathcal{O}_v\setminus\{0\}} |a_v|_v^{1-s} d_v^\times a &= \mathrm{vol}(\mathcal{O}_v^\times, d_v^\times x)\sum_{n=-\mathrm{ord}(\theta_v)}^{\infty} q_v^{-n(1-s)} \\ &= \mathrm{vol}(\mathcal{O}_v^\times, d_v^\times x)\frac{q_v^{\mathrm{ord}(\theta_v)(1-s)}}{1-q_v^{1-s}} \end{aligned}$$

So that:

$$\mathrm{RHS} = q^{2-2g}\zeta_X(1-s)\cdot\prod_v q_v^{\mathrm{ord}(\theta_v)(1-s)} = q^{2-2g}\zeta_X(1-s)q^{(1-s)(2g-2)}$$

We thus have:

$$q^{1-g}\zeta_X(s) = q^{2-2g}\zeta_X(1-s)q^{(1-s)(2g-2)} \iff \zeta_X(1-s) = q^{(g-1)(2s-1)}\zeta_X(s)$$

and recover the standard functional equation.

□

In the next two exercises, let F be a non-archimedean local field.

Exercise B.8. *Let $d^\times x$ be any Haar measure on $F^\times$. Let $\chi : F^\times \to \mathbb{C}$ be a continuous character. Let $f \in \mathcal{S}(F)$. Show that:*

1. *When χ is ramified, $\int_{F^\times} f(x)\chi(x)|x|^s d^\times x$ is a Laurent polynomial in q^{-s}.*
2. *When χ is unramified,*

$$\int_{F^\times} f(x)\chi(x)|x|^s d^\times x = \frac{P(q^{-s})}{1-\chi(\varpi)q^{-s}}$$

for some Laurent polynomial P and $\varpi \in \mathcal{O}$ a uniformizer.

Proof. 1. Since $f(x)$ is a finite sum of characteristic functions of open balls, we may assume:

$$f(x) = \mathbf{1}_{c+\varpi^n\mathcal{O}_F}$$

for some $c \in F, n \in \mathbb{Z}$. By the invariance property of $d^\times x$, we may also assume $c = 0, 1$ (namely we can consider the change of variable: $y = cx$ for $c \neq 0$). For $c = 1$, the above integral becomes

$$\int_{1+\varpi^n\mathcal{O}_F\backslash\{0\}} \chi(x)|x|^s d^\times x = \sum_{d=-\infty}^{\infty} q^{-ds} \int_{(1+\varpi^n\mathcal{O}_F\backslash\{0\})\cap\varpi^d\mathcal{O}_F^\times} \chi(x) d^\times x$$

Note that for $y \in \mathcal{O}_F$, I have:

$$|1+\varpi^n y| \leq |1| + |\varpi^n y| \leq 1 + q^{-n} < q^{-d}$$

for d sufficiently negative. So that for d sufficiently negative the above integral has zero contribution. Now for d sufficiently positive, there are two possibilities:

(a) If $n > 0$, so that $(1+\varpi^n\mathcal{O}_F\backslash\{0\}) \cap \varpi^d\mathcal{O}_F^\times = 0$, the integral has no contribution.

(b) If $n < 0$, so that $1+\varpi^n\mathcal{O}_F\backslash\{0\} = \varpi^n\mathcal{O}_F\backslash\{0\}$, so that:

$$(1+\varpi^n\mathcal{O}_F\backslash\{0\}) \cap \varpi^d\mathcal{O}_F^\times = 0 = \varpi^d\mathcal{O}_F^\times$$

The contribution of the integral is

$$\int_{\varpi^d\mathcal{O}_F^\times} \chi(x) d^\times x = \chi(\varpi^d) \int_{\mathcal{O}_F^\times} \chi(x) d^\times x = 0$$

by the invariance property of the measure.

Therefore we indeed get a Laurent polynomial in this case. The remaining case is $c = 0$, this computes:

$$\int_{\varpi^n\mathcal{O}_F\backslash\{0\}} \chi(x)|x|^s d^\times x = \sum_{d=-\infty}^{\infty} q^{-ds} \int_{(\varpi^n\mathcal{O}_F\backslash\{0\})\cap\varpi^d\mathcal{O}_F^\times} \chi(x) d^\times x$$

For d sufficiently positive, the inner integral is:

$$\int_{\varpi^d \mathcal{O}_F^\times} \chi(x) d^\times x = 0$$

as before. For d sufficiently negative, the intersection is again empty and we have no contribution. So again we obtain a Laurent polynomials in q^{-s}.

2. For unramified characters the discussion is almost the same. The only difference is the following two cases:

   (a) For $c = 1, n < 0$ and $d \geq N$ sufficiently positive, we need to compute:

   $$\begin{aligned}\sum_{d=N}^{\infty} q^{-ds} \int_{\varpi^d \mathcal{O}_F^\times} \chi(x) d^\times x &= \sum_{d=N}^{\infty} q^{-ds} \chi(\varpi^d) \mathrm{vol}(\mathcal{O}_F^\times, d^\times x) \\ &= \mathrm{vol}(\mathcal{O}_F^\times, d^\times x) \sum_{d=N}^{\infty} (q^{-s}\chi(\varpi))^d \\ &= \frac{\mathrm{vol}(\mathcal{O}_F^\times, d^\times x) q^{-sN} \chi(\varpi)^{dN}}{1 - q^{-s}\chi(\varpi)}\end{aligned}$$

   has our desired form.

   (b) For the case $c = 0$ and $d \geq N$ sufficiently positive, we carry essentially the same computation.

   This concludes that the result must be of the required form.

□

Exercise B.9. *Fix a nontrivial additive character $\psi : F \to \mathbb{C}^\times$, and use it to identify F with its Pontryagin dual: $y \in F$ is viewed as a character on F by $x \mapsto \psi(xy)$. Let dx be the self-dual Haar measure on F. The conductor of ψ is defined to be the maximal $\mathcal{O}$ submodule $\Lambda \subset F$ such that $\psi|_\Lambda = 1$. Now let $d^\times x = \frac{dx}{|x|}$ a Haar measure on $F^\times$. Let $\chi : F^\times \to \mathbb{C}^\times$ be a tamely ramified character (i.e. it is trivial on $\ker(\mathcal{O}^\times \to k^\times)$). Compute the gamma factor:*

$$\gamma(s, \chi, \psi) = Z(1 - s, \chi, f)/Z(s, \chi^{-1}, \hat{f})$$

in terms of s, χ, ψ.

Proof. Suppose $f = \mathbf{1}_{-1+\varpi\mathcal{O}_F}$, it has Fourier transform:

$$\begin{aligned}\hat{f}(x) &= \int_{a \in F} f(a)\psi(xa) da \\ &= \int_{-1+\varpi\mathcal{O}_F} \psi(xa) da \\ &= \int_{z \in \varpi\mathcal{O}_F} \psi(x(-1+z)) dz \\ &= \psi(x)^{-1} \int_{z \in \varpi\mathcal{O}_F} \psi(xz) dz \\ &= \psi(x)^{-1} \cdot \mathrm{vol}(\varpi\mathcal{O}_F, da) \cdot \mathbf{1}_{x\varpi \in \Lambda}\end{aligned}$$

So let's instead set $f = \psi^{-1}\text{vol}(\varpi\mathcal{O}_F, da)\mathbf{1}_{x\varpi\in\Lambda}$, by Fourier inversion formula we know:

$$\hat{f}(x) = \mathbf{1}_{-1+\varpi\mathcal{O}_F}(-x) = \mathbf{1}_{1+\varpi\mathcal{O}_F}(x)$$

This significantly simplify the denominator:

$$\begin{aligned} Z(s, \chi^{-1}, \hat{f}) &= \int_{1+\varpi\mathcal{O}_F} \chi^{-1}(a)|a|^s d^\times a \\ &= \text{vol}(1+\varpi\mathcal{O}_F, d^\times a) = \text{vol}(1+\varpi\mathcal{O}_F, da) \\ &= \text{vol}(\varpi\mathcal{O}_F, da) \end{aligned}$$

because χ is tamely and $da = d^\times a$ on $1+\varpi\mathcal{O}_F$.

Now let's deal with the numerator:

$$Z(1-s, \chi, f) = \int_{a:a\varpi\in\Lambda\setminus\{0\}} \psi(a)^{-1}\text{vol}(\varpi\mathcal{O}_F, da)|a|^{1-s}\chi(a)d^\times a$$

It follows that:

$$\gamma(s, \chi, \psi) = \int_{a:a\varpi\in\Lambda\setminus\{0\}} \psi(a)^{-1}|a|^{1-s}\chi(a)d^\times a$$

Now assume $\Lambda = \varpi^k\mathcal{O}$, so that we can break the above expression into:

$$\gamma(s, \chi, \psi) = \sum_{d=k-1}^{\infty} \int_{a:v(a)=d} \psi(a)^{-1}q^{d(s-1)}\chi(a)d^\times a$$

If $d \geq k$ then we have $\psi(a) \equiv 1$ and the inner integral becomes:

$$q^{d(s-1)} \int_{a:v(a)=d} \chi(a)d^\times a = q^{d(s-1)} \int_{u\in\mathcal{O}^\times} \chi(u\varpi^d)d^\times u = \chi(\varpi^d)q^{d(s-1)} \int_{u\in\mathcal{O}^\times} \chi(u)d^\times u = 0$$

since χ ramifies. Therefore:

$$\begin{aligned} \gamma(s, \chi, \psi) &= q^{(k-1)(s-1)} \int_{u\in\mathcal{O}^\times} \psi(u\varpi^{k-1})^{-1}\chi(u\varpi^{k-1})d^\times a \\ &= (q^{s-1}\chi(\varpi))^{k-1} \int_{u\in\mathcal{O}^\times} \psi(-u\varpi^{k-1})\chi(u)d^\times u \end{aligned}$$

Let us pick $u_1, u_2, \cdots, u_{q-1}$ represents a complete list of elements in $\mathcal{O}^\times/1+\varpi\mathcal{O}$, so we can break the above integral into:

$$\begin{aligned} \int_{u\in\mathcal{O}^\times} \psi(u\varpi^{k-1})\chi(u)d^\times u &= \sum_{i=1}^{q-1} \int_{u'\in 1+\varpi\mathcal{O}} \psi(-u_iu'\varpi^{k-1})\chi(u_iu')d^\times u' \\ &= \sum_{i=1}^{q-1} \chi(u_i) \int_{u'\in 1+\varpi\mathcal{O}} \psi(-u_i\varpi^{k-1})\psi((1-u')u_i\varpi^{k-1})\chi(u')d^\times u' \\ &= \sum_{i=1}^{q-1} \chi(u_i)\psi(-u_i\varpi^{k-1}) \int_{u'\in 1+\varpi\mathcal{O}} d^\times u' \end{aligned}$$

since $\chi|_{1+\mathcal{O}^\times} = 1, (1-u')u_i\varpi^{k-1} \in \Lambda$. As before, we see:

$$\int_{u'1+\varpi\mathcal{O}} d^\times u' = \mathrm{vol}(\varpi\mathcal{O}, da) = q^{-1}\mathrm{vol}(\mathcal{O}, da) = q^{\frac{k}{2}-1}$$

by a similar argument as in Homework 1, Q3. Putting everything together, we see:

$$\gamma(s, \chi, \psi) = q^{(s-\frac{1}{2})k-s}\chi(\varpi)^{k-1}\sum_{i=1}^{q-1}\chi(u_i)\psi(-u_i\varpi^{k-1})$$

□

B.3 Homework 3

We assume G to be a totally disconnected locally compact topological group: G has a neighborhood basis of 1 consisting of open and compact subgroups.

Exercise B.10. *1. Let $K \subset G$ be a compact open subgroup. Show that G has a neighborhood basis of 1 consisting of open and compact normal subgroups of K.*

2. Let F be a global field, and denote $\mathbb{A}_F^\infty$ be the restricted product of F_v for all non-archimedean places. Show that:

$$\mathrm{GL}_n(\mathbb{A}_F^\infty)$$

is a totally disconnected locally compact topological group. Here the topology comes from:

$$\mathrm{GL}_n(\mathbb{A}_F^\infty) \hookrightarrow (\mathbb{A}_F^\infty)^{n^2+1}$$

Proof. 1. Suppose G has a compact open neighborhood basis: $O_i, i \in I$ for some index set I. Let's consider: $O_i \cap K$. This is clearly open compact subgroup of K, hence of finite index. Altough it is not necessarily normal, we can adjust in the following way. Let g_{ij} be the (finite) complete representatives of elements in $K/O_i \cap K$, then we consider:

$$N_i = \bigcap g_{ij}(O_i \cap K)g_{ij}^{-1}$$

This is by design an open compact normal subgroup of K. Moreover, they clearly form a neighborhood basis of 1 since $O_i \cap K$ already form a neighborhood basis.

2. Consider:

$$O_{S,\underline{c}} = (\prod_{v \notin S} \mathcal{O}_v) \times (\prod_{v \in S} 1 + \varpi^{c_v}\mathcal{O}_v)$$

for S runs through all finite subsets of S, and all $\underline{c} \in \mathbb{N}^S$. These form a compact open neighborhood basis for $1 \in \mathbb{A}_F^\infty$. We consider:

$$\mathrm{GL}_n(O_{S,\underline{c}}) \hookrightarrow \mathrm{GL}_n(\mathbb{A}_F^\infty)$$

We claim these are our open and compact subgroups forming a neighborhood basis for $\mathrm{GL}_n(\mathbb{A}_F^\infty)$. Note that we have pullback diagram:

$$\begin{array}{ccc} \mathrm{GL}_n(O_{S,\underline{c}}) & \hookrightarrow & O_{S,\underline{c}}^{n^2+1} \\ \downarrow & & \downarrow \\ \mathrm{GL}_n(\mathbb{A}_F^\infty) & \hookrightarrow & (\mathbb{A}_F^\infty)^{n^2+1} \end{array}$$

$\mathrm{GL}_n(O_{S,\underline{c}})$ is open since its preimage of an open subset $O_{S,\underline{c}}^{n^2+1}$. It is compact since $\mathrm{GL}_n(\mathbb{A}_F^\infty)$ is a closed in $(\mathbb{A}_F^\infty)^{n^2+1} \implies \mathrm{GL}_n(O_{S,\underline{c}})$ is closed in $O_{S,\underline{c}}^{n^2+1}$, hence is compact. It forms a neighborhood basis since $O_{S,\underline{c}}^{n^2+1}$ form a neighborhood basis for $(1, 1, \cdots, 1) \in (\mathbb{A}_F^\infty)^{n+1}$

□

Exercise B.11. *Let V be a smooth representation of G. Let $K \subset G$ be a compact open subgroup.*

1. *Show that every vector v is locally K-finite.*
2. *Let $\widehat{K}$ be the set of isomorphism classes of finite dimensional irreducible smooth representations of K, with corresponding space V_ρ. Show that there will be canonical decomposition:*
$$V = \bigoplus_{\rho \in \widehat{K}} V_\rho \otimes M_\rho$$
3. *Show that V is admissible if and only if $\dim M_\rho < \infty$ for each $\rho \in \widehat{K}$.*

Proof. 1. From definition, we know $\mathrm{stab}_G(v)$ is open in $G \implies \mathrm{stab}_K(v)$ is open in K. Since K is compact, so that $\mathrm{stab}_K(v)$ has finite index, therefore we can write:
$$K \cdot v = (K/\mathrm{stab}_K(v)) \cdot v$$
is evidentally finite dimensional.

2. Since K has a bi-invariant Haar measure, an averaging argument shows that K-representations are semisimple. Therefore we know V can be written as direct sum of irreducible K-modules. But the first part implies that each irreducible module must be finite dimensional, hence we get the desired result.

3. We claim that for each $K' \subset K$ compact open subgroup of K, suppose we have a decomposition:
$$V = \bigoplus_{\rho \in \widehat{K}} V_\rho \otimes M_\rho$$
and
$$V = \bigoplus_{\eta \in \widehat{K'}} V'_\eta \otimes M'_\eta$$
Then $\dim M_\rho < \infty, \forall \rho \implies \dim M'_\eta < \infty, \forall \eta$. Indeed we have:
$$\dim M'_\eta = \sum_{\rho \in \widehat{K}} \mathrm{mult}(\eta, \rho) \dim M_\rho$$
where $\mathrm{mult}(\eta, \rho)$ is the multiplicity of representation η appearing in ρ. Therefore it is finite dimensional iff η appears in only finitely many K-types. But these are precisely those appearing in: $\mathrm{Ind}_{K'}^K \eta$, which is finite-dimensional since K' has finite index inside K. This proves the claim. Now if $\dim M_\rho < \infty$ for each ρ, we know that for each $K' \subset K$ compact open subgroup, we have:
$$\dim V^{K'} = \dim M_{\mathrm{triv}_{K'}} < \infty$$

from the claim. Now these K' form a neighborhood basis for 1, this checks that V is admissible. Conversely, assume there exists some ρ with $\dim M_\rho = \infty$. Look at K-representation V_ρ, pick any nonzero $v \in V_\rho$ and set $K' = \mathrm{stab}_K(v)$ is another compact open subgroup. So that:

$$\dim V^{K'} \geq \dim M_\rho = \infty$$

This means V is not admissible.

□

Exercise B.12. *1. Show that the smooth dual $V^\vee$ admits a canonical decomposition of K-representations:*

$$V^\vee = \bigoplus_{\rho \in \widehat{K}} V_\rho^* \otimes M_\rho^*$$

2. Show that if V is admissible, then the natural map $V \to (V^\vee)^\vee$ is an isomorphism.

Proof. 1. There is a natural map:

$$\bigoplus V_\rho^* \otimes M_\rho^* \hookrightarrow V^*$$

$$v^* \otimes m^* \mapsto (v^* \otimes m^*) \circ \mathrm{pr}_\rho$$

for $\rho : V \to V_\rho \otimes M_\rho$ the projection map. This map is injective because of linear algebra reason. We first need to see it has image lies in the image of smooth vectors. It suffices to check for the pure tensors $v^* \otimes m^* \in V_\rho^* \otimes M_\rho^*$. Pick a basis $v_1, \cdots, v_n$ for V_ρ, choose:

$$K' = \bigcap \mathrm{stab}_K(v_i)$$

So V_ρ has trivial K'-action. Therefore K' fixes the vector $v^* \otimes m^*$ by design so that it is a smooth vector. On the other hand, given: $f \in V^*$ a smooth vector. Suppose it is fixed by $K' \subset K$. Then we know f actually descends to a linear functional on the coinvariant:

$$f : V_{K'} \to \mathbb{C}$$

But now we have a diagram:

$$\begin{array}{ccccc} (V_{K'})^* & \twoheadleftarrow & V_{\mathrm{triv}}^* \otimes M_{\mathrm{triv}}^* & \twoheadleftarrow & \bigoplus_{\rho \in \widehat{K'}} V_\eta^* \otimes M_\eta^* \\ \downarrow & & & & \uparrow \simeq \\ V^* & \xleftarrow{\quad\phi\quad} & & & \bigoplus_{\rho \in \widehat{K}} V_\rho^* \otimes M_\rho^* \end{array}$$

Therefore the surjectivity of the top arrows implies that f is in the image of the natural map ϕ. We are therefore done.

2. With the previous description, we know the natural map $V \to (V^\vee)^\vee$ has form (view as K-representations):
$$\bigoplus_{\rho \in \widehat{K}} V_\rho \otimes M_\rho \to \bigoplus_{\rho \in \widehat{K}} V_\rho^{**} \otimes M_\rho^{**}$$
This is an isomorphism since both V_ρ and M_ρ are finite dimensional.

□

Exercise B.13. *Let F be a non-archimedean local field, $G = \mathrm{GL}_n(F)$, $P \subset G$ a parabolic subgroup with Levi quotient L. Let π be an admissible smooth representation of L. Show that the parabolic induction: $\mathrm{Ind}_P^G(\pi)$ is an admissible representation of G.*

Proof. Let $\pi : L \to \mathrm{Aut}(V)$. Recall we defined:
$$\mathrm{Ind}_P^G(\pi) = \{f : G \to V \text{smooth} \mid f(pg) = \delta(p)^{\frac{1}{2}} \pi(\overline{p}) f(g)\}$$
Let K be any open compact subgroup of G, we must check $(\mathrm{Ind}_P^G(\pi))^K$ is finite dimensional. Note that $P\backslash G$ is compact and $K \subset G$ is open, so that:
$$P\backslash G/K$$
is in fact a finite set. Let's pick any collection of representative $g_1, \cdots, g_m$ for it. So that we have:
$$(\mathrm{Ind}_P^G(\pi))^K \to V^{\oplus m}$$
$$f \mapsto (f(g_1), \cdots, f(g_m))$$
This is injective by design. Now let $K_i = g_i K g_i^{-1} \cap P$ is a compact open subgroup inside P, then for each $h \in K_i$, we write $h = g_i k g_i^{-1}$ for $k \in K$, we see:
$$h \cdot f(g_i) = \delta(h)^{-\frac{1}{2}} f(h \cdot g_i) = \delta(h)^{-\frac{1}{2}} f(g_i k) = \delta(h)^{-\frac{1}{2}} f(g_i)$$
Therefore for any f, we must have:
$$f(g_i) \in \delta^{-\frac{1}{2}}|_{K_i} \otimes \mathrm{Hom}_{K_i}(\delta^{-\frac{1}{2}}|_{K_i}, V) \subset V$$
which is finite dimensional. It follows that:
$$(\mathrm{Ind}_P^G(\pi))^K \hookrightarrow \prod_{i=1}^m \delta^{-\frac{1}{2}}|_{K_i} \otimes \mathrm{Hom}_{K_i}(\delta^{-\frac{1}{2}}|_{K_i}, V) \subset V$$
is finite dimensional. □

B.4 Homework 4

Let F be a non-archimedean local field. Let $\mathcal{O}$ be its ring of integers and $\varpi \in \mathcal{O}$ a uniformizer.

Exercise B.14. *Let $n = n_1 + n_2 + \cdots + n_r$ be a partition of $n \in \mathbb{N}$ into positive integers, and $P \subset \mathrm{GL}_n(F)$ be hte parabolic subgroup of type $(n_1, \cdots, n_r)$. Compute the modulus character $\delta_P : P \to \mathbb{C}^\times$*

Proof. Let $L = \mathrm{GL}_{n_1}(F) \oplus \cdots \mathrm{GL}_{n_r}(F)$ be the Levi and U the unipotent radical for P, so that $LU = P$. We know:

$$\delta_P(p) = \det(\overline{p} \mid \mathfrak{u})$$

where $\overline{p}$ is the image of p under the projection $P \to L$. Note that $\mathfrak{u}$ also have decomposition:

$$\mathfrak{u} = \oplus_{1 \le i < j \le r} \mathfrak{u}_{ij}, \quad \mathfrak{u}_{\mathrm{ij}} \simeq \mathrm{Mat}_{n_i \times n_J}(F)$$

Now given $\overline{p} = \mathrm{diag}(A_1, \cdots, A_r)$, then $\overline{p}$ acts on $\mathfrak{u}_{ij}$ as:

$$\mathrm{Ad}(\overline{p})(M) = A_i M A_j^{-1}, \quad M \in \mathfrak{u}_{ij}$$

In particular $\det(\overline{p} \mid \mathfrak{u}_{ij}) = \det(A_i)\det(A_j)^{-1}$. Therefore:

$$\delta_P(p) = \prod_{1 \le i < j \le r} \det(\overline{p} \mid \mathfrak{u}_{ij}) = \prod_{i<j} \det(A_i)\det(A_j)^{-1} = \prod_{i=1}^{r} \det(A_i)^{r+1-2i}$$

where $\overline{p} = \mathrm{diag}(A_1, \cdots, A_r)$. □

Exercise B.15. *Let $G = \mathrm{PGL}_2(F)$. Suppose the residue field of F is $\mathbb{F}_q$.*

1. *Write down details how to construct a $q+1$-regular tree $\mathbb{T}$ with $\mathrm{PGL}_2(F)$-action whose vertices are in bijection with $\mathrm{PGL}_2(F)/\mathrm{PGL}_2(\mathcal{O})$. Prove that you get a tree.*

2. *Let v_1, v_2, v_1', v_2' be vertices of $\mathbb{T}$. Show that there exists a $g \in \mathrm{PGL}_2(F)$ such that*

$$gv_1 = v_1', gv_2 = v_2'$$

   if and only if $d(v_1, v_2) = d(v_1', v_2')$.

3. *Let L_1, L_2, L_3 be three $\mathcal{O}$-lattices in F^2, corresponding to vertices v_1, v_2, v_3 of $\mathbb{T}$. Show that v_2 lies on the geodesic connecting v_1, v_3 if and only if one can find F-basis e_1, e_2 of F^2 and integers $0 \le a \le b$ such that*

$$L_1 \sim \mathcal{O}e_1 + \mathcal{O}e_2, \quad L_2 \sim \mathcal{O}e_1 + \varpi^a \mathcal{O}e_2, \quad L_3 \sim \mathcal{O}e_1 + \varpi^b \mathcal{O}e_2$$

   where $\sim$ means in the same scaling class of lattices.

Proof. 1. It will be helpful to have a parametrization for $\mathrm{PGL}_2(F)/\mathrm{PGL}_2(\mathcal{O})$. Note that:

$$\mathrm{PGL}_2(F)/\mathrm{PGL}_2(\mathcal{O}) \simeq (\mathrm{GL}_2(F)/\mathrm{GL}_2(\mathcal{O}))/F^\times$$

And we have a surjection:

$$N_n(F)\varpi^{\mathbb{Z}^2} \to \mathrm{GL}_2(F)/\mathrm{GL}_2(\mathcal{O})$$

by Iwasawa decomposition. More concretely, we know every $\mathcal{O}$-lattice takes form:

$$\begin{pmatrix} 1 & a \\ & 1 \end{pmatrix}\begin{pmatrix} \varpi^{\lambda_1} & \\ & \varpi^{\lambda_2} \end{pmatrix} = \begin{pmatrix} \varpi^{\lambda_1} & a\varpi^{\lambda_2} \\ & \varpi^{\lambda_2} \end{pmatrix}$$

where the columns are the generators of the lattice. Now we can rescale to assume $\lambda_2 = 0$, and set $\lambda = \lambda_1$. So every lattice up to rescaling takes form:

$$\mathcal{O}\begin{pmatrix} \varpi^\lambda \\ 0 \end{pmatrix} \oplus \mathcal{O}\begin{pmatrix} a \\ 1 \end{pmatrix}$$

we may also adjust a by certain integral multiple of ϖ^λ, so that:

$$a = \sum_{i>0} a_i \varpi^{\lambda - i}$$

We claim the pair (λ, a) parametrizes the lattices up to rescaling. Indeed, if $(\lambda, a), (\lambda', b)$ define the same lattice up to rescaling, then first of all we must have $\lambda = \lambda'$ by looking at their volumes. But then we need:

$$\begin{pmatrix} \varpi^\lambda & a \\ & 1 \end{pmatrix}^{-1}\begin{pmatrix} \varpi^\lambda & b \\ & 1 \end{pmatrix} = \begin{pmatrix} 1 & (b-a)\varpi^{-\lambda} \\ & 1 \end{pmatrix} \in \mathrm{GL}_2(\mathcal{O}_F)$$

This happens iff b and a represents the same element in $F/\varpi^\lambda\mathcal{O}_F$.

Let's now describe the edges. Indeed, for a vertex (lattice) given by matrix:

$$\begin{pmatrix} \varpi^\lambda & a \\ & 1 \end{pmatrix}$$

It must connect to the superlattice by adding a new vector:

$$c_1\varpi^{-1} \cdot \begin{pmatrix} \varpi^\lambda \\ 0 \end{pmatrix} + c_2\varpi^{-1}\begin{pmatrix} a \\ 1 \end{pmatrix}$$

for $[c_1 : c_2] \in \mathbb{P}^1(\mathbb{F}_q)$. In the case $c_2 \neq 0$, we may assume $c_2 = 1$ (and set $c_1 = c$), so that the new lattice will be given by matrix:

$$\begin{pmatrix} \varpi^\lambda & c\varpi^{\lambda-1} + a \cdot \varpi^{-1} \\ 0 & \varpi^{-1} \end{pmatrix} \sim \begin{pmatrix} \varpi^{\lambda+1} & c\varpi^\lambda + a \\ 0 & 1 \end{pmatrix}$$

since we have the freedom to rescale. The remaining case is $[c_1 : c_2] = [1 : 0]$, and the lattice is given by:

$$\begin{pmatrix} \varpi^{\lambda-1} & a \\ 0 & 1 \end{pmatrix}$$

Therefore, we need to connect the vertex (λ, a) with $q+1$-vertices:

$$(\lambda+1, c\varpi^\lambda + a), (\lambda-1, a) \quad c \in \mathbb{F}_q = \mathcal{O}/\varpi\mathcal{O}$$

In summary, we obtain a graph with vertex

$$\{(\lambda, a) \mid \lambda \in \mathbb{Z}, a \in F/\varpi^\lambda \mathcal{O}_F\}$$

and edges:

$$(\lambda, a) \sim (\lambda+1, c\varpi^\lambda + a), (\lambda-1, a)$$

One can easily check that $\sim$ is reflexive and therefore indeed define a graph. To see this is indeed a tree, let's arrange these vertices according their λ-values. Let's call it depth. Note that the vertices at depth λ are indexed by $F/\varpi^\lambda\mathcal{O}_F$, and each vertex (λ, a) has a unique father $(\lambda, a-1)$ and q offsprings given by elements in $F/\varpi^{\lambda+1}$ that reduce to a module ϖ^λ. From this structure we see that our graph is indeed a $q+1$-regular tree.

Finally, let's deal with the $\mathrm{PGL}_2(F)$-actions. Obviously $\mathrm{PGL}_2(F)$ will act transitively on the vertices. The difficulty for proving the group action preserves edges is that our parametrization of vertices is refering to the relative position of a vertex with the "origin" $(0, 0)$. An alternative way to specify the edges is that:

$$[L_1] \sim [L_2] \iff \varpi L_1 \subsetneq L_2 \subsetneq L_1 \text{ for certain representatives } L_1, L_2$$

This is equivalent to our previous description since essentially what we did to construct to the adjacent vertices for $L_1 = \begin{pmatrix} \varpi^\lambda & a \\ & 1 \end{pmatrix}$ is by picking a nonzero element in $L_1/\varpi L_1$, and add this vector to the lattice ϖL_1 to obtain a sublattice for L_1, which actually exhausts all possible choices of L_2. Now with this description, the existence of $\mathrm{PGL}_2(F)$ action is easy. Indeed:

$$[L_1] \sim [L_2] \iff \varpi L_1 \subsetneq L_2 \subsetneq L_1 \iff \varpi(gL_1) \subsetneq gL_2 \subsetneq gL_1 \iff g[L_1] \sim g[L_2]$$

We're therefore done.

2. Since $g \in \mathrm{PGL}_2(F)$ acts as automorphism, so the direction $\implies$ is clear. Conversely, assume $d(v_1, v_2) = d(v_1', v_2')$. We can pick g_1, g_2 such that:

$$tv_1 = t'v_1' = 1$$

Now that $d(1, tv_2) = d(1, t'v_2')$. Suppose I can find h such that $h \in \mathrm{GL}_2(\mathcal{O}_F)$ such that: $htv_2 = t'v_2'$, then I can set: $g = t'^{-1}ht$, then:

$$gv_1 = t'^{-1}htv_1 = t'^{-1}h1 = t'^{-1}1 = v_1'$$

$$gv_2 = t'^{-1}htv_2 = v_2'$$

and we're done. Therefore, it remains to show if $d(1, v_2) = d(1, v_2') = d$, then there exists $g \in \mathrm{GL}_2(\mathcal{O}_F)$ such that $gv_2 = v_2'$. Fix $d > 0$, let V_d be the vertices of distance d from 1. Consider the action map:

$$\mathrm{GL}_2(\mathcal{O}_F) \to V_d, \quad g \mapsto g \cdot \begin{pmatrix} \varpi^d & \\ & 1 \end{pmatrix}$$

We know $|V_d| = (q+1)q^{d-1}$. To see this map is surjective, we only need to bound the index of the stablizer. Indeed the stablizer is:

$$\mathrm{stab} = \mathrm{GL}_2(\mathcal{O}_F) \cap \begin{pmatrix} \varpi^d & \\ & 1 \end{pmatrix} \mathrm{GL}_2(\mathcal{O}_F) \begin{pmatrix} \varpi^{-d} & \\ & 1 \end{pmatrix} = \{A = (a_{ij}) \mid \mathrm{GL}_2(\mathcal{O}_F) \mid \varpi^d \mid a_{12}\}$$

Consider the natural map:

$$\mathrm{GL}_2(\mathcal{O}_F) \to \mathrm{GL}_2(\mathcal{O}_F/\varpi^d)$$

Then $\mathrm{GL}_2(\mathcal{O}_F)/\mathrm{stab}$ will project to:

$$\mathrm{GL}_2(\mathcal{O}_F/\varpi^d)/B(\mathcal{O}_F/\varpi^d) \simeq \mathbb{P}^1(\mathcal{O}_F/\varpi^d)$$

where B denotes the lower triangular Borel. In particular, we see that:

$$|\mathrm{GL}_2(\mathcal{O}_F)/\mathrm{stab}| \geq |\mathbb{P}^1(\mathcal{O}_F/\varpi^d)| = (q+1)q^{d-1}$$

This proves the action map must be surjective.

3. Assume we're given v_1, v_2, v_3 with $v_2 \in [v_1, v_3]$. We wish to find the explicit basis presentation as claimed. Thanks to the second part, we may assume $v_1 = \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}, v_3 = \begin{pmatrix} \varpi^d & \\ & 1 \end{pmatrix}$, by apply $g \in \mathrm{PGL}_2(F)$ where $d = d(v_1, v_3)$. Now very explicitly the path connecting v_1, v_3 consisting of vertices of form:

$$\begin{pmatrix} \varpi^i & \\ & 1 \end{pmatrix}$$

for $0 \leq i \leq d$, which is precisely what we want. Conversely, suppose they three lattices are of given form, then we can find g such that:

$$ge_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, ge_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Then it will send $v_1 \mapsto \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}, v_2 \mapsto \begin{pmatrix} \varpi^a & \\ & 1 \end{pmatrix}, v_3 \mapsto \begin{pmatrix} \varpi^b & \\ & 1 \end{pmatrix}$. So that v_2 obviously lies on $[v_1, v_3]$.

□

Exercise B.16. *Continuing from the previous exercise.*

1. *A ray in the tree $\mathbb{T}$ is a path starting from a vertex and going on infinitely in one direction without backtracking. Two rays r, r' are equivalent if $r \cap r'$ is also a ray. Show that $\mathrm{PGL}_2(F)$ acts transitively on the set of equivalence classes of rays, and compute a stablizer.*

2. *A line in the tree $\mathbb{T}$ is a path that is infinite in both directions. Show that $\mathrm{PGL}_2(F)$ acts transitively on the set of lines, and compute the stablizer. What can you say about vertices lying on the same line.*

Proof. 1. Let's use $\mathrm{GL}_2(F)$ action for simplicity. Let $r = (v_0, v_1, \cdots), r' = (w_0, w_1, \cdots)$ be two rays. For the sake of transitivity, we may assume:

$$v_i = \begin{pmatrix} \varpi^i & \\ & 1 \end{pmatrix}$$

gives a specific ray. There exists g_0 such that $g_0(v_0) = w_0$. We will inductively construct g_i such that $g_i([v_0, v_i]) = [w_0, w_i]$ and that:

$$g_i \equiv g_{i-1} (\mathrm{mod}\ \varpi^{i-1})$$

We alreay have our g_0, and suppose we've already constructed g_{i-1}. Let's pick any g_i' with the property that $g_i'([v_0, v_i] = [w_0, w_i])$, and we will adjust from this. Note that g_i' must also send $[v_0, v_{i-1}] \to [w_0, w_{i-1}]$. Now let $K_i = \pi_i^{-1}(B(\mathcal{O}_F/\varpi^i))$, where $\pi_i : \mathrm{GL}_2(\mathcal{O}_F) \to \mathrm{GL}_2(\mathcal{O}_F/\varpi^i)$, to be the stablizer of the line $[v_0, v_i]$. Then we know:

$$g_{i-1}^{-1} g_i' \in K_{i-1}$$

But I have the freedom to adjust my g_i by an element in K_i. Note that: $\mathrm{GL}_2(F)/K_i = \mathbb{P}^1(\mathcal{O}/\varpi^i)$ has cardinality $q^{i-1}(q+1)$, $\mathrm{GL}_2(F)/K_{i-1}$ has cardinality $q^{i-2}(q+1)$, so that

$$K_i \subset K_{i-1}$$

has index q. On the other hand we have:

$$\begin{pmatrix} 1 & c\varpi^{i-1} \\ & 1 \end{pmatrix}, \quad c \in \mathbb{F}_q$$

they represent distincts elements in K_{i-1}/K_i. So they are in fact a complete collection of representives of K_{i-1}/K_i. In particular, we can find $h_i \in K_i$ such that:

$$g_{i-1}^{-1} g_i' h_i = \begin{pmatrix} 1 & c_i\varpi^{i-1} \\ & 1 \end{pmatrix}$$

for some c_i. Let's then set $g_i = g_i' h_i$, we then deduce:

$$g_i \equiv g_{i-1} (\mathrm{mod}\ \varpi^{i-1})$$

This concludes the induction process. Now that the sequence g_i will converge to some $g \in \mathrm{GL}_2(F)$ by our construction, and g will certainly map r to r'. Therfore, we in fact prove that $\mathrm{PGL}_2(F)$ acts transitively on all rays (not only on the equivalence classes).

Now let's compute the stablizer at $r = (v_i)$ with the choice v_i as in the previous discussion. Indeed the stablizer will consists of all $g \in \mathrm{GL}_2(F)$ such that $g[v_i, \infty] = [v_j, \infty]$. Let's first deal with the case $i = j = 0$. Then we see that

$$g \in \bigcap K_i = F^\times \cdot B(\mathcal{O}_F)$$

the lower triangular matrices with coefficients in $\mathcal{O}_F$. Now the matrices that will send $[v_i, \infty] \to [v_j, \infty]$ consists of

$$\begin{pmatrix} \varpi^j & \\ & 1 \end{pmatrix} F^\times \cdot B(\mathcal{O}_F) \begin{pmatrix} \varpi^{-i} & \\ & 1 \end{pmatrix}$$

Altogether they will generate $B(F)$ the Borel with coefficient in F. Therefore the stablizer will be $B_{\mathrm{PGL}_2}(F)$ the Borel of $\mathrm{PGL}_2(F)$.

2. Let's pick a distinguished line:

$$\ell = (\cdots, v_{-1}, v_0, v_1, \cdots)$$

with $v_i = \begin{pmatrix} \varpi^i & \\ & 1 \end{pmatrix}$, with $\ell' = (\cdots, w_{-1}, w_0, w_1, \cdots)$ be another line. We must find $g \in \mathrm{GL}_2(F)$ sending $\ell \mapsto \ell'$. As before we inductively choose g_i maps $[v_{-i}, v_i]$ to $[w_{-i}, w_i]$, such that:

$$g_i \equiv g_{i-1} (\mathrm{mod}\ \varpi^{i-1})$$

The proof is very similar to the previous argument. Indeed we can pick g_0, suppose we've constructed g_{i-1}. Note that now the stablizer of the line $[v_{-i}, v_i]$ is:

$$T_d = \begin{pmatrix} \varpi^{-d} & \\ & 1 \end{pmatrix} \begin{pmatrix} \mathcal{O}_F & \varpi^{2d}\mathcal{O}_F \\ \mathcal{O}_F & \mathcal{O}_F \end{pmatrix} \begin{pmatrix} \varpi^d & \\ & 1 \end{pmatrix} = \begin{pmatrix} \mathcal{O}_F & \varpi^d\mathcal{O}_F \\ \varpi^d\mathcal{O}_F & \mathcal{O}_F \end{pmatrix}$$

It is evident that T_d is the preimage of $T(\mathcal{O}_F/\pi^d)$ under: $\pi_d : \mathrm{GL}_2(\mathcal{O}_F) \to \mathrm{GL}_2(\mathcal{O}_F/\pi^d)$. Again we pick any g_i sending $[v_{-i}, v_i] \mapsto [w_{-i}, w_i]$, then we know:

$$g_{i-1}^{-1} g_i' \in T_{i-1}$$

As before we can find $h_i \in T_i$ such that

$$g_{i-1}^{-1} g_i' h_i = \begin{pmatrix} 1 & \alpha_i \varpi^{i-1} \\ \beta_i \varpi^{i-1} & 1 \end{pmatrix}$$

and we simply take $g_i = g_i' h_i$ will do the job. Now as before g_i will converge to some element g, and it will send $\ell \mapsto \ell'$. This shows the transitivity. Now let's compute the stablizer for the distinguished line ℓ. A very similar argument as above will show that the stablizer will be:

$$T_{\mathrm{PGL}_2}(F)$$

is a maximal torus inside $\mathrm{PGL}_2(F)$. As we see, a choice of a line is equivalent to a choice of a maximal torus inside $\mathrm{PGL}_2(F)$, and a vertex on the line corresponds to a point in the cocharacter lattice for this torus.

□

Exercise B.17. *Let $G = \mathrm{GL}_2(F)$, and $K = \mathrm{GL}_2(\mathcal{O})$. Let $\chi : F^\times \to \mathbb{C}^\times$ be an unramified character. Consider:*

$$V = \mathrm{Ind}_B^G(\chi, 1)$$

be the normalized induction.

1. *Write down a nonzero K-fixed vector in V.*
2. *Let dg be the Haar measure on G such that $\int_K dg = 1$, and use it to identify the Hecke algebra $\mathcal{H}_K$ with functions $C_c(K \setminus G/K)$. Let $\phi \in \mathcal{H}_K$ be the characteristic function of $K \begin{pmatrix} \varpi & 0 \\ 0 & 1 \end{pmatrix} K$. Compute the eigenvalue of ϕ on V^K.*

Proof. 1. A K-fixed vector is given by:

$$f(\begin{pmatrix} a & b \\ & d \end{pmatrix} \cdot k) = |\frac{a}{d}|_F^{\frac{1}{2}} \chi(a)$$

for $\begin{pmatrix} a & b \\ & d \end{pmatrix} \in B(F), k \in K$. Note this is indeed a K-fixed vector by design, and it is well-defined by Iwasawa decomposition and δ_B, χ are both unramified (this is checked in lecture).

2. Let's start by computing a complete representatives for the cosets:

$$K \begin{pmatrix} \varpi & \\ & 1 \end{pmatrix} K/K$$

It suffices to compute for:

$$\begin{pmatrix} \varpi^{-1} & \\ & 1 \end{pmatrix} K \begin{pmatrix} \varpi & \\ & 1 \end{pmatrix} K/K \simeq \begin{pmatrix} \varpi^{-1} & \\ & 1 \end{pmatrix} K \begin{pmatrix} \varpi & \\ & 1 \end{pmatrix} /K \cap \begin{pmatrix} \varpi^{-1} & \\ & 1 \end{pmatrix} K \begin{pmatrix} \varpi & \\ & 1 \end{pmatrix}$$

$$\simeq K/\begin{pmatrix} \varpi & \\ & 1 \end{pmatrix} K \begin{pmatrix} \varpi^{-1} & \\ & 1 \end{pmatrix} \cap K \simeq \mathbb{P}^1(\mathcal{O}_F/\varpi)$$

as we seen in previous questions. Therefore, there are $q+1$ representative:

$$\begin{pmatrix} 1 & b \\ & 1 \end{pmatrix} \cdot \begin{pmatrix} \varpi & \\ & 1 \end{pmatrix} = \begin{pmatrix} \varpi & b \\ & 1 \end{pmatrix}, \begin{pmatrix} & 1 \\ 1 & \end{pmatrix} \cdot \begin{pmatrix} \varpi & \\ & 1 \end{pmatrix} = \begin{pmatrix} & 1 \\ \varpi & \end{pmatrix}$$

for each $b \in \mathbb{F}_q$. Now we're ready to compute the eigenvalue.

$$(\phi \cdot f)(g) = \int_{h \in G} \phi(h) f(gh) dh = \int_{h \in K \begin{pmatrix} \varpi & \\ & 1 \end{pmatrix} K} f(gh) dh$$

$$= \sum_{h \in K \begin{pmatrix} \varpi & \\ & 1 \end{pmatrix} /K} f(gh)$$

Let's specialize to $g = 1$, we get:

$$(\phi \cdot f)(1) = \sum_{b \in \mathbb{F}_q} f(\begin{pmatrix} \varpi & b \\ & 1 \end{pmatrix}) + f(\begin{pmatrix} & 1 \\ \varpi & \end{pmatrix})$$

But $f(\begin{pmatrix} \varpi & b \\ & 1 \end{pmatrix}) = q^{-\frac{1}{2}}\chi(\varpi)$, and:

$$f(\begin{pmatrix} & 1 \\ \varpi & \end{pmatrix}) = f(\begin{pmatrix} 1 & \\ & \varpi \end{pmatrix} \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}) = q^{\frac{1}{2}}$$

To sum up we see

$$(\phi \cdot f)(1) = q^{\frac{1}{2}}(\chi(\varpi) + 1)$$

which is the eigenvalue.

□

B.5 Homework 5

Exercise B.18. *Let F be a non-archimedean local field. Let $\mathcal{O}$ be its ring of integers and $\varpi \in \mathcal{O}$ be a uniformizer. Let $G = \mathrm{GL}_2(F), K = \mathrm{GL}_2(\mathcal{O})$. Let dg be the Haar measure on G such that $\int_K dg = 1$, and use it to identify the Hecke algebra $\mathcal{H}_K$ with $C_c(K \setminus G/K)$. Then the Satake transform takes form:*

$$S : \mathcal{H}_K \to \mathbb{C}[u, u^{-1}, v, v^{-1}]$$

$$Sf = \sum_{a,b \in \mathbb{Z}} |\varpi^{a-b}|^{\frac{1}{2}} \left(\int_F f\left(\begin{pmatrix} \varpi^a & \\ & \varpi^b \end{pmatrix} \begin{pmatrix} 1 & x \\ & 1 \end{pmatrix} \right) dx \right) u^a v^b$$

where dx is chosen such that $\int_{\mathcal{O}} dx = 1$. Let us verify Satake isomorphism explicitly in this case.

1. *Let $T_{m,n}$ be the characteristic function of $K \begin{pmatrix} \varpi^m & \\ & \varpi^n \end{pmatrix} K$. Compute $S(T_{m,n})$. Conclude that the image of S lands in symmetric Laurent polynomials in u and v.*
2. *Show that S is a ring homomorphism.*
3. *Show that S induces an isomorphism:*

$$\mathcal{H}_K \simeq \mathbb{C}[x, y, y^{-1}]$$

Proof. 1. Let's assume F has residue field $\mathbb{F}_q$. Fix a, b, x. Let's try to reduce

$$\begin{pmatrix} \varpi^a & \\ & \varpi^b \end{pmatrix} \begin{pmatrix} 1 & x \\ & 1 \end{pmatrix}$$

in the form of Cartan decomposition. If $v(x) \geq 0$, then it already has the form of Cartan decomposition. Now let $x = u\varpi^{-n}$ for some $n > 0$. Then:

$$\begin{aligned} \begin{pmatrix} \varpi^a & \\ & \varpi^b \end{pmatrix} \begin{pmatrix} 1 & u\varpi^{-n} \\ & 1 \end{pmatrix} &= \begin{pmatrix} \varpi^a & \\ & \varpi^b \end{pmatrix} \begin{pmatrix} 0 & u\varpi^{-n} \\ -u^{-1}\varpi^n & 1 \end{pmatrix} \begin{pmatrix} 1 & \\ u^{-1}\varpi^n & 1 \end{pmatrix} \\ &= \begin{pmatrix} \varpi^a & \\ & \varpi^b \end{pmatrix} \begin{pmatrix} \varpi^{-n} & \\ u^{-1} & \varpi^n \end{pmatrix} \begin{pmatrix} u & \\ & -u^{-1} \end{pmatrix} \begin{pmatrix} & 1 \\ 1 & \end{pmatrix} \begin{pmatrix} 1 & \\ u^{-1}\varpi^n & 1 \end{pmatrix} \end{aligned}$$

We can discard all terms in K, so we need to look at:

$$\begin{pmatrix} \varpi^a & \\ & \varpi^b \end{pmatrix} \begin{pmatrix} \varpi^{-n} & \\ u^{-1} & \varpi^n \end{pmatrix} = \begin{pmatrix} \varpi^{a-n} & \\ u^{-1}\varpi^b & \varpi^{b+n} \end{pmatrix} = \begin{pmatrix} 1 & \\ u^{-1}\varpi^{b-a+n} & 1 \end{pmatrix} \begin{pmatrix} \varpi^{a-n} & \\ & \varpi^{b+n} \end{pmatrix}$$

This is valid when $b - a + n \geq 0$. On the other hand, we can write:

$$\begin{pmatrix} \varpi^a & \\ & \varpi^b \end{pmatrix} \begin{pmatrix} 1 & u\varpi^{-n} \\ & 1 \end{pmatrix} = \begin{pmatrix} \varpi^a & u\varpi^{a-n} \\ & \varpi^b \end{pmatrix} = \begin{pmatrix} 1 & u\varpi^{a-b-n} \\ & 1 \end{pmatrix} \begin{pmatrix} \varpi^a & \\ & \varpi^b \end{pmatrix}$$

This is valid when $b-a+n\leq 0$. Therefore, we have:

$$\begin{aligned}\alpha(a,b,m,n)&=\int_F T_{m,n}(\begin{pmatrix}\varpi^a & \\ & \varpi^b\end{pmatrix}\begin{pmatrix}1 & x\\ & 1\end{pmatrix})dx\\ &=\sum_{k=-\infty}^{\max\{0,a-b\}} T_{m,n}(\begin{pmatrix}\varpi^a & \\ & \varpi^b\end{pmatrix})\mathrm{vol}(\varpi^{-k}\mathcal{O}^\times)+\\ &\quad\sum_{k=\max\{0,a-b\}+1}^{\infty} T_{m,n}(\begin{pmatrix}\varpi^{a-k} & \\ & \varpi^{b+k}\end{pmatrix})\mathrm{vol}(\varpi^{-k}\mathcal{O}^\times)\end{aligned}$$

Without loss let's assume $m\geq n$. Then:

$$\alpha(a,b,m,n)=\begin{cases}q^{a-b}, & \text{if } a\geq b, \{a,b\}=\{m,n\};\\ 1, & \text{if } a\leq b, \{a,b\}=\{m,n\};\\ q^{a-n}(1-\frac{1}{q}), & \text{if } n<b\leq a<m, a+b=m+n;\\ q^{a-n}(1-\frac{1}{q}), & \text{if } n<a\leq b<m, a+b=m+n;\\ 0, & \text{otherwise.}\end{cases}$$

We're now ready to compute ($m\geq n$):

$$\begin{aligned}ST_{m,n}&=\sum_{a,b\in\mathbb{Z}} q^{-\frac{1}{2}(a-b)}\alpha(a,b,m,n)u^av^b\\ &=q^{-\frac{1}{2}(m-n)}q^{m-n}u^mv^n+q^{-\frac{1}{2}(n-m)}u^nv^m+\\ &\sum_{a=n+1}^{m-1} q^{-\frac{1}{2}(a-(m+n-a))}q^{a-n}(1-\frac{1}{q})u^av^{m+n-a}\\ &=q^{\frac{1}{2}(m-n)}(u^mv^n+u^nv^m)+(1-\frac{1}{q})\sum_{a=n+1}^{m-1} q^{\frac{1}{2}(m-n)}u^av^{m+n-a}\\ &=q^{\frac{1}{2}(m-n)}(u^mv^n+u^nv^m)+q^{\frac{1}{2}(m-n)}(1-\frac{1}{q})\sum_{a=n+1}^{m-1} u^av^{m+n-a}\\ &=q^{\frac{1}{2}(m-n)}(u^mv^n+u^nv^m+(1-\frac{1}{q})\frac{u^{n+1}v^m-u^mv^{n+1}}{v-u})\end{aligned}$$

This is clearly symmetric.

2. Note that abstractly, $\mathcal{H}_K\simeq\mathbb{C}[T_{1,0},T_{1,1}^{\pm 1}]$. This is because:

$$T_{m,0}=T_{1,0}^{*m}+\text{lower terms}$$

$$T_{m,n}*T_{1,1}=T_{m+1,n+1}$$

Since we already know $\mathcal{H}_K$ is commutative, let's define:

$$S' : \mathcal{H}_K \to \mathbb{C}[u, v, u^{-1}, v^{-1}]$$

$$S'(T_{1,0}) = q^{\frac{1}{2}}(u+v), \quad S'(T_{1,1}) = uv$$

This is a ring homomorphism. Our goal is of course to show $S = S'$. First:

$$S(T_{m,n}) = S(T_{m-n,0})(uv)^n$$

by our explicit formula above. Similar relation holds for S' by design. So it suffices to show:

$$S(T_{m,0}) = S'(T_{m,0})$$

Let's prove by induction. The case for $m = 1$ is for design. Suppose the statement is true for $\leq m-1$. Then, note that:

$$T_{m-1,0} * T_{1,0} = T_{m,0} + qT_{m-1,1}$$

by the geometry of Bruhat-Tit's tree. Therefore:

$$S(T_{m,0}) = S(T_{m-1,0} * T_{1,0} - qT_{m-1,1}) = S(T_{m-1,0} * T_{1,0}) - qS(T_{m-1,1})$$

and similarly:

$$S'(T_{m,0}) = S'(T_{m-1,0} * T_{1,0}) - qS'(T_{m-1,1}) = S'(T_{m-1,0}) * S'(T_{1,0}) - qS'(T_{m-1,1})$$

By induction hypothesis:

$$S'(T_{m,0}) = S(T_{m-1,0}) * S(T_{1,0}) - qS(T_{m-1,1})$$

Therefore it suffices to show:

$$S(T_{m-1} * T_{1,0}) = S(T_{m-1}) * S(T_{1,0})$$

Let's replace $m-1$ by m to simplify the notation. Then:

$$\begin{aligned}
\text{LHS} &= q^{\frac{m}{2}}(u^m + (1 - \frac{1}{q})u^{m-1}v + \cdots + (1 - \frac{1}{q})uv^{m-1} + v^m) \\
&\quad + quvq^{\frac{m-2}{2}}(u^{m-2} + (1 - \frac{1}{q})u^{m-3}v + \cdots + v^{m-2}) \\
&= q^{\frac{m}{2}}(u^m + (2 - \frac{1}{q})u^{m-1}v + (2 - \frac{2}{q})u^{m-2}v^2 + \cdots + v^m)
\end{aligned}$$

On the other hand:

$$\begin{aligned}
\text{RHS} &= q^{\frac{m-1}{2}}(u^{m-1} + (1 - \frac{1}{q})u^{m-2}v + (1 - \frac{1}{q})u^{m-3}v^2 + \cdots + v^{m-1}) \cdot q^{\frac{1}{2}}(u+v) \\
&= q^{\frac{m}{2}}(u^m + (2 - \frac{1}{q})u^{m-1}v + (2 - \frac{2}{q})u^{m-2}v^2 + \cdots + v^m)
\end{aligned}$$

which agrees with LHS. This finishes the induction process. Hence $S = S'$ is a ring homomorphism.

3. From previous part we see abstractly $\mathcal{H}_K \simeq \mathbb{C}[x, y, y^{-1}]$. To show this is obtained by S we only need to show S is injective. But this is clear from our formula for S' (note that $uv, u+v$ are algebraically independent).

□

Exercise B.19. *Let G be a Lie group with Lie algebra $\mathfrak{g}$, and $K \subset G$ a maximal compact subgroup. Recall a $(\mathfrak{g}, K)$-module V is a $\mathbb{C}$-vector space with an action of $\mathfrak{g}$ and K satisfying:*

- *Any vector in V is K-finite.*
- *For any $X \in \mathfrak{k} = \mathrm{Lie}K$ and $v \in V$, $Xv = \frac{d}{dt}|_{t=0}(e^{tX}v)$*
- *For any $X \in \mathfrak{g}$, $k \in K$:*

$$k \circ X \circ k^{-1} = \mathrm{Ad}(k)X \in \mathrm{End}(V)$$

Show that:

1. *When K is connected, the third condition is redundant.*
2. *Give an example of $(\mathfrak{g}, K, V)$ satisfying the first two conditions but not the third.*

Proof. 1. It's equivalent to show:

$$k \circ X = \mathrm{Ad}(k)X \circ k$$

Let's fix $Y \in \mathrm{Lie}K, v \in V$, and set:

$$\gamma(t) = e^{tY} \circ X(v) - (\mathrm{Ad}(e^{tY})X \circ e^{tY})(v) \in V$$

Note that $\gamma(t)$ has image lies in $W = KXv + \mathfrak{g} \cdot K \cdot v$ a finite dimensional subspace, by the first property. Our goal is to show $\gamma(t) \equiv 0 \in W$. Since γ is analytic, it suffices to show:

$$\frac{d^k}{dt^k}|_{t=0}\gamma(t) = 0$$

Note the second condition essentially reads:

$$Y \cdot_{\mathfrak{g}} v = Y \cdot_{\mathfrak{k}} v$$

for $Y \in \mathfrak{k}$. Therefore we can suppress the subscript and just to denote as Yv. Therefore, we compute:

$$\frac{d^k}{dt^k}|_{t=0}e^{tY}(Xv) = Y^k \cdot_{\mathfrak{k}} Xv = Y^k Xv$$

And recall that:

$$\mathrm{Ad}(e^{tY})X = e^{t \cdot \mathrm{ad}_Y} X$$

Hence:

$$\frac{d^k}{dt^k}|_{t=0}(\mathrm{Ad}(e^{tY})X \circ e^{tY})(v) = \sum_{k_1+k_2=k} \binom{k}{k_1} \mathrm{ad}_Y^{k_1}(X)Y^{k_2}v$$

Therefore, we must check:

$$Y^k X = \sum_{k_1+k_2=k} \binom{k}{k_1} \mathrm{ad}_Y^{k_1}(X) Y^{k_2} \in \mathfrak{g}$$

This can be done by induction. For $k = 1$ this is clear. Now assume we know the case for $k-1$, we can write:

$$\begin{aligned} Y^k X &= Y(Y^{k-1}X) = \mathrm{ad}_Y(Y^{k-1}X) + (Y^{k-1}X)Y \\ &= \sum_{k_1+k_2=k-1} \binom{k}{k_1} \mathrm{ad}_Y^{k_1+1}(X) X^{k_2} + \mathrm{ad}_Y^{k_1}(X) Y^{k_2+1} \\ &= \sum_{k_1+k_2=k} \binom{k}{k_1} \mathrm{ad}_Y^{k_1}(X) Y^{k_2} \end{aligned}$$

as desired. Now vary Y, t, we see $k \circ X \circ k^{-1} = \mathrm{Ad}(k)X$ for $k \in U$ in some open neighborhood of $1 \in K$. But now U generates K because K is connected, so the formula holds for all $k \in K$.

2. Let $G = \mathbb{R}, K = \{\pm 1\}, \mathfrak{g} = \mathbb{R}$. Let $V = \mathbb{C}^2$ and:

$$\mathfrak{g} = \mathbb{R} \to \mathfrak{gl}(V), \quad 1 \mapsto \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}$$

We let $-1 \in K \mapsto \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$. Note condition (i) is clear and condition (ii) is vacuous since $\mathfrak{k} = 0$. But now let's choose $X = 1 \in \mathfrak{g}, k = -1 \in K$, then:

$$k \circ X \circ k^{-1} = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix} \begin{pmatrix} & 1 \\ 1 & \end{pmatrix} \begin{pmatrix} 1 & \\ & -1 \end{pmatrix} = \begin{pmatrix} & -1 \\ -1 & \end{pmatrix}$$

But:

$$\mathrm{Ad}(k)(X) = X = \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}$$

This contradicts the third condition.

□

Exercise B.20. *Let $\widetilde{V}$ be the space of smooth $\mathbb{C}$-valued functions on $\mathbb{R}^2 \setminus 0$ with the standard action of $\mathrm{SL}_2(\mathbb{R})$:*

$$(g \cdot f)(x) = f(g^{-1}x)$$

Let $\chi : \mathbb{R}^\times \to \mathbb{C}^\times$ be a continuous character, and $\widetilde{V}_\chi$ be the subspace of $f \in \widetilde{V}$ such that:

$$f(tx) = \chi(t) f(x)$$

Let $V_\chi \subset \widetilde{V}_\chi$ be the subspace of K-finite vectors.

1. *Describe V_χ as a $(\mathfrak{sl}_2, \mathrm{SO}(2))$-module explicitly.*

2. *Compute the action of* $\Omega = \frac{1}{2}h^2 + ef + fe$ *on* V_χ.

Proof. 1. The good news here is that SO(2) is actually a torus, and its irreducibles are classified by $n \in \mathbb{Z}$:

$$e^{i\theta} \mapsto e^{in\theta}$$

Therefore we can decompose:

$$V_\chi = \bigoplus_{n\in\mathbb{Z}} V_{\chi,n}$$

$$V_{\chi,n} = \{f : \mathbb{R}^2 \setminus 0 \to \mathbb{C} \mid f(re^{i\theta}) = \chi(r)e^{in\theta}f(1), r \in \mathbb{R}, \theta \in \mathbb{R}/2\pi\mathbb{Z}\}$$

This is one-dimensional if:

$$\chi(-1) = e^{in\theta} = (-1)^n$$

and zero dimensional otherwise. Therefore, if we set:

$$v_{\chi,n} : \mathbb{R}^2 \setminus 0 \to \mathbb{C}, \quad re^{i\theta} \mapsto \chi(r)e^{in\theta}$$

for $\chi(-1) = (-1)^n$, then:

$$V_\chi = \begin{cases} \bigoplus_{n\in 2\mathbb{Z}} \mathbb{C} \cdot v_{\chi,n}, & \text{if } \chi(-1) = 1; \\ \bigoplus_{n\in 2\mathbb{Z}+1} \mathbb{C} \cdot v_{\chi,n}, & \text{if } \chi(-1) = -1. \end{cases}$$

Now let's describe the $\mathfrak{sl}_2$-action. Note we have exponential map:

$$\exp : \mathfrak{so}_2 = \mathbb{R} \cdot J \to \mathrm{SO}(2)$$

$$\theta J \mapsto \exp(\theta J) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \simeq e^{-i\theta}$$

where $J = \begin{pmatrix} & 1 \\ -1 & \end{pmatrix}$ is a basis element for $\mathfrak{so}_2$. Note than:

$$J \cdot v_{\chi,n}(x) = \frac{d}{d\theta}|_{\theta=0}\exp(\theta J)v_{\chi,n}(x) = \frac{d}{d\theta}|_{\theta=0}v_{\chi,n}(e^{i\theta}x) = inv_{\chi,n}(x)$$

This motivates me to choose $h = -iJ = \begin{pmatrix} & -i \\ i & \end{pmatrix}$, so that $hv_{\chi,n} = nv_{\chi,n}$. We wish to extend this into a $\mathfrak{sl}_2$-triple. Indeed we have seen in class:

$$e = \frac{1}{2}\begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}, \quad f = \frac{1}{2}\begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}$$

Now let's compute their actions. Note for any element in the Lie algebra $X \in \mathfrak{g}$, we have:

$$\begin{aligned}(X \cdot f)(u, v) &= \frac{d}{dt}|_{t=0}f(e^{-tX}(u, v)) \\ &= -(\partial_x f(u, v), \partial_y f(u, v)) \cdot X \begin{pmatrix} u \\ v \end{pmatrix}\end{aligned}$$

As a sanity check, let's plug in $X = h$, then we find:

$$(X \cdot f)(u,v) = -(\partial_x f(u,v), \partial_y f(u,v)) \begin{pmatrix} -iv \\ iu \end{pmatrix} \implies X = iy\partial_x - ix\partial_y$$

But recall that:

$$\partial_x = \cos\theta \partial_r - \frac{\sin\theta}{r}\partial_\theta$$

$$\partial_y = \sin\theta \partial_r + \frac{\cos\theta}{r}\partial_\theta$$

So that

$$h = iy\partial_x - ix\partial_y = -i\partial_\theta$$

and indeed:

$$-i\partial_\theta v_{\chi,n} = n v_{\chi_n}$$

Let's now compute the action of $X = e$:

$$(e \cdot f)(u,v) = -\frac{1}{2}(\partial_x f(u,v), \partial_y f(u,v)) \cdot \begin{pmatrix} u+iv \\ iu - v \end{pmatrix}$$

$$e = -\frac{1}{2}(x+iy)\partial_x - \frac{1}{2}(ix-y)\partial_y = -\frac{r}{2}e^{2i\theta}\partial_r - \frac{i}{2}e^{2i\theta}\partial_\theta$$

To explicitly compute e-action, we must explicitly write:

$$\chi(r) = |r|^s \mathsf{sgn}(r)^\varepsilon$$

for some $s \in \mathbb{C}, \varepsilon \in \{0,1\}$. So that for $r > 0$:

$$e \cdot v_{\chi,n} = (-\frac{1}{2}e^{2i\theta})(r\partial r + i\partial\theta)(r^s e^{in\theta}) = \frac{n-s}{2} r^s e^{i(n+2\theta)} = \frac{n-s}{2} v_{\chi,n+2}$$

Similarly:

$$f \cdot v_{\chi,n} = \frac{-n-s}{2} v_{\chi,n-2}$$

To check we did the correct computation, let's compute the scalar of $ef - fe$ on $v_{\chi,n}$:

$$ef - fe = \frac{-n-s}{2} \cdot \frac{n-2-s}{2} - \frac{n-s}{2}\frac{-n-2-s}{2} = n$$

which is precisely the scalar that h acts by. Therefore in summary, we have two possibilites:

(a) If $\chi(r) = |r|^s$, we have:

$$V_\chi = \bigoplus_{n \in 2\mathbb{Z}} \mathbb{C} \cdot v_{\chi,n}$$

with

$$h \cdot v_{\chi,n} = n v_{\chi,n}; \quad e \cdot v_{\chi,n} = \frac{n-s}{2} v_{\chi,n+2}; \quad f \cdot v_{\chi,n} = \frac{-n-s}{2} v_{\chi,n-2}$$

(b) If $\chi(r) = |r|^s \mathsf{sgn}(r)$, then:

$$V_\chi = \bigoplus_{n \in 2\mathbb{Z}} \mathbb{C} \cdot v_{\chi,n}$$

with the same formula.

2. Now we need to compute the action of $\Omega = ef + fe + \frac{1}{2}h^2 = \frac{1}{2}h^2 + h + 2fe$. On $V_{\chi,n}$ it acts via:

$$\frac{1}{2}n^2 + n + 2\frac{n-s}{2}\frac{-n-2-s}{2} = \frac{1}{2}(n^2 + 2n + (s+2+n)(s-n))$$

$$= \frac{1}{2}(n^2 + 2n + s^2 + 2s - 2n - n^2) = \frac{1}{2}(s^2 + 2s)$$

Notably it only depends on s.

□

B.6 Homework 6

I consulted the fact that $\mathrm{GL}_n(\mathbb{A}_{F,f})$ will help to classify all algebraic vector bundles on Spec $\mathcal{O}_F$. I also consulted the Steinitz's trick.

Exercise B.21. *We give a geometric interpretation of the double coset groupoid:*

$$\mathrm{GL}_n(\mathbb{Q}) \setminus \mathrm{GL}_n(\mathbb{A})/K_\infty \mathrm{GL}_n(\widehat{\mathbb{Z}}) \tag{1}$$

where $K_\infty \subset \mathrm{GL}_n(\mathbb{R})$ is a maximal compact subgroup.

1. *A rank n* **metrized lattice** *is a pair: (Λ, q) where Λ is a free rank n $\mathbb{Z}$-module and q is a positive definite quadratic form on $\Lambda_\mathbb{R} = \Lambda \otimes \mathbb{R}$. Let $\mathcal{L}_n$ be the groupoid of rank n metrized lattices (with isometries as isomorphisms). Show that:*

   $$\mathcal{L}_n \simeq \mathrm{GL}_n(\mathbb{Z}) \setminus \mathrm{GL}_n(\mathbb{R})/K_\infty$$

   as groupoid.

2. *Construct an equivalence of groupoids between $\mathcal{L}_n$ and 1.*

3. *Let N be a positive integer, and let:*

   $$K(N) = \ker(\mathrm{GL}_N(\widehat{\mathbb{Z}}) \to \mathrm{GL}_n(\mathbb{Z}/N\mathbb{Z})), \quad \Gamma(N) = \ker(\mathrm{GL}_n(\mathbb{Z}) \to \mathrm{GL}_n(\mathbb{Z}/N\mathbb{Z}))$$

   Give a similar interpretation of the double cosets:

   $$\Gamma(N) \setminus \mathrm{GL}_n(\mathbb{R})/K_\infty, \quad \mathrm{GL}_n(\mathbb{Q}) \setminus \mathrm{GL}_n(\mathbb{A})/K_\infty K(N)$$

Proof. 1. Fix a base standard metrized lattice (Λ_0, q_0). Let's construct a functor:

$$F : \mathrm{GL}_n(\mathbb{Z}) \setminus \mathrm{GL}_n(\mathbb{R})/K_\infty \to \mathcal{L}_n$$

On objects it maps:

$$g \mapsto (\Lambda_0, g \cdot q_0)$$

On morphisms, given $g \in \mathrm{GL}_n(\mathbb{R}), h \in \mathrm{GL}_n(\mathbb{Z}), k \in K_\infty$:

$$e_{g,(h,k)} : g \to hgk^{-1}$$

I map it to:

$$F(e_{g,(h,k)}) : (\Lambda_0, g \cdot q_0) \to (\Lambda_0, hgk^{-1}q_0) = (\Lambda_0, hg \cdot q_0)$$
$$v \mapsto hv$$

Note this is indeed an isometry. So our functor F is well-defined. We must check:

(a) F is essentially surjective. Consider any metrized lattice (Λ, q), and pick an isomorphism of inner product spaces:

$$(\Lambda_\mathbb{R}, q) \xrightarrow{\sim} (\mathbb{R}^n, q_0)$$

The image of Λ under this isomorphism is a rank n lattice in $\mathbb{R}^n$, so it equals $g^{-1} \cdot \Lambda_0$ for some $g \in \mathrm{GL}_n(\mathbb{R})$. It follows that:

$$(\Lambda, q) \simeq (g^{-1}\Lambda_0, q_0) \simeq (\Lambda_0, g \cdot q_0)$$

This proves the essential surjection.

(b) F is fully faithful. Let $g, g' \in \mathrm{GL}_n(\mathbb{R})$, we must show:

$$\{(h,k) \mid g' = hgk^{-1}\} = \mathrm{Hom}(g,g') \simeq \mathrm{Hom}((\Lambda_0, g\cdot q_0),(\Lambda_0, g'\cdot q_0))$$

Indeed, if we have a

$$\phi : (\Lambda_0, g\cdot q_0) \to (\Lambda_0, g'\cdot q_0)$$

an isometry, it must be given by some $g'kg^{-1}$ for $k \in K_\infty$. Moreover, we need this to preserve the lattice Λ_0, so that the condition reads:

$$g'kg^{-1} \in \mathrm{GL}_n(\mathbb{Z})$$

In summary, k defines a isometry iff there exists $h \in \mathrm{GL}_n(\mathbb{Z})$ with $g' = hgk^{-1}$. This proves the fully faithfulness.

2. Let $\mathcal{L}'_n$ be the set rigidified metrized lattices containing a third entry of $\mathbb{Q}$-basis of $\Lambda_\mathbb{Q}$. Then

$$\mathrm{GL}_n(\mathbb{Q}) \setminus \mathcal{L}'_n \simeq \mathcal{L}_n$$

as quotient groupoids. Therefore to prove the statement it suffices to construct a bijection:

$$\Phi : \mathrm{GL}_n(\mathbb{A})/K_\infty \mathrm{GL}_n(\widehat{\mathbb{Z}}) \to \mathcal{L}'_n$$

We still have the frame object:

$$(\Lambda_0 = \mathbb{Z}^n, q_0, \underline{e}_0) \in \mathcal{L}'_n$$

and we will let:

$$\Phi(g_f, g_\infty) = (g_f \cdot \widehat{\mathbb{Z}}^n \cap \mathbb{Q}^n, g_\infty \cdot q_0, \underline{e}_0)$$

where $g_f \in \mathrm{GL}_n(\mathbb{A}_f).g_\infty \in \mathrm{GL}_n(\mathbb{R})$. This makes sense because:

(a) $N\widehat{\mathbb{Z}}^n \subset g_f \cdot \widehat{\mathbb{Z}}^n \subset \frac{1}{N}\widehat{\mathbb{Z}}^n$ for some large integer N, so $g_f \cdot \widehat{\mathbb{Z}}^n \cap \mathbb{Q}^n$ is a lattice;

(b) Our lattice always lie in $\mathbb{Q}^n$, so we always have a prefered choice of $\mathbb{Q}$-basis element.

Let's check:

(a) Φ is surjective. Start from any rigidified

$$(\Lambda, q, \underline{e})$$

By applying certain $h \in \mathrm{GL}_n(\mathbb{Q})$ such tht $h(\underline{e}) = \underline{e}_0$, we get:

$$(\Lambda, q, \underline{e}) = (\lambda', q', \underline{e}_0)$$

but the later can obviously be obtained because we have freedom in the first two entries.

(b) Φ is injective. Since Φ comes from a group action, it suffices to check the stablizer at $(\Lambda_0, q_0, \underline{e_0})$ is $K_\infty \cdot \mathrm{GL}_n(\widehat{\mathbb{Z}})$. But this is clear since we need to preserve both Λ_0 and q_0.

3. Consider $\mathcal{L}_n(N)$ the groupoid of metrize lattice together with an ordered basis for the modulo N module:

$$(\Lambda, q, \underline{v})$$

where $\underline{v}$ is a basis for the $\mathbb{Z}/N\mathbb{Z}$-module $\Lambda/N\mathbb{Z}$, where the morphisms are isometries preserving the ordered basis. Then a similar argument as in (1) shows that the functor:

$$\Gamma(N) \setminus \mathrm{GL}_n(\mathbb{R})/K_\infty \to \mathcal{L}_n(N)$$

$$g \mapsto (\Lambda_0, g \cdot q_0, \underline{v}_0)$$

On the other hand, we can consider rigidified version $\mathcal{L}'_n(N)$ and we have a bijection:

$$\mathrm{GL}_n(\mathbb{A})/K_\infty K(N) \to \mathcal{L}'_n(N)$$

$$(g_f, g_\infty) \mapsto (g_f \cdot \widehat{\mathbb{Z}}^n, g_\infty \cdot q_0, g_f \cdot \underline{v}_0, \underline{e}_0)$$

The formula $g \cdot \underline{v_0}$ makes sense, since we have an isomorphism of $\mathbb{Z}/N\mathbb{Z}$-modules:

$$(g_f \cdot \widehat{\mathbb{Z}}^n \cap \mathbb{Q}^n)/N(g_f \cdot \widehat{\mathbb{Z}}^n \cap \mathbb{Q}^n) \simeq g_f \cdot \widehat{\mathbb{Z}}^n/N(g_f \cdot \widehat{\mathbb{Z}}^n)$$

via inclusion (this is clearly injective, and both have cardinality $(N+1)^n$). Now $g \cdot \underline{v_0}$ makes sense in RHS, so can be transported to LHS. Now quotienting out by $\mathrm{GL}_n(\mathbb{Q})$, we obtain:

$$\Gamma(N) \setminus \mathrm{GL}_n(\mathbb{R})/K_\infty \simeq \mathcal{L}_n(N) \simeq \mathrm{GL}_n(\mathbb{Q}) \setminus \mathrm{GL}_n(\mathbb{A})/K_\infty K(N)$$

And by construction these isomorphisms are compatible with the natural inclusion:

$$\mathrm{GL}_n(\mathbb{R}) \hookrightarrow \mathrm{GL}_n(\mathbb{A})$$

□

Exercise B.22. *In this exercise, we replace $\mathbb{Q}$ with an arbitrary number field F. Let $K_\infty \subset \mathrm{GL}_n(F_\infty)$ be a maximal compact subgroup.*

1. *Define a notion of metrized $\mathcal{O}_F$-lattices such that the groupoid $\mathcal{L}_n(\mathcal{O}_F)$ of rank n metrized $\mathcal{O}_F$-lattices is equivalent to the groupoid*

$$\mathrm{GL}_n(F) \setminus \mathrm{GL}_n(\mathbb{A}_F)/K_\infty \mathrm{GL}_n(\widehat{\mathcal{O}}_F) \tag{2}$$

2. *Give a lattice interpretation of the groupoid*

$$\mathrm{GL}_n(\mathcal{O}_F) \setminus \mathrm{GL}_n(F_\infty)/K_\infty \tag{3}$$

3. *Define a map from double coset 2 to the class group $\mathrm{Cl}(F)$, such that 3 is the kernel.*

Proof. 1. Let's denote:

$$F_\infty = F \otimes_{\mathbb{Q}} \mathbb{R} = \mathcal{O}_F \otimes_{\mathbb{Z}} \mathbb{R} = \prod_{v \text{ real}} \mathbb{R} \times \prod_{v \text{ complex}} \mathbb{C}$$

Let's define a rank n metrized $\mathcal{O}_F$-lattice to be a pair (Λ, q) where Λ is a rank r vector bundle on Spec $\mathcal{O}_F$ (equivalently, a projective module over $\mathcal{O}_F$ that $\Lambda \otimes_{\mathcal{O}_F} F \simeq F^n$) and q an positive definitie quadratic form on $\Lambda \otimes_{\mathcal{O}_F} \mathbb{R}$, namely choose a quadratic form on each $\mathbb{R}$ and $\mathbb{C}$. We again have a fram lattice:

$$(\mathcal{O}_F^n, q_0)$$

where q_0 is the quadratic form defining K_∞. A rigidified metrized lattice is again by adding a F-basis for $\Lambda_F = \Lambda \otimes_{\mathcal{O}_F} F$. We again have an action isomorphism:

$$\mathrm{GL}_n(\mathbb{A}_F)/K_\infty \mathrm{GL}_n(\widehat{\mathcal{O}_F}) \to \mathcal{L}_n(\mathcal{O}_F)'$$

Note here

$$g_f \cdot \widehat{\mathcal{O}_F}^n \cap F^n$$

actually exhausts all vector bundles over Spec $\mathcal{O}_F$.

$$(g_f, g_\infty) \mapsto (g_f \cdot \widehat{\mathcal{O}_F}^n \cap F^n, g_\infty \cdot q, \underline{e}_0)$$

As in the previous question, this descends to an isomorphism of groupoids:

$$\mathrm{GL}_n(F) \setminus \mathrm{GL}_n(\mathbb{A}_F)/K_\infty \mathrm{GL}_n(\widehat{\mathcal{O}_F}) \simeq \mathcal{L}_n(\mathcal{O}_F)$$

2. We still have a groupoid morphism:

$$\mathrm{GL}_n(\mathcal{O}_F) \setminus \mathrm{GL}_n(F_\infty)/K_\infty \to \mathcal{L}_n(\mathcal{O}_F)$$

$$g \mapsto (\mathcal{O}_F^n, g \cdot q)$$

The image is precisely the sub-groupoid of all metrized lattices with *trivial* vector bundle. Call this sub-groupoid $\mathcal{L}_n(\mathcal{O}_F)_0$

3. The determinat map $\det : \mathrm{GL}_n(\mathbb{A}_F) \to \mathbb{A}^\times$ induces a surjection:

$$\det : \mathrm{GL}_n(F) \setminus \mathrm{GL}_n(\mathbb{A}_F)/K_\infty \mathrm{GL}_n(\widehat{\mathcal{O}}_F) \to F^\times \setminus \mathbb{A}_F^\times / \prod_{v \text{ fin}} \mathcal{O}_v^\times \det(K_\infty)$$

Now $\det(K_\infty) \subset F_\infty^\times$, so I can further quotient and get a map:

$$\mathrm{GL}_n(F) \setminus \mathrm{GL}_n(\mathbb{A}_F)/K_\infty \mathrm{GL}_n(\widehat{\mathcal{O}}_F) \to F^\times \setminus \mathbb{A}_F^\times / \prod_{v \text{ fin}} \mathcal{O}_v^\times \cdot F_\infty^\times \simeq \mathrm{Cl}(F)$$

It is clear from definition that 3 lies in the kernel. On the other hand, by Steinitz's trick every rank n-vector bundle on Spec $\mathcal{O}_F$ is isomorphic to:

$$\mathcal{O}_F^{\oplus n-1} \oplus \mathfrak{a}$$

for some fractional ideal $\mathfrak{a}$. It follows that $\mathcal{L}_n(\mathcal{O}_F)_0$ has index h_F inside $\mathcal{L}_n(\mathcal{O}_F)$. Comparing cardinality shows that the kernel is precisely 3.

□

Exercise B.23. *In this exercise we construct a subset of* $\mathrm{GL}_n(\mathbb{R})$ *that maps surjectively to the double coset 1. Let* (Λ, q) *be a metrized lattice. A basis* $v_1, \cdots, v_n$ *for* (Λ, q) *is called is called* **optimal** *if it is inductively produced using the following algorithm: take* v_1 *to be the shortest vector in* $\Lambda - \{0\}$*; let* $\overline{v_2}, \cdots, \overline{v_n}$ *be an optimal basis for the rank* $n-1$ *lattice* $\overline{\Lambda}$ *that is the image of the orthogonal projection* $\mathrm{pr}_1 : \Lambda \to v_1^{\perp}$*. Then for* $i \geq 2$*, let* v_i *be a shortest vector in* $\mathrm{pr}_1^{-1}(\overline{v_i})$

1. *Let* $v_1, \cdots, v_n$ *be an optimal basis for* (Λ, q)*, and let* $v_1' = v_1, v_2', \cdots, v_n'$ *be the result of Gram-Schmidt process (without normalizing each vector). Write:*

$$(v_1', v_2', \cdots, v_n') = (v_1, \cdots, v_n)U$$

where $U \in \mathrm{GL}_n(\mathbb{R})$ *is an upper triangular unipotent matrix. Show that there exists a constant* $c > 0$ *(depending on* n *but not on* (Λ, q)*) such that all entries of* U *are bounded by* c *in absolute value.*

2. *Continuing with the notation from the previous part, show that:*

$$|v_2'| \geq \frac{\sqrt{3}}{2}|v_1'|, \quad |v_3'| \geq \frac{\sqrt{3}}{2}|v_2'|, \cdots$$

3. *Show that the following subset of upper triangular matrices in* $\mathrm{GL}_n(\mathbb{R})$ *maps surjectively onto the double coset* $\mathrm{GL}_n(\mathbb{Z}) \setminus \mathrm{GL}_n(\mathbb{R})/K_\infty$*:*

$$\{g = ua \mid u = (u_{ij}) \text{ unip.}, |u_{ij}| \leq c \text{ for } i > j, a = \mathrm{diag}(a_1, \cdots, a_n), a_i > 0, \frac{a_i}{a_{i+1}} \geq \frac{\sqrt{3}}{2}\}$$

4. *When* $n = 2$*, compute* c *explicitly in the first problem and draw the image of the subset on the upper half plane (identified with the quotient* $\mathrm{GL}_2(\mathbb{R})/\mathbb{R}^{>0}K_\infty$ *)*

Proof. 1. This is a linear algebra exercise. Recall the Gram-Schmidt process reads:

$$(v_1', v_2, \cdots, v_n) = (v_1, \cdots, v_n)U^1$$
$$(v_1', v_2', \cdots, v_n) = (v_1', v_2, \cdots, v_n)U^2$$
$$\cdots$$
$$(v_1', v_2', \cdots, v_n') = (v_1', \cdots, v_{n-1}', v_n)U_n$$

where each U_i is a unipotent matrix concentrated in column i, and $U = U^1U^2 \cdots U^n$. We claim that each entry of U^i has absolute value $\leq \frac{1}{2}$. To prove this we proceed by induction. For each entry not on the first row we're done from induction (observe that $\overline{v_i}' = v_i'$ for all $i \geq 2$). The entry at position $(1, j)$ equals:

$$\frac{\langle v_1, v_j \rangle}{\langle v_1, v_1 \rangle}$$

which has absolute value $\leq \frac{1}{2}$, since by assumption we must have:

$$|v_j + nv_1| \geq |v_j|$$

for all $n \in \mathbb{Z}$.

2. We only need to prove the first equality, and the rest will follow from induction and the observation that $\overline{v_i'} = v_i'$ for $i \geq 2$. But then we can write:

$$v_2 = v_2' + \lambda v_1, \quad \lambda \in [-\frac{1}{2}, \frac{1}{2}]$$

We need $v_1 = v_1'$ to have minimal length:

$$|v_2' + \lambda v_1|^2 \geq |v_1|^2 \implies |v_2'|^2 \geq (1-\lambda^2)|v_1|^2 = \frac{3}{4}|v_1|^2 \implies |v_2'| \geq \frac{\sqrt{3}}{2}|v_1'|$$

3. Given any element g in the double coset, we can pick $v = (v_1, \cdots, v_n)$ an optimal basis for $g^{-1} \cdot \mathbb{Z}^n$, so that:

$$(\mathbb{Z}^n, gq_0) \sim (g^{-1}\mathbb{Z}^n, q_0) \simeq (v\mathbb{Z}^n, q_0) \simeq (\mathbb{Z}^n, v^{-1}q_0)$$

where we also view v as a matrix, which differs from g^{-1} by some element in $\mathrm{GL}_n(\mathbb{Z})$. Now that by previous question, we know there exists u, a satisfying the corresponding condition such that: vua is an orthonormal matrix, with respect to the standard inner product q_0. So we have:

$$(\mathbb{Z}^n, q_0) \sim (\mathbb{Z}^n, vuaq_0) \implies (\mathbb{Z}^n, uaq_0) \sim (\mathbb{Z}^n, v^{-1}q_0) \sim (\mathbb{Z}^n, gq_0)$$

In other words, ua and g represent the same element in the double coset.

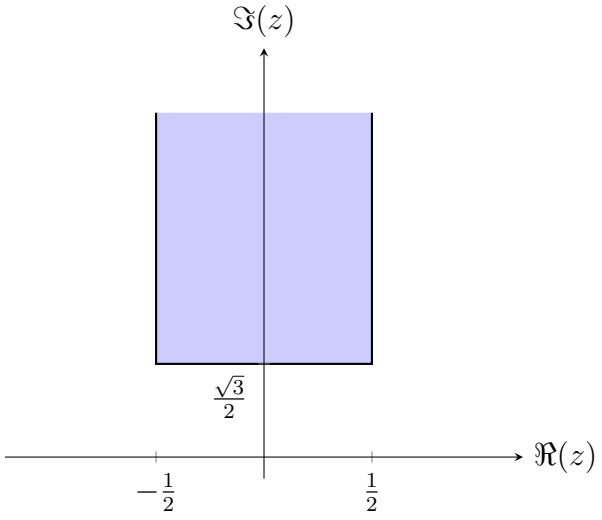
4. In this case:

$$U = \begin{pmatrix} 1 & \lambda \\ & 1 \end{pmatrix}$$

for $|\lambda| \leq \frac{1}{2}$ so that $c = \frac{1}{2}$. We identify the coset with upper-half plane via:

$$ua \mapsto ua \cdot i = \begin{pmatrix} 1 & \lambda \\ & 1 \end{pmatrix} \begin{pmatrix} a_1 & \\ & a_2 \end{pmatrix} i = \begin{pmatrix} 1 & \lambda \\ & 1 \end{pmatrix} \frac{a_1}{a_2} i = \frac{a_1}{a_2} i + \lambda$$

This corresponds to the region:



□

B.7 Homework 7

Exercise B.24. *Let G be a Lie group with Lie algebra $\mathfrak{g}$, and K a maximal compact subgroup with Lie algebra $\mathfrak{k}$. Let dk be the Haar measure on K with $\int_K dk = 1$.*

1. *Let $\mathcal{H}_K \subset C^\infty(K)dk$ be the $\mathbb{C}$-subspace of matrix coefficients (from finite-dimensional $\mathbb{C}$-representations of K) times a Haar measure dk. Show that $\mathcal{H}_K$ is an associative algebra under convolution.*

2. *Consider the $\mathbb{C}$-vector space:*

$$\mathcal{H}_{\mathfrak{g},K} = U(\mathfrak{g}_\mathbb{C}) \otimes_{U(\mathfrak{k}_\mathbb{C})} \mathcal{H}_K$$

Here in forming the tensor product, we are using the right $U(\mathfrak{k}_\mathbb{C})$-module structure on $U(\mathfrak{g}_\mathbb{C})$ be right multiplication, and the left $U(\mathfrak{k}_\mathbb{C})$-module structure on $\mathcal{H}_K$ given by the derivative of the left translation action of K on matrix coefficient. Construct a (not necessarily unital) associative algebra structure on $\mathcal{H}_{\mathfrak{g},K}$ such that each $(\mathfrak{g}, K)$-module V carries a unique $\mathcal{H}_{\mathfrak{g},K}$-module structure, with $\mathcal{H}_K$ and $U(\mathfrak{g}_\mathbb{C})$ act in the usual way coming from the $(\mathfrak{g}, K)$-module structure.

3. *Construct a fully faithful embedding*

$$(\mathfrak{g}, K)\text{-mod} \hookrightarrow \mathcal{H}_{\mathfrak{g},K}\text{-mod}$$

and characterize the image.

Proof. 1. Associativity is clear. It suffices to show it is closed under convolution. Let V, W be two (unitary) irreducible representations of K, and let $v_1, v_2 \in V$, $w_1, w_2 \in W$. We must compute:

$$m_{v_1,v_2} * m_{w_1,w_2}$$

The fundamental property we will use is the orthogonality relation. We let $g \in K$, we compute:

$$\begin{aligned} m_{v_1,v_2} * m_{w_1,w_2}(g) &= \int_{k\in K} m_{v_1,v_2}(k) m_{w_1,w_2}(k^{-1}g)dk \\ &= \int_{k\in K} m_{v_1,v_2}(k) m_{gw_1,w_2}(k^{-1})dk \\ &= \int_{k\in K} m_{v_1,v_2}(k)\overline{m_{gw_1,w_2}(k)}dk \end{aligned}$$

This vanishes in $V \not\simeq W$. Otherwise if $V \simeq W$ we know it equals:

$$\frac{1}{\dim V}\langle v_1, gw_1\rangle_V \cdot \langle v_2, w_2\rangle_V$$

which is again a matrix coefficient.

2. Let $\{X_i\}$ be a basis of $\mathfrak{g}_{\mathbb{C}}$ with dual basis $\{X^i\}$, and $f_i^j = \langle X^j, \mathrm{Ad}(-)X_i\rangle$ be the matrix coefficients of the adjoint action of K on $\mathfrak{g}_{\mathbb{C}}$. Then for $\phi \in \mathcal{H}_K$, we impose the commutation relation:

$$\phi \cdot X_i = \sum_j X_j \otimes (f_i^j \phi)$$

where $f_i^j \phi$ is pointwise multiplication. Let $V = C_c^\infty(G)$, we claim that one can realize $\mathcal{H}_{\mathfrak{g},K} \subset \mathrm{End}(V)$ as a subalgebra. Indeed, consider:

$$\alpha : \mathcal{H}_{\mathfrak{g},K} \to \mathrm{End}(V)$$

$$X \otimes \phi \mapsto (f \mapsto X(\int_K \phi(k) k \cdot f))$$

To check this is well-defined, we need to check for $Y \in U(\mathfrak{k}_{\mathbb{C}}), \phi \in \mathcal{H}_K$, $1 \otimes Y\phi$ and $Y \otimes \phi$ has the same image. In other words:

$$Y(\int_K \phi(k) k \cdot f) = \int_K Y(\phi)(k) k \cdot v$$

Now we compute for $g \in G$:

$$\begin{aligned} Y(\int_K \phi(k) k \cdot f)(g) &= \frac{d}{dt}|_{t=0} \int_K \phi(k) k \cdot f(\exp(-tY)g) \\ &= \frac{d}{dt}|_{t=0} \int_K \phi(k) f(k^{-1} \exp(-tY) g) \end{aligned}$$

and on the other hand

$$\begin{aligned} (\int_K Y(\phi)(k) k \cdot f)(g) &= \int_K \frac{d}{dt}|_{t=0} \phi(\exp(-tY)k) f(k^{-1}g) \\ &= \frac{d}{dt}|_{t=0} \int_K \phi(\exp(-tY)k) f(k^{-1}g) \\ &= \frac{d}{dt}|_{t=0} \int_{w \in K} \phi(w) f(w^{-1} \exp(-tY) g) \quad (\text{set } w = \exp(-tY)k) \end{aligned}$$

This checks our claim, and therefore our map α is indeed a map of vector spaces. Now let's check this map is in fact injective. Indeed as vector spaces we can identify:

$$\mathcal{H}_{\mathfrak{g},K} \simeq U(\mathfrak{p}_{\mathbb{C}}) \otimes \mathcal{H}_K$$

Consider $\sum D_i \otimes \phi_i$ for $D_i \in U(\mathfrak{p}_{\mathbb{C}})$ linearly independent, and $\phi_i \in \mathcal{H}_K$ linearly independent. Recall globally: $G \simeq K \times \mathfrak{p}$. From distribution theory we know there exists some function $f \in C_c^\infty(G)$ such that $D_1(f)|_K = \overline{\phi_1}$, $D_i(f)|_K = 0$ for $i \geq 2$. Now if $T = \sum D_i \phi_i$, viewed as a measure on K, then:

$$\langle T, f \rangle = \sum_i -\langle \phi_i, D_i f \rangle = -\langle \phi_1, \phi_1 \rangle \neq 0$$

This show $T \neq 0$, and hence $\sum D_i \otimes \phi_i \neq 0$. This proves the injectivity. Now it remains to show that the induced multiplication is given by the above formula. In other words:

$$(\phi \cdot X_i) \cdot v = \sum_j X_j \otimes (f_i^j \phi) \cdot v$$

for $v \in V$. But this is a formal verification:

$$\begin{aligned}(\phi \cdot X_i) \cdot v &= \int_{k \in K} \phi(k) k \cdot X_i \cdot v \\ &= \int_{k \in K} \phi(k) \mathrm{Ad}(k)(X_i) k \cdot v \\ &= \int_{k \in K} \phi(k) \sum_j f_i^j(k) X_j \cdot k \cdot v \\ &= \sum_j X_j \otimes (f_i^j \phi) \cdot v\end{aligned}$$

as desired. From this formula, we see $\mathcal{H}_{\mathfrak{g},K}$ is indeed closed under multiplication, hence is indeed a subalgebra of $\mathrm{End}(V)$.

3. Given a $(\mathfrak{g}, K)$-module V, the formula:

$$X \otimes \phi \mapsto (v \mapsto X \int_K \phi(k) k v)$$

realizes V as a $\mathcal{H}_{\mathfrak{g},K}$-module. To check this is indeed defines $\mathcal{H}_{\mathfrak{g},K}$-module, it suffices to realize that our checking for the specific choice $V = \mathbb{C}_c^\infty(G)$ being a $\mathcal{H}_{\mathfrak{g},K}$-module, only depends on the formal property that V is a $(\mathfrak{g}, K)$-module. This defines a functor:

$$(\mathfrak{g}, K)\text{-mod} \to \mathcal{H}_{\mathfrak{g},K}\text{-mod}$$

Now this morphism is fully faithful simply because morphisms on both sides are just linear maps that commute with both $\mathfrak{g}$ and K-actions (note that K-actions determines $\mathcal{H}_K$-actions). It's essential image is just those V such that each $\mathcal{H}_K$-isotypical component has finite dimensional. In class we denoted this space as:

$$\mathcal{H}_{\mathfrak{g},K}\text{-mod}_{\mathrm{sm}}$$

□

In the following exercises, let X be a smooth projective and geometrically connected curve over a finite field $\mathbb{F}_q$ with function field F.

Exercise B.25. *Let $X = \mathbb{P}^1_{\mathbb{F}_q}$ and $G = \mathrm{GL}_n$ with $n > 1$, and $K = \prod_{x \in |X|} G(\mathcal{O}_x) \subset G(\mathbb{A}_F)$. Show that there is no nonzero cusp form on $G(F) \setminus G(\mathbb{A}_F)/K$.*

Proof. An automorphic form at level K can be interpreted as a function on the set of rank n vector bundles on X. Fix $\mathcal{E}$, a rank n-vector bundle on X. We know $\mathcal{E}$ must split completeley:

$$\mathcal{E} = \mathcal{O}(k_1) \oplus \cdots \mathcal{O}(k_n)$$

we may assume $k_1 \geq k_2 \geq \cdots k_n$. Let $\mathcal{E}' = \mathcal{O}(k_2) \oplus \cdots \oplus \mathcal{O}(k_n)$, it is well-known:

$$\mathrm{Ext}^1(\mathcal{O}(k_1), \mathcal{E}') = 0$$

Now if f is cuspidal, we must have:

$$\sum_{0 \to \mathcal{O}(k_1) \to \mathcal{V} \to \mathcal{E}' \to 0} f(\mathcal{V}) = 0$$

But the only choice of $\mathcal{V}$ is $\mathcal{V} = \mathcal{E}$. It follows that $f(\mathcal{E}) = 0$. □

Exercise B.26. *In class we learned that cusp forms are compactly supported modulo the center. This exercise is about the converse. Let $G = \mathrm{GL}_2$, and f be a $\mathbb{C}$-valued function on $G(F) \setminus G(\mathbb{A}_F)/K$ that has compact support modulo $Z(\mathbb{A}_F)$. Let $x \in |X|$. Recall that the spherical Hecke algebra*

$$\mathcal{H}_x = C_c(G(\mathcal{O}_x) \setminus G(F_x)/G(\mathcal{O}_x))$$

acts on $C(G(F) \setminus G(\mathbb{A}_F)/K)$ by right convolution. Assume f is an eigenvector for the action of $\mathcal{H}_x$ for some $x \in |X|$. Show that f is a cusp form. (Hint: the assignment $c_f : t \mapsto \int_{N(F)\setminus N(\mathbb{A}_F)} f(nt)dn$ gives a function on $T(F) \setminus T(\mathbb{A}_F)/K_T$. From the eigenproperty of f, deduce a functional equation for c_f).

Proof. Let $T_x = \mathbf{1}_{K\begin{pmatrix}\varpi_x & \\ & 1\end{pmatrix}K}$ be the main generator of the Hecke algebra $\mathcal{H}_x$. Let's write $\varpi = \varpi_x$ to ease notation. The eigenproperty reads: there exist eigenvalue λ such that:

$$T_x \cdot f = \lambda \cdot f$$

But we also know for $g \in G(\mathbb{A})$:

$$(T_x \cdot f)(g) = \sum_{a \in \mathbb{F}_q} f(g \cdot \begin{pmatrix}\varpi & a \\ & 1\end{pmatrix}) + f(g \cdot \begin{pmatrix}1 & \\ & \varpi\end{pmatrix})$$

Now we have:

$$\lambda c_f(t) = c_{T_x \cdot f}(t) = \int_{n \in N(F)\setminus N(\mathbb{A})} f(nt\begin{pmatrix}1 & \\ & \varpi\end{pmatrix})dn + \sum_{a \in \mathbb{F}_q} \int_{n \in N(F)\setminus N(\mathbb{A})} f(nt\begin{pmatrix}\varpi & a \\ & 1\end{pmatrix})dn$$

Now let's look at each term on the right hand side closely. Indeed:

$$\int_{n \in N(F)\setminus N(\mathbb{A})} f(nt\begin{pmatrix}1 & \\ & \varpi\end{pmatrix}) = c_f(t\begin{pmatrix}1 & \\ & \varpi\end{pmatrix})$$

and if $t = \begin{pmatrix} t_1 & \\ & t_2 \end{pmatrix}$

$$nt\begin{pmatrix} \varpi & a \\ & 1 \end{pmatrix} = n\begin{pmatrix} t_1 & \\ & t_2 \end{pmatrix}\begin{pmatrix} \varpi & a \\ & 1 \end{pmatrix}\begin{pmatrix} \varpi^{-1}t_1^{-1} & \\ & t_2^{-1} \end{pmatrix}\begin{pmatrix} \varpi t_1 & \\ & t_2 \end{pmatrix} = nn'\begin{pmatrix} \varpi t_1 & \\ & t_2 \end{pmatrix}$$

for some fixed $n' \in N(\mathbb{A}_F)$. It follows that:

$$\int_{n\in N(F)\backslash N(\mathbb{A})} f(nt\begin{pmatrix} \varpi & a \\ & 1 \end{pmatrix})dn = c_f(t\begin{pmatrix} \varpi & \\ & 1 \end{pmatrix})$$

Therefore, if we identify $T(F) \setminus T(\mathbb{A}_F)/K_T \simeq (\mathrm{Pic}X)^2$, then:

$$\lambda c_f(\mathcal{L}_1, \mathcal{L}_2) = qc_f(\mathcal{L}_1(-x), \mathcal{L}_2) + c_f(\mathcal{L}_1, \mathcal{L}_2(-x))$$

which can be rewritten as:

$$c_f(\mathcal{L}_1, \mathcal{L}_2) = \lambda c_f(\mathcal{L}_1, \mathcal{L}_2(x)) - qc_f(\mathcal{L}_1(-x), \mathcal{L}_2(x))$$

Now f has compact support modulo $Z(\mathbb{A})$ implies that c_f has finite support modulo the diagonal of $(\mathrm{Pic}X)^2$. In particular $c_f(\mathcal{L}_1, \mathcal{L}_2) = 0$ for $\deg \mathcal{L}_2 - \deg \mathcal{L}_1 \ll 0$. But if there exists $\mathcal{L}_1, \mathcal{L}_2$ with $c_f(\mathcal{L}_1, \mathcal{L}_2) \neq 0$, then one of $c_f(\mathcal{L}_1, \mathcal{L}_2(x))$ and $c_f(\mathcal{L}_1(-x), \mathcal{L}_2(x))$ will be nonzero. But either of these have smaller value of $\deg \mathcal{L}_2 - \deg \mathcal{L}_1$. This is a contradiciton. Therefore $c_f \equiv 0$ and f is cuspidal. □

B.8 Homework 8

Exercise B.27. *Let F be a local non-archimedean field with a uniformizer ϖ. Let $\psi : F \to \mathbb{C}^\times$ be a character with conductor $\mathcal{O}_F$. Let dx be a Haar measure on F with $\int_{\mathcal{O}_F} dx = 1$. Let $G = \mathrm{GL}_2(F)$, $\chi_1, \chi_2 : F^\times \to \mathbb{C}^\times$ unramified characters with $\chi_i(\varpi) = \alpha_i$. Let $\pi : \mathrm{Ind}_B^G(\chi_1, \chi_2)$ with space V_π and spherical vector $\phi_0 \in V_\pi$ such that $\phi_0(1) = 1$. Consider the Whittaker functional: $\Lambda : V_\pi \to \mathbb{C}$ given by:*

$$\Lambda(\phi) = \int_F \phi(\begin{pmatrix} 0 & -1 \\ 1 & x \end{pmatrix})\psi(-x)dx$$

Let W_0 be the element in the Whittaker model for π corresponding to ϕ_0. Compute the value of W_0 on the diagonal matrices $\begin{pmatrix} \varpi^m & \\ & 1 \end{pmatrix}$ for all $m \in \mathbb{Z}$ in terms of α_1, α_2.

Proof. First note that:

$$W_0(\begin{pmatrix} \varpi^m & \\ & 1 \end{pmatrix}) = \int_F \phi_0(\begin{pmatrix} & -1 \\ 1 & x \end{pmatrix} \begin{pmatrix} \varpi^m & \\ & 1 \end{pmatrix})\psi(-x)dx$$

Therefore we need to write:

$$\begin{pmatrix} & -1 \\ 1 & x \end{pmatrix} \begin{pmatrix} \varpi^m & \\ & 1 \end{pmatrix}$$

in terms of Iwasawa decomposition. Let $t \in F$ to be determined. Let $x = u\varpi^k$ for some $u \in \mathcal{O}_F^\times$. We compute:

$$\begin{pmatrix} 1 & t \\ & 1 \end{pmatrix} \begin{pmatrix} & -1 \\ 1 & x \end{pmatrix} \begin{pmatrix} \varpi^m & \\ & 1 \end{pmatrix} = \begin{pmatrix} \varpi^m t & tu\varpi^k - 1 \\ \varpi^m & u\varpi^k \end{pmatrix}$$

For $m \geq k$, we can choose $t = x^{-1}$ and then:

$$\begin{pmatrix} \varpi^m t & tu\varpi^k - 1 \\ \varpi^m & u\varpi^k \end{pmatrix} = \begin{pmatrix} u^{-1}\varpi^{m-k} & 0 \\ \varpi^m & u\varpi^k \end{pmatrix} = \begin{pmatrix} \varpi^{m-k} & \\ & \varpi^k \end{pmatrix} \begin{pmatrix} u^{-1} & 0 \\ \varpi^{m-k} & u \end{pmatrix}$$

For $m < k$, we choose $t = \varpi^{-m}$ and therefore:

$$\begin{pmatrix} \varpi^m t & tu\varpi^k - 1 \\ \varpi^m & u\varpi^k \end{pmatrix} = \begin{pmatrix} 1 & u\varpi^{k-m} - 1 \\ \varpi^m & u\varpi^k \end{pmatrix} = \begin{pmatrix} 1 & \\ & \varpi^m \end{pmatrix} \begin{pmatrix} 1 & u\varpi^{k-m} - 1 \\ 1 & u\varpi^{m-k} \end{pmatrix}$$

In summary, for $x = u\varpi^k$, we have Iwasawa decomposition:

$$\begin{pmatrix} & -1 \\ 1 & x \end{pmatrix} \begin{pmatrix} \varpi^m & \\ & 1 \end{pmatrix} = \begin{cases} \begin{pmatrix} 1 & -x^{-1} \\ & 1 \end{pmatrix} \begin{pmatrix} \varpi^{m-k} & \\ & \varpi^k \end{pmatrix} \begin{pmatrix} u^{-1} & 0 \\ \varpi^{m-k} & u \end{pmatrix}, & \text{if } m \geq k; \\ \begin{pmatrix} 1 & -\varpi^{-m} \\ & 1 \end{pmatrix} \begin{pmatrix} 1 & \\ & \varpi^m \end{pmatrix} \begin{pmatrix} 1 & u\varpi^{k-m} - 1 \\ 1 & u\varpi^{m-k} \end{pmatrix}, & \text{if } m < k. \end{cases}$$

It follows that:

$$R = W_0(\begin{pmatrix} \varpi^m & \\ & 1 \end{pmatrix}) = \sum_{k=-\infty}^{m-1} q^{k-\frac{m}{2}} \chi_1(\varpi^{m-k})\chi_2(\varpi^k) \int_{x\in\varpi^k\mathcal{O}_F^\times} \psi(-x)dx + \sum_{k=m}^{\infty} q^{\frac{m}{2}} \chi_2(\varpi^m) \int_{x\in\varpi^k\mathcal{O}_F^\times} \psi(-x)dx$$
$$= q^{-\frac{m}{2}}\alpha_1^m \sum_{k=-\infty}^{m-1} (\frac{q\alpha_2}{\alpha_1})^k \int_{x\in\varpi^k\mathcal{O}_F^\times} \psi(-x)dx + q^{\frac{m}{2}}\alpha_2^m \sum_{k=m}^{\infty} \int_{x\in\varpi^k\mathcal{O}_F^\times} \psi(-x)dx$$

But we know:

$$\int_{x\in\varpi^k\mathcal{O}_F^\times} \psi(-x)dx = \begin{cases} 0, & \text{if } k \le -2; \\ -1, & \text{if } k = -1; \\ \dfrac{q-1}{q^{k+1}}, & \text{otherwise.} \end{cases}$$

Therefore if $m \le -1$, $R = 0$. If $m \ge 0$, then:

$$R = q^{-\frac{m}{2}}\alpha_1^m(-\frac{\alpha_1}{q\alpha_2} + \sum_{k=0}^{m-1} \frac{q-1}{q}(\frac{\alpha_2}{\alpha_1})^k) + q^{\frac{m}{2}}\alpha_2^m \frac{1}{q^m}$$
$$= -q^{-\frac{m}{2}-1}\alpha_1^{m+1}\alpha_2^{-1} + q^{-\frac{m}{2}-1}(q-1)\frac{\alpha_1^m - \alpha_2^m}{1-\alpha_2/\alpha_1} + q^{-\frac{m}{2}}\alpha_2^m$$
$$= q^{-\frac{m}{2}-1}\frac{(\alpha_1^{m+1} - \alpha_2^{m+1})(q - \frac{\alpha_1}{\alpha_2})}{\alpha_1 - \alpha_2}$$

In summary:

$$W_0(\begin{pmatrix} \varpi^m & \\ & 1 \end{pmatrix}) = \begin{cases} q^{-\frac{m}{2}-1}\dfrac{(\alpha_1^{m+1} - \alpha_2^{m+1})(q - \frac{\alpha_1}{\alpha_2})}{\alpha_1 - \alpha_2}, & \text{if } m \ge 0; \\ 0, & \text{otherwise.} \end{cases}$$

□

Exercise B.28. *Let F be a global funciton field. Let $\psi : F \setminus \mathbb{A} \to \mathbb{C}^\times$ be a nontrivial character. Let $N_n \subset G_n = \mathrm{GL}_n$ be the subgroup of upper triangular unipotent matrices, and $\psi_n : N_n(F) \setminus N_n(\mathbb{A}) \to \mathbb{C}^\times$ be given by:*

$$\psi_n(u) = \psi(u_{1,2} + u_{2,3} + \cdots + u_{n-1,n}), u \in N_n(\mathbb{A})$$

Let $\mathcal{C}((N_n(\mathbb{A}), \psi_n) \setminus G_n(\mathbb{A}))$ be the global Whittaker space consisting of functions ϕ on $G_n(\mathbb{A})$ such that:

$$\phi(ug) = \psi_n(u)\phi(g)$$

and ϕ is right invariant under some compact open subgroup of $G_n(\mathbb{A})$. For a place v of F, we similarly have the local Whittaker space $\mathcal{C}((N_n(F_v), \psi_{n,v}) \setminus G_n(F_v))$.

1. *Let $\phi_v \in \mathcal{C}((N_n(F_v), \psi_{n,v}) \setminus G_n(F_v))$. Show that there exists $C > 0$ (depending on ϕ_v) such that the support of ϕ_v is contained in the matrices of the form*

$$u \cdot \mathrm{diag}(a_1, \cdots, a_n) \cdot k$$

where $u \in N_n(F_v), k \in G_n(\mathcal{O}_v)$ and $a_i \in F_v^\times$ satisfy $|\frac{a_i}{a_{i+1}}| \leq C$ for $i = 1, \cdots, n-1$.

2. *Let $\phi \in \mathcal{C}(N_n(\mathbb{A}, \psi_n) \setminus G_n(\mathbb{A}))$. Let ϕ_{n-1} be the restriction of ϕ to $G_{n-1}(\mathbb{A}) \subset G_n(\mathbb{A})$ (embedded as $g \mapsto \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix}$), so that $\phi_{n-1} \in \mathcal{C}((N_{n-1}(\mathbb{A}), \psi_{n-1}) \setminus G_{n-1}(\mathbb{A}))$. Show that the restriction of ϕ_{n-1} to $N_{n-1}(F) \setminus G_{n-1}(\mathbb{A})^1$ has compact support, where*

$$G_{n-1}(\mathbb{A})^1 = \ker(G_{n-1}(\mathbb{A}) \xrightarrow{\det} \mathbb{A}^\times \xrightarrow{|-|} \mathbb{R}_{>0})$$

Proof. 1. First of all we since ψ_n is non-constant, so that $\psi_{n,v}$ is non-constant for each place v. Now let's fix v, and choose $u \in N_n(F_v)$ such that $\psi_{n,v}(u) \neq 1$. Let $a = \mathrm{diag}(a_1, \cdots, a_n) \in A$. Let's consider the following equation:

$$\phi_v(a \cdot (a^{-1}ua)) = \phi_v(u \cdot a) = \psi_{n,v}(u)\phi_v(a)$$

If $a^{-1}ua \in G_n(\mathcal{O})$, then the equation becomes:

$$\phi_v(a) = \psi_{n,v}(u)\phi_v(a) \implies \phi_v(a) = 0$$

since $\psi_{n,v}(u) \neq 1$. Now let's try to fulfill the condition $a^{-1}ua \in G_n(\mathcal{O})$. This is equivalent to:

$$|a_i^{-1}a_j u_{ij}|_v \geq 1, \quad \forall i < j$$

Now let's choose $C > |u_{i,i+1}|_v$. Then if I have a such that:

$$|a_i a_{i+1}^{-1}| > C \implies 1 > |a_i^{-1}a_{i+1}|_v C > |a_i^{-1}a_{i+1}u_{i,i+1}|_v$$

which implies $\phi_a = 0$ by previous argument. Using the eigenproperty of $N_n(F_v)$ and $G_n(\mathcal{O}_v)$-actions, we see the support is indeed as hoped.

2. This follows almost immediately from the previous part. Note previous part tells me each local component $\phi_{n-1,v}$ has support on:

$$\{n \cdot a \cdot k \mid n \in N_{n-1}(F_v), |a_i|_v \leq C_v^{n-i}, k \in G_{n-1}(\mathcal{O}_v)\}$$

Therefore, ϕ_{n-1} as a section of line bundle on $N_{n-1}(\mathbb{A}) \setminus G_{n-1}(\mathbb{A})^1$ has support:

$$\{\mathrm{diag}(a_1, \cdots, a_{n-1}) \mid |a_i|_v \leq C_v^{n-i}, \prod_i \prod_v |a_i|_v = 1\} \cdot K$$

which is compact. We conclude by noting that $N(F) \setminus N(\mathbb{A}) \simeq (F \setminus \mathbb{A})^{\oplus m}$ for some m, which is compact.

□

Exercise B.29. *Continuing from the previous exercise. Let $Q_n \subset G_n = \mathrm{GL}_{n-1}$ be the mirabolic subgroup:*

$$\begin{pmatrix} \mathrm{GL}_{n-1} & * \\ 0 & 1 \end{pmatrix}$$

Let $\mathcal{C}_{\mathrm{cusp}}(Q_n(F) \setminus G_n(\mathbb{A}))$ be the space of cuspidal functions on $Q_n(F) \setminus G_n(\mathbb{A})$ that are right invariant under some compact open subgroup of $G_n(\mathbb{A})$. Here $f \in \mathcal{C}(Q_n(F) \setminus G_n(\mathbb{A}))$ is called cuspidal if:

$$\int_{U(F)\setminus U(\mathbb{A})} f(ug)du = 0, \quad \forall g \in G_n(\mathbb{A})$$

for $U = U_{m,n-m} = \begin{pmatrix} I_m & * \\ 0 & I_{n-m} \end{pmatrix}$, for all $1 \leq m \leq n-1$.

1. *Show that the map:*

$$\mathcal{C}((N_n(\mathbb{A}), \psi_n) \setminus G_n(\mathbb{A})) \to \mathcal{C}(Q_n(F) \setminus G_n(\mathbb{A}))$$
$$\phi \mapsto (f_\phi : g \mapsto \sum_{\gamma \in N_{n-1}(F)\setminus G_{n-1}(F)} \phi(\gamma g))$$

   is well-defined, i.e. the summation is finite.

2. *Show that f_ϕ defined above is cuspidal.*

3. *Show that the map $\phi \mapsto f_\phi$ defines an isomorphism of $G(\mathbb{A})$-representations:*

$$\mathcal{C}((N_n(\mathbb{A}), \psi_n) \setminus G_n(\mathbb{A})) \xrightarrow{\sim} \mathcal{C}_{\mathrm{cusp}}(Q_n(F) \setminus G_n(\mathbb{A}))$$

4. *Deduce that each cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb{A})$ appears with multiplicity one in the space of automorphic forms $\mathcal{A}_G$.*

Proof. 1. Note that $g \cdot \phi \in \mathcal{C}(N_n(\mathbb{A}, \psi_n) \setminus G_n(\mathbb{A}))$ for any $g \in G(\mathbb{A})$, therefore its restriciton to $N_{n-1}(F) \setminus G_{n-1}(\mathbb{A})^1$ has compact support. Namely, there exists compact subset $K \subset N_{n-1}(F) \setminus G_{n-1}(\mathbb{A})^1$ such that: for $\gamma \in N_{n-1} \setminus G_{n-1}(\mathbb{A})$ with $f(\gamma g) \neq 0 \implies \gamma \in K$. Therefore we're summing over the space:

$$N_{n-1}(F) \setminus G_{n-1}(F) \cap K$$

which is finite (Note that $G_{n-1}(F) \subset G_{n-1}(\mathbb{A})^1$). Moreover, for $q = \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix} \in Q_n(F), \gamma = \begin{pmatrix} B & \\ & 1 \end{pmatrix} \in G_{n-1}(F)$ we have:

$$\gamma q = \begin{pmatrix} BA & Bv \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & Bv \\ & 1 \end{pmatrix} \begin{pmatrix} BA & \\ & 1 \end{pmatrix}$$

Suppose we call $\gamma' = \begin{pmatrix} BA & \\ & 1 \end{pmatrix}$, then $f(\gamma q g) = f(\gamma' g)$ for all $g \in G_n(\mathbb{A})$. But for fixed q, then $\gamma \mapsto \gamma'$ is a bijection on $N_{n-1}(F) \setminus G_{n-1}(F)$. This shows the invariance of $Q_{n-1}(F)$-action of the target. It is also clear that this map is $G(\mathbb{A})$-equivariant.

2. We claim there is an isomorphism:

$$\mathcal{C}_{\text{cusp}}(Q_n(F)\setminus G_n(\mathbb{A}))\to\mathcal{C}_{\text{cusp}}\left(Q_{n-1}(F)\cdot\left(\begin{pmatrix}I_{n-1}&*\\&1\end{pmatrix},\psi_n\right)\setminus G_n(\mathbb{A})\right)$$

For RHS the cuspidality condition means We only impose the condition for $U=U_{m,n-m}$ with $m\leq n-2$. The map is given by:

$$f\mapsto\left(f_1^{\psi_n}:g\mapsto\int_{(F\setminus\mathbb{A})^{n-1}}f\left(\begin{pmatrix}1&0&\cdots&x_{n-1}\\0&1&\cdots&\vdots\\&&\ddots&x_1\\&&&1\end{pmatrix}g\right)\psi_n(-x_1)dx_1\cdots dx_{n-1}\right)$$

Conceptually, if we define:

$$\theta_{f,g}:(F\setminus\mathbb{A})^{n-1}\to\mathbb{C}$$

$$\begin{pmatrix}x_{n-1}\\\vdots\\x_1\end{pmatrix}\mapsto f\left(\begin{pmatrix}1&0&\cdots&x_{n-1}\\0&1&\cdots&\vdots\\&&\ddots&x_1\\&&&1\end{pmatrix}g\right)$$

Then $f_1^{\psi_n}(g)=\hat{\theta}_{f,g}(\psi_n)$. Note any nonzero $\psi:(F\setminus\mathbb{A})^{n-1}\to\mathbb{C}$ can be written as:

$$\psi=\psi_n\circ\gamma$$

for a unique $\gamma\in\mathrm{GL}_{n-1}(F)/Q_{n-1}(F)$. Now by a direct change of variable we see $f_1^{\psi_n}(\gamma g)=f_1^{\psi_n\circ\gamma}(g)=\hat{\theta}_{f,g}(\psi_n\circ\gamma)$. In particular f_1 is $Q_{n-1}(F)$-invariant. It is also easy to check the cuspidality of $f_1^{\psi_n}$ coming from the cuspidality of f.

Note this also checks our map is well-defined (namely the $Q_{n-1}(F)$-invariance property). Therefore, by Fourier inversion formula:

$$f(g)=\theta_{f,g}(0)=\sum_{\gamma\in\mathrm{GL}_{n-1}(F)/Q_{n-1}(F)}\hat{\theta}_{f,g}(\psi_n\circ\gamma)+\hat{\theta}_{f,g}(0)$$

Recall that we established that:

$$\hat{\theta}_{f,g}(\psi_n\circ\gamma)=f_1^{\psi_n}(\gamma g)$$

and by cuspidality of f:

$$\hat{\theta}_{f,g}(0)=\int_{u\in U(F)\setminus U(\mathbb{A})}f(ug)du=0$$

for $U=U_{n-1,1}$. It follows that:

$$f(g)=\sum_{\gamma\in\mathrm{GL}_{n-1}(F)/Q_{n-1}(F)}f_1^{\psi_n}(\gamma g)$$

Namely we can recover f from $f_1^{\psi_n}$. So the map we defined is indeed a bijection, with inverse:

$$\lambda \mapsto \sum_{\gamma \in \mathrm{GL}_{n-1}(F)/Q_{n-1}(F)} \lambda(\gamma g)$$

From this we can now inductively establish isomorphsims:

$$\mathcal{C}((N_n(\mathbb{A}), \psi_n) \setminus G_n(\mathbb{A})) \xrightarrow{\sim} \cdots \xrightarrow{\sim} \mathcal{C}_{\mathrm{cusp}}\left(Q_{n-k}(F) \cdot \left(\begin{pmatrix} I_{n-k} & * \\ & I_k \end{pmatrix}, \psi_n\right) \setminus G_n(\mathbb{A})\right) \xrightarrow{\sim}$$

$$\mathcal{C}_{\mathrm{cusp}}\left(Q_{n-k+1}(F) \cdot \left(\begin{pmatrix} I_{n-k+1} & * \\ & I_{k-1} \end{pmatrix}, \psi_n\right) \setminus G_n(\mathbb{A})\right) \xrightarrow{\sim} \cdots \xrightarrow{\sim} \mathcal{C}_{\mathrm{cusp}}(Q_n(F) \setminus G_n(\mathbb{A}))$$

Now for the k-th step we're mapping:

$$\phi_{n-k} \mapsto (\phi_{n-k+1} : g \mapsto \sum_{\gamma \in Q_{n-k+1}(F)/Q_{n-k}(F)} \phi_{n-k}(\gamma g))$$

Composing all these isomorphisms, we see the image of $\phi = \phi_0$ is indeed f_ϕ as desired.

3. It is established in the previous part.

4. Note now I have $G(\mathbb{A})$-equivariant embeddings:

$$\mathcal{A}_{\mathrm{cusp}} \hookrightarrow \mathcal{C}_{\mathrm{cusp}}(Q_n(F) \setminus G_n(\mathbb{A})) \xrightarrow{\sim} \mathcal{C}((N_n(\mathbb{A}), \psi_n) \setminus G_n(\mathbb{A}))$$

Therefore if π is a cuspidal representation, then:

$$\mathrm{Hom}_{G(\mathbb{A})}(\pi, \mathcal{A}_{\mathrm{cusp}}) \hookrightarrow \mathrm{Hom}_{G(\mathbb{A})}(\pi, \mathcal{C}((N_n(\mathbb{A}), \psi_n) \setminus G(\mathbb{A}))) \simeq \mathrm{Hom}_{N(\mathbb{A})}(\pi, \mathbb{C}_{\psi_n})$$

Note the last isomorphism is Frobenius reciprocity, and we have no moderate growth condition since we're working in function fields. Moreover, the last space has dimension one by uniqueness of global Whittaker model. Therefore the first space has dimension ≤ 1. Hence the multiplicity one theorem is proved.

□